Per-sequence summary: mean UWB epoch interval (= real-time budget of one solve) and 3D position RMSE [m]

dataset	sequence	budget [ms]	desktop solve mean/med/p95 [ms]	KIMS	LM	SFUISE	filter
MILUV	default_1_remoteControlHighPace_1	14.4	9.7 / 5.3 / 53.2	0.205	0.239	div.	1.319
MILUV	default_1_random3_1	14.5	-	0.231	0.269	0.488	0.255
MILUV	default_1_random3_0	14.1	7.6 / 5.2 / 13.2	0.165	0.682	0.704	0.217
MILUV	default_1_circular3D_0	14.2	-	0.093	0.093	0.123	0.303
MILUV	default_1_remoteControlLowPace_0_v2	14.4	-	0.086	0.089	0.114	0.219
MILUV	default_1_remoteControlLowPace_0	14.3	-	0.202	0.219	0.955	0.247
MILUV	default_1_circular2D_0	14.7	9.1 / 5.3 / 49.6	0.134	0.133	0.192	0.335
MILUV	default_1_remoteControlHighPace_0	14.6	-	0.100	0.097	div.	1.358
MILUV	obstacles_1_random3_0b	14.0	-	0.130	0.595	0.639	0.252
MILUV	default_1_random3_2	15.2	-	0.135	0.140	div.	0.265
MILUV	default_1_remoteControlHighPace_2	15.0	-	0.244	0.241	0.763	1.414
MILUV	noAprilTags_1_random3_2b	14.6	-	0.130	0.133	0.368	0.241
MILUV	noAprilTags_1_remoteControlHighPace_2b	15.3	-	0.191	0.197	1.144	0.273
NTU VIRAL	eee_01	17.0	3.8 / 3.8 / 3.9	0.204	0.200	0.126	-
NTU VIRAL	eee_02	18.0	-	0.367	0.363	2.537	-
NTU VIRAL	eee_03	16.6	-	1.845	1.861	1.144	-
NTU VIRAL	nya_01	21.4	-	0.791	0.785	div.	-
NTU VIRAL	nya_02	17.0	-	3.084	3.083	0.395	-
NTU VIRAL	nya_03	22.1	-	0.903	0.997	div.	-
NTU VIRAL	sbs_01	15.8	3.8 / 3.7 / 3.9	0.190	0.183	0.207	-
NTU VIRAL	sbs_02	19.9	-	0.409	0.633	3.109	-
NTU VIRAL	sbs_03	15.8	-	0.435	0.433	3.146	-
IULUI	SquareTurn	10.0	5.3 / 5.3 / 5.4	0.120	0.548	0.106	-
IULUI	CircleSpin	10.0	-	0.118	0.418	0.235	-
IULUI	AC3_125000	10.0	-	0.128	0.175	0.300	-
UTIL	const1-trial1-tdoa2	20.0	6.1 / 4.1 / 10.2	0.096	0.096	0.095	0.108
UTIL	const1-trial1-tdoa3	20.0	4.6 / 4.2 / 6.8	0.100	0.100	0.094	0.131
UTIL	const1-trial2-tdoa2	20.0	-	0.112	0.112	0.115	0.122
UTIL	const1-trial2-tdoa3	20.0	-	0.110	0.111	0.104	0.143
UTIL	const1-trial3-tdoa2	20.0	-	0.098	0.098	0.094	0.108
UTIL	const1-trial3-tdoa3	20.0	-	0.097	0.097	0.090	0.120
UTIL	const1-trial4-tdoa2	20.0	-	0.101	0.102	0.099	0.111
UTIL	const1-trial4-tdoa3	20.0	-	0.097	0.097	0.090	0.124
UTIL	const1-trial5-tdoa2	20.0	-	0.107	0.107	0.103	0.114
UTIL	const1-trial5-tdoa3	20.0	-	0.112	0.112	0.108	0.139
UTIL	const1-trial6-tdoa2	20.0	-	0.111	0.111	0.104	0.123
UTIL	const1-trial6-tdoa3	20.0	-	0.108	0.109	0.103	0.134
UTIL	const2-trial1-tdoa2	20.0	-	0.095	0.095	0.094	0.103

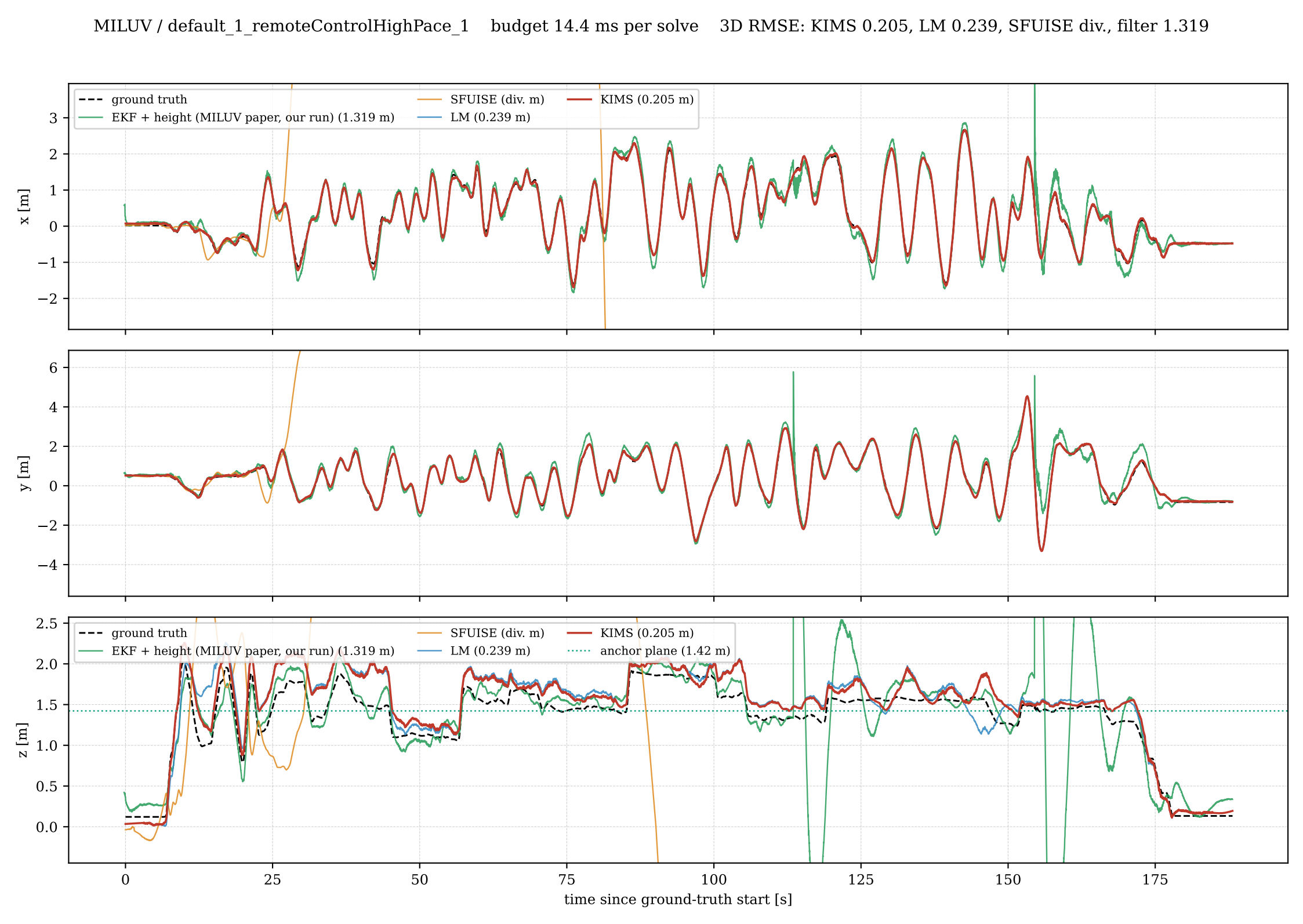
budget: mean interval between UWB epochs (one KIMS solve per epoch); a solve whose mean time stays below it keeps up with the sensor.
desktop solve: RTX 5070 Ti + i7-14700K, FP32 rollout, 8 host threads, run alone (paper Table VII; measured on these 8 flights only).
filter: MILUV = our run of the MILUV paper's EKF with the laser height sensor (Table 10 values are within 5 mm per axis); UTIL = our run of the devkit ESKF.
NTU VIRAL: Leica prism, yaw + translation fit per run, 10 s trimmed at both ends; other datasets: whole span, no alignment. div. = RMSE > 100 m.

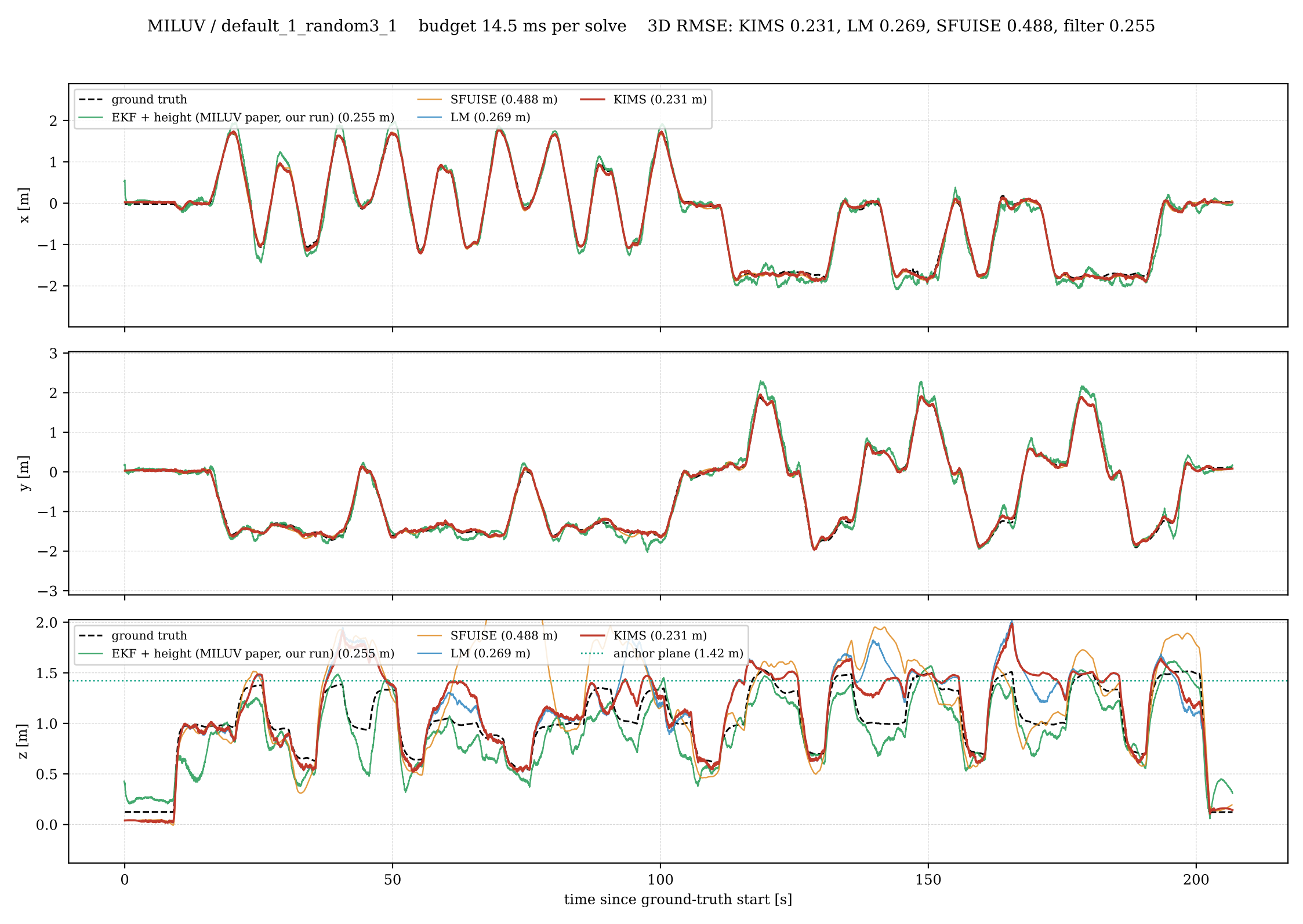
Per-sequence summary: mean UWB epoch interval (= real-time budget of one solve) and 3D position RMSE [m]

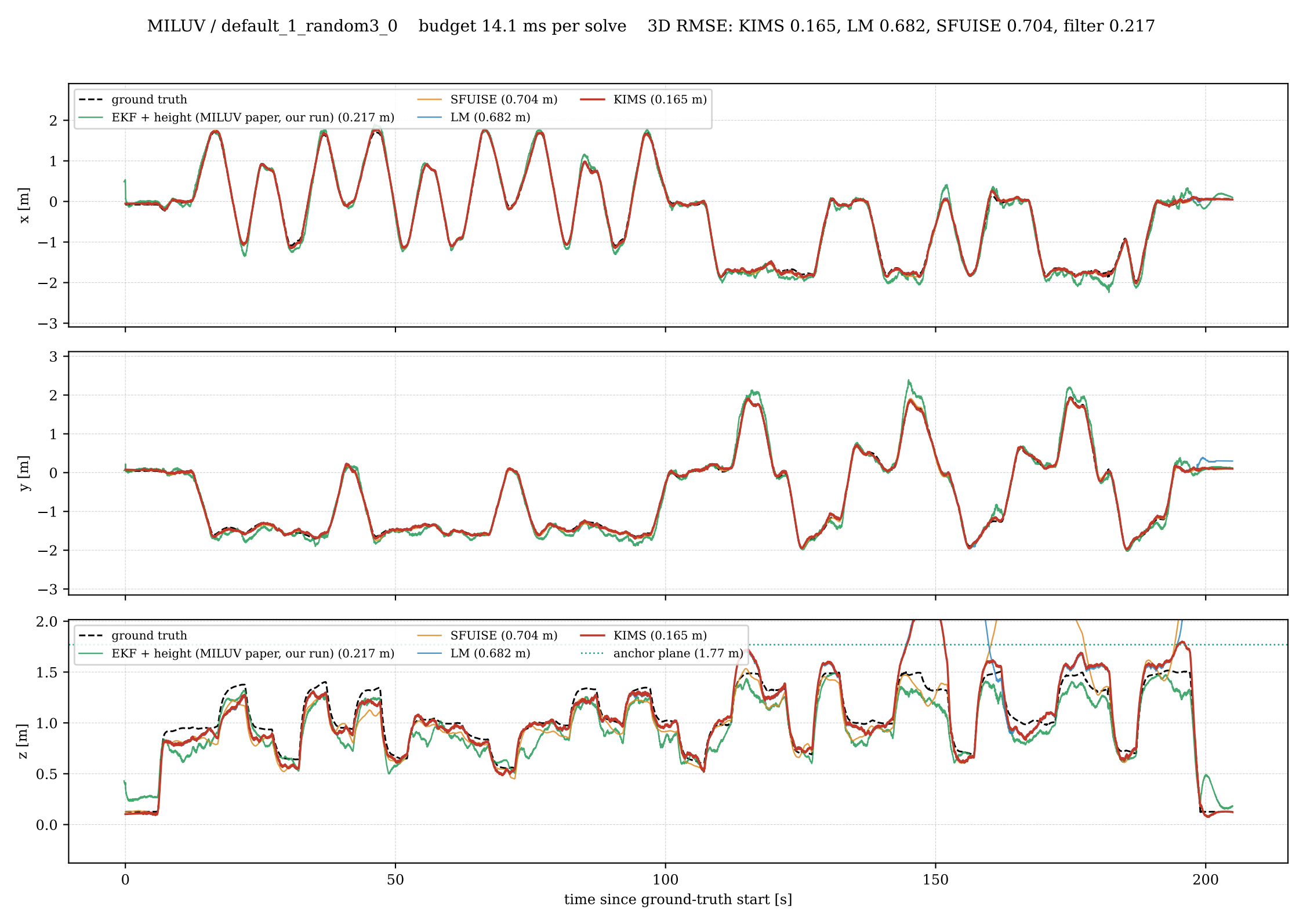
dataset	sequence	budget [ms]	desktop solve mean/med/p95 [ms]	KIMS	LM	SFUISE	filter
UTIL	const2-trial1-tdoa3	20.0	-	0.095	0.096	0.096	0.113
UTIL	const2-trial2-tdoa2	20.0	-	0.106	0.106	0.103	0.114
UTIL	const2-trial2-tdoa3	20.0	-	0.088	0.088	0.084	0.113
UTIL	const2-trial3-tdoa2	20.0	-	0.107	0.107	0.110	0.108
UTIL	const2-trial3-tdoa3	20.0	-	0.078	0.077	0.083	0.112
UTIL	const2-trial4-tdoa2	20.0	-	0.088	0.088	0.096	0.093
UTIL	const2-trial4-tdoa3	20.0	-	0.088	0.088	0.086	0.117
UTIL	const2-trial5-tdoa2	20.0	-	0.111	0.111	0.111	0.121
UTIL	const2-trial5-tdoa3	20.0	-	0.096	0.096	0.089	0.118
UTIL	const2-trial6-tdoa2	20.0	-	0.101	0.101	0.101	0.106
UTIL	const2-trial6-tdoa3	20.0	-	0.100	0.100	0.094	0.117
UTIL	const3-trial1-tdoa2	20.0	-	0.230	0.234	0.330	0.245
UTIL	const3-trial1-tdoa3	20.0	-	0.199	0.200	0.207	0.226
UTIL	const3-trial2-tdoa2	20.0	-	0.187	0.193	0.207	0.197
UTIL	const3-trial2-tdoa3	20.0	-	0.169	0.170	0.182	0.219
UTIL	const3-trial3-tdoa2	20.0	-	0.187	0.190	0.175	0.189
UTIL	const3-trial3-tdoa3	20.0	-	0.175	0.176	0.184	0.208
UTIL	const3-trial4-tdoa2	20.0	-	0.197	0.201	0.183	0.217
UTIL	const3-trial4-tdoa3	20.0	-	0.160	0.160	0.174	0.199
UTIL	const3-trial5-tdoa2	20.0	-	0.202	0.206	0.186	0.221
UTIL	const3-trial5-tdoa3	20.0	-	0.182	0.184	0.177	0.230
UTIL	const3-trial6-tdoa2	20.0	-	0.169	0.174	0.171	0.195
UTIL	const3-trial6-tdoa3	20.0	-	0.157	0.158	0.165	0.202
UTIL	const3-trial7-tdoa2-manual1	20.0	-	0.234	0.238	0.138	0.258
UTIL	const3-trial7-tdoa2-manual2	20.0	-	0.302	0.305	0.162	0.320
UTIL	const3-trial7-tdoa3-manual3	20.0	-	0.219	0.221	0.224	0.301
UTIL	const3-trial7-tdoa3-manual4	20.0	-	0.216	0.218	0.187	0.323
UTIL	const4-trial1-tdoa2-traj1	20.0	-	0.198	0.199	0.202	0.201
UTIL	const4-trial1-tdoa2-traj2	20.0	-	0.156	0.156	0.159	0.168
UTIL	const4-trial1-tdoa2-traj3	20.0	-	0.173	0.174	0.184	0.194
UTIL	const4-trial1-tdoa3-traj1	20.0	-	0.159	0.159	0.155	0.195
UTIL	const4-trial1-tdoa3-traj2	20.0	-	0.139	0.139	0.131	0.160
UTIL	const4-trial1-tdoa3-traj3	20.0	-	0.156	0.157	0.153	0.187
UTIL	const4-trial2-tdoa2-traj1	20.0	-	0.235	0.235	0.240	0.231
UTIL	const4-trial2-tdoa2-traj2	20.0	-	0.183	0.185	0.195	0.199
UTIL	const4-trial2-tdoa2-traj3	20.0	-	0.214	0.215	0.230	0.221
UTIL	const4-trial2-tdoa3-traj1	20.0	-	0.183	0.183	0.177	0.217
UTIL	const4-trial2-tdoa3-traj2	20.0	-	0.181	0.181	0.189	0.212

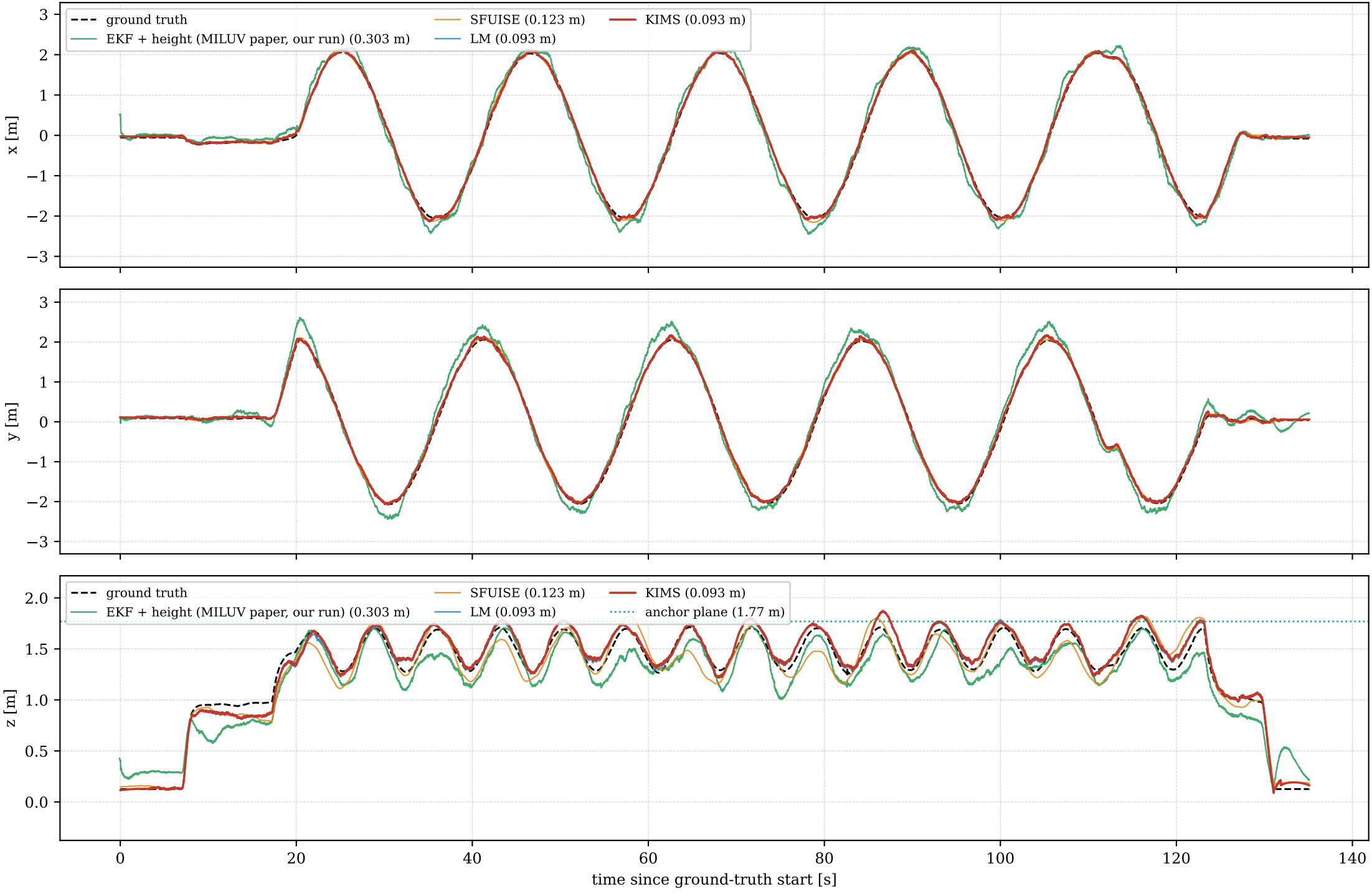
Per-sequence summary: mean UWB epoch interval (= real-time budget of one solve) and 3D position RMSE [m]

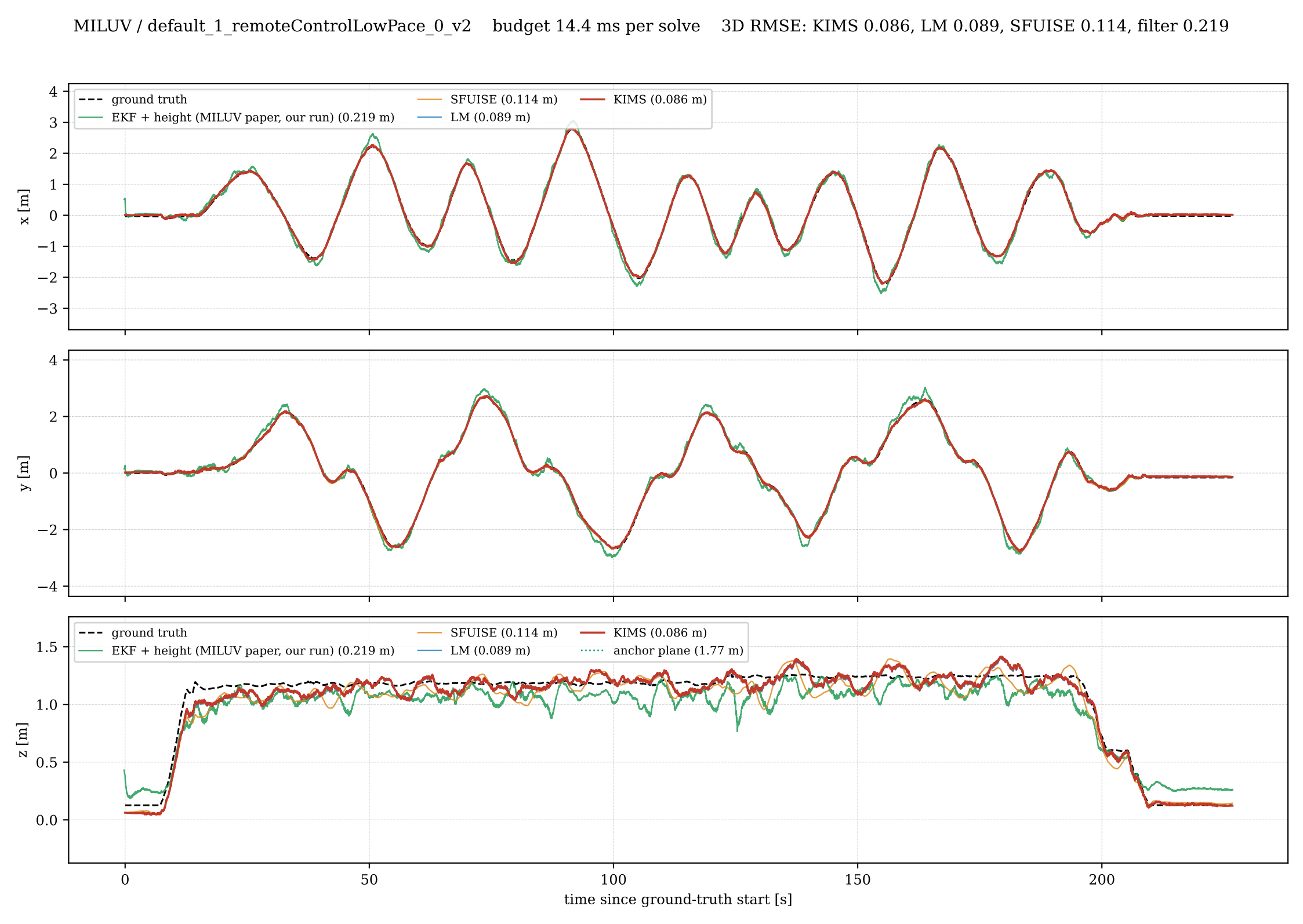
dataset	sequence	budget [ms]	desktop solve mean/med/p95 [ms]	KIMS	LM	SFUISE	filter
UTIL	const4-trial2-tdoa3-traj3	20.0	-	0.191	0.191	0.199	0.227
UTIL	const4-trial3-tdoa2-traj1	20.0	-	0.292	0.296	0.504	0.615
UTIL	const4-trial3-tdoa2-traj2	20.0	-	0.299	0.299	div.	0.592
UTIL	const4-trial3-tdoa2-traj3	20.0	-	0.307	0.308	0.449	0.514
UTIL	const4-trial3-tdoa3-traj1	20.0	-	0.242	0.242	0.216	0.431
UTIL	const4-trial3-tdoa3-traj2	20.0	-	0.233	0.233	0.722	0.405
UTIL	const4-trial3-tdoa3-traj3	20.0	-	0.244	0.245	0.221	0.555
UTIL	const4-trial4-tdoa2-traj1	20.0	-	0.595	0.597	div.	0.693
UTIL	const4-trial4-tdoa2-traj2	20.2	-	0.505	0.508	div.	0.627
UTIL	const4-trial4-tdoa2-traj3	20.0	-	0.489	0.491	1.296	0.562
UTIL	const4-trial4-tdoa3-traj1	20.0	-	0.433	0.433	0.399	0.617
UTIL	const4-trial4-tdoa3-traj2	20.0	-	0.396	0.396	0.349	0.631
UTIL	const4-trial4-tdoa3-traj3	20.0	-	0.403	0.403	0.447	0.650
UTIL	const4-trial5-tdoa2-traj1	20.0	-	0.380	0.393	0.272	0.411
UTIL	const4-trial5-tdoa2-traj2	20.0	-	0.262	0.264	0.218	0.426
UTIL	const4-trial5-tdoa2-traj3	20.0	-	0.286	0.287	0.227	0.363
UTIL	const4-trial5-tdoa3-traj1	20.2	-	0.274	0.275	div.	0.387
UTIL	const4-trial5-tdoa3-traj2	20.0	-	0.258	0.259	0.206	0.366
UTIL	const4-trial5-tdoa3-traj3	20.0	-	0.255	0.256	0.210	0.384
UTIL	const4-trial6-tdoa2-traj1	20.2	-	0.412	0.413	div.	0.454
UTIL	const4-trial6-tdoa2-traj2	20.0	-	0.352	0.353	0.314	0.384
UTIL	const4-trial6-tdoa2-traj3	20.0	-	0.394	0.395	0.351	0.458
UTIL	const4-trial6-tdoa3-traj1	20.0	-	0.380	0.381	0.320	0.556
UTIL	const4-trial6-tdoa3-traj2	20.0	-	0.401	0.402	0.320	0.540
UTIL	const4-trial6-tdoa3-traj3	20.0	-	0.391	0.392	0.347	0.523
UTIL	const4-trial7-tdoa2-manual1	20.0	-	0.204	0.218	0.184	0.281
UTIL	const4-trial7-tdoa2-manual2	20.0	-	0.206	0.281	0.159	0.302
UTIL	const4-trial7-tdoa2-manual3	20.0	-	0.211	0.212	0.190	0.270

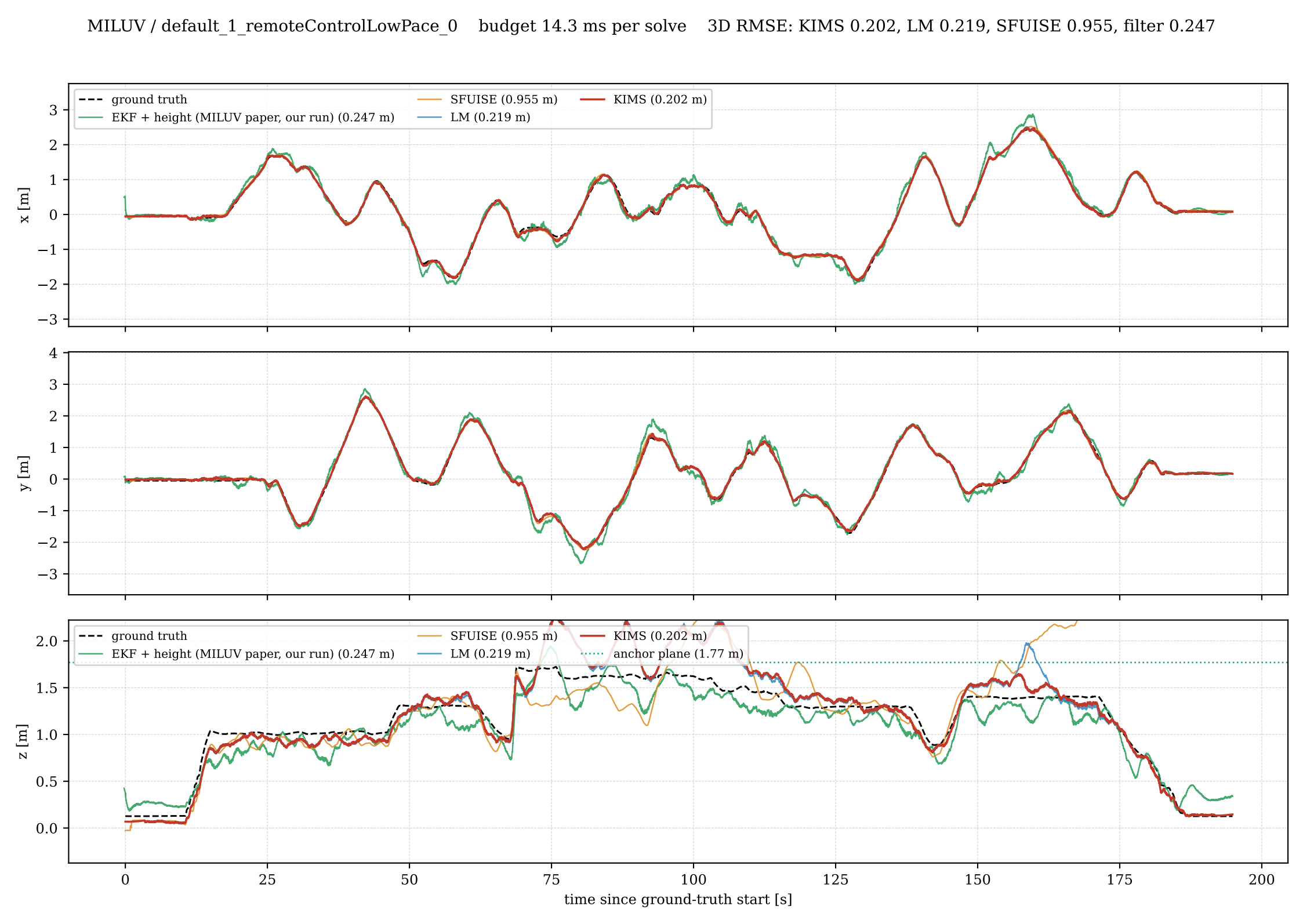

MILUV / default_1_remoteControlHighPace_1 budget 14.4 ms per solve 3D RMSE: KIMS 0.205, LM 0.239, SFUISE div., filter 1.319
ground truth
EKF + height (MILUV paper, our run) (1.319 m)
SFUISE (div. m)
LM (0.239 m)
KIMS (0.205 m)
anchor plane (1.42 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

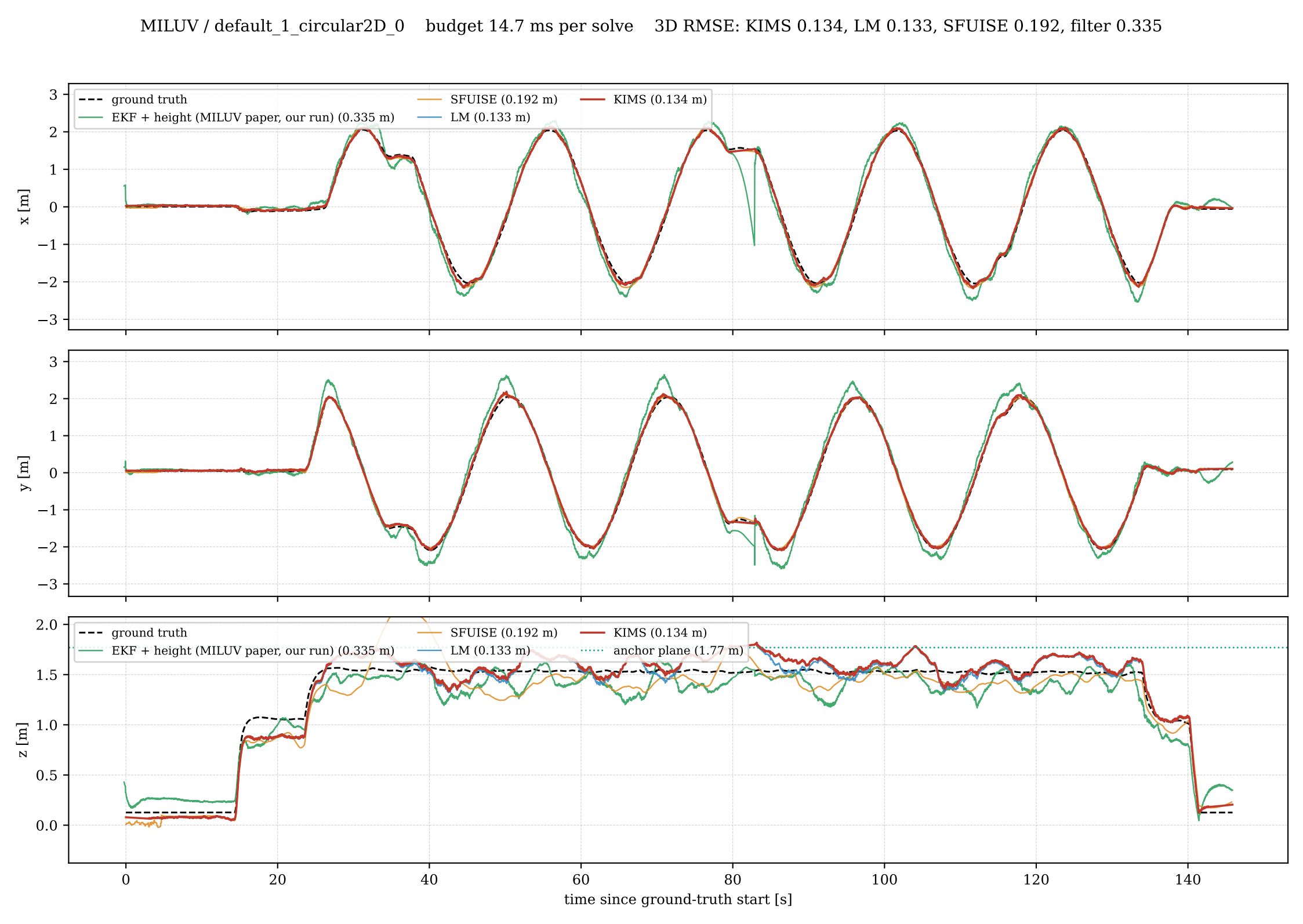

MILUV / default_1_random3_1 budget 14.5 ms per solve 3D RMSE: KIMS 0.231, LM 0.269, SFUISE 0.488, filter 0.255
ground truth
EKF + height (MILUV paper, our run) (0.255 m)
SFUISE (0.488 m)
LM (0.269 m)
KIMS (0.231 m)
anchor plane (1.42 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

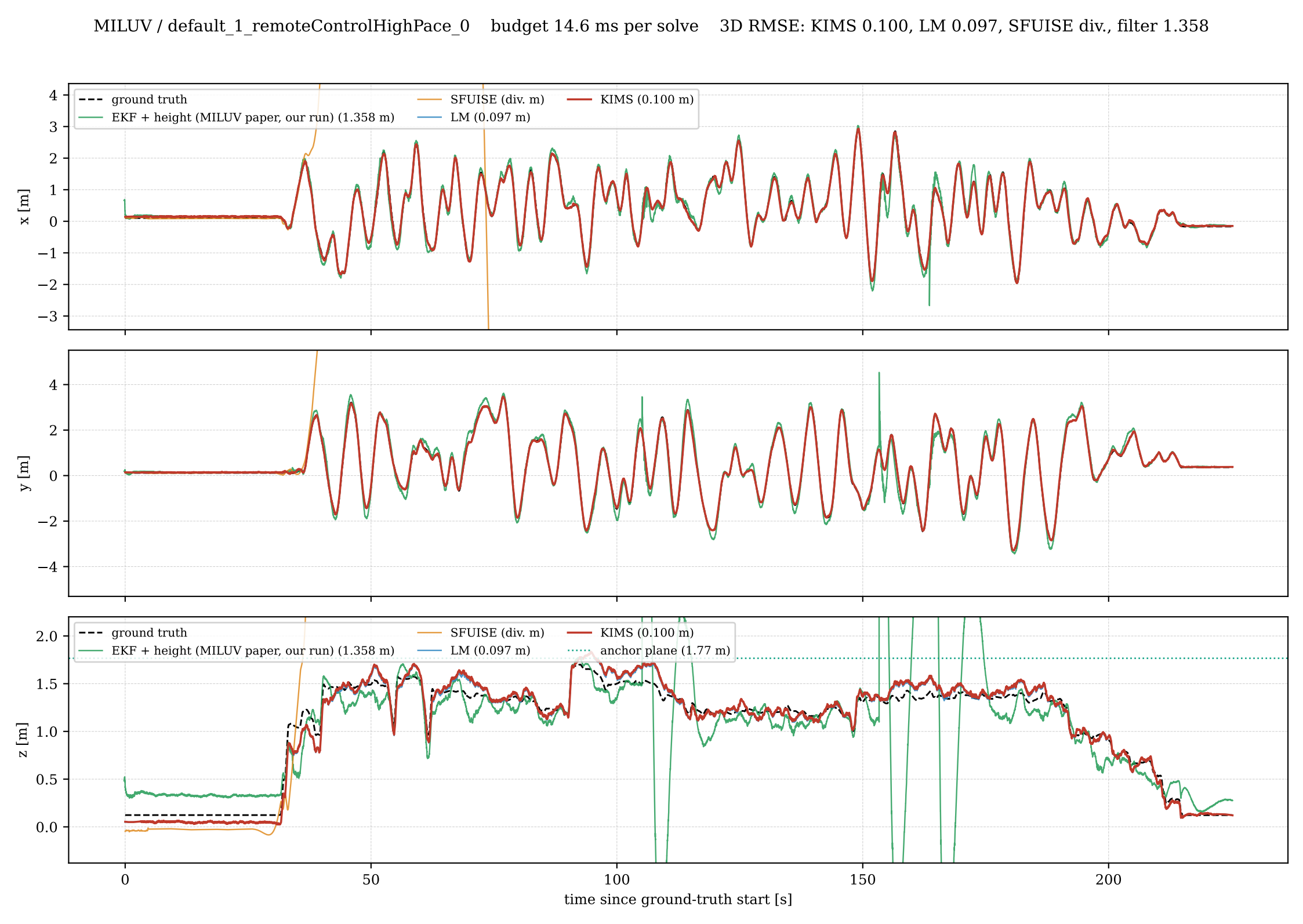

MILUV / default_1_random3_0 budget 14.1 ms per solve 3D RMSE: KIMS 0.165, LM 0.682, SFUISE 0.704, filter 0.217
ground truth
EKF + height (MILUV paper, our run) (0.217 m)
SFUISE (0.704 m)
LM (0.682 m)
KIMS (0.165 m)
x [m]
y [m]
z [m]
anchor plane (1.77 m)
time since ground-truth start [s]

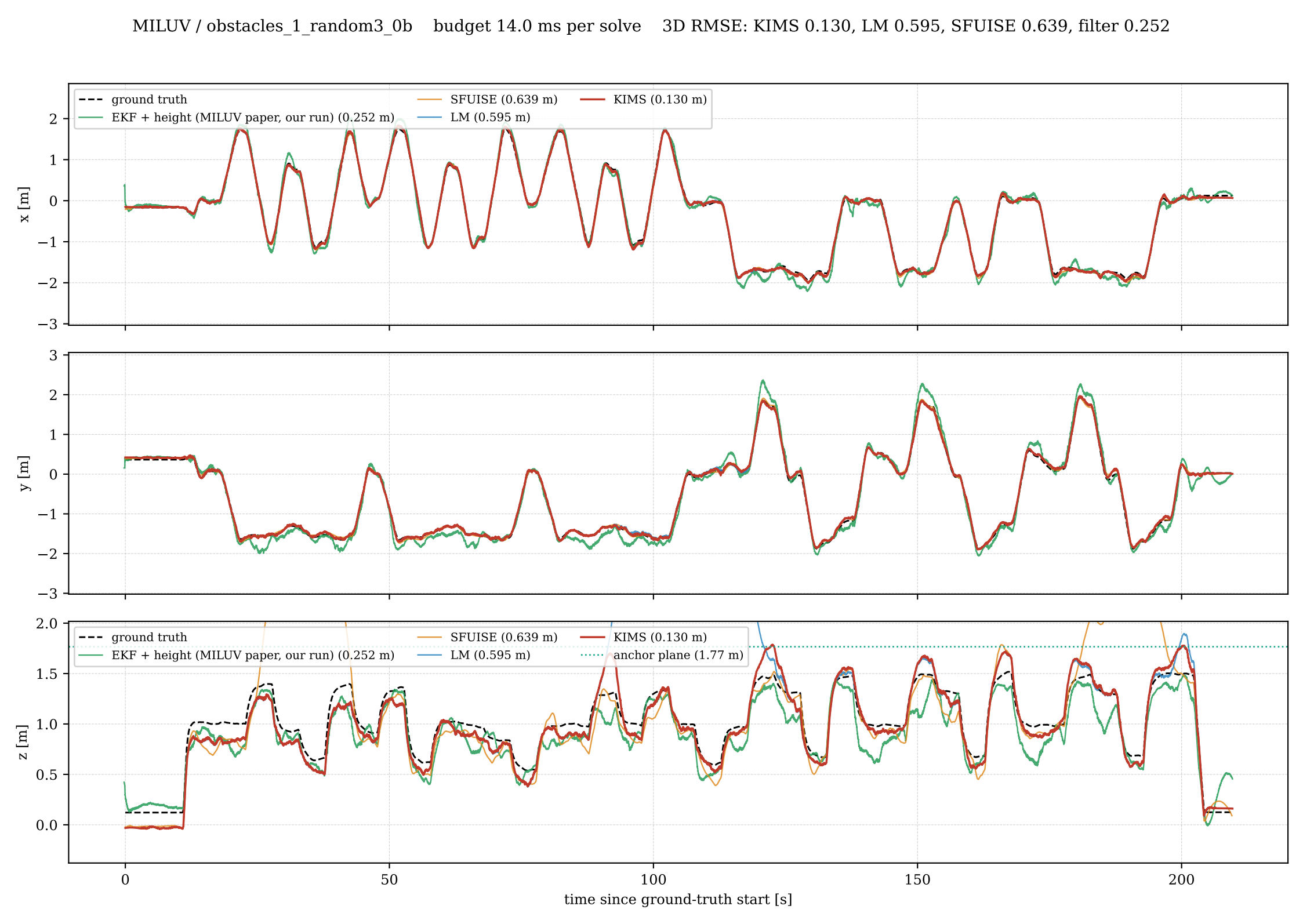

MILUV / default_1_circular3D_0 budget 14.2 ms per solve 3D RMSE: KIMS 0.093, LM 0.093, SFUISE 0.123, filter 0.303
ground truth
EKF + height (MILUV paper, our run) (0.303 m)
SFUISE (0.123 m)
LM (0.093 m)
KIMS (0.093 m)
anchor plane (1.77 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

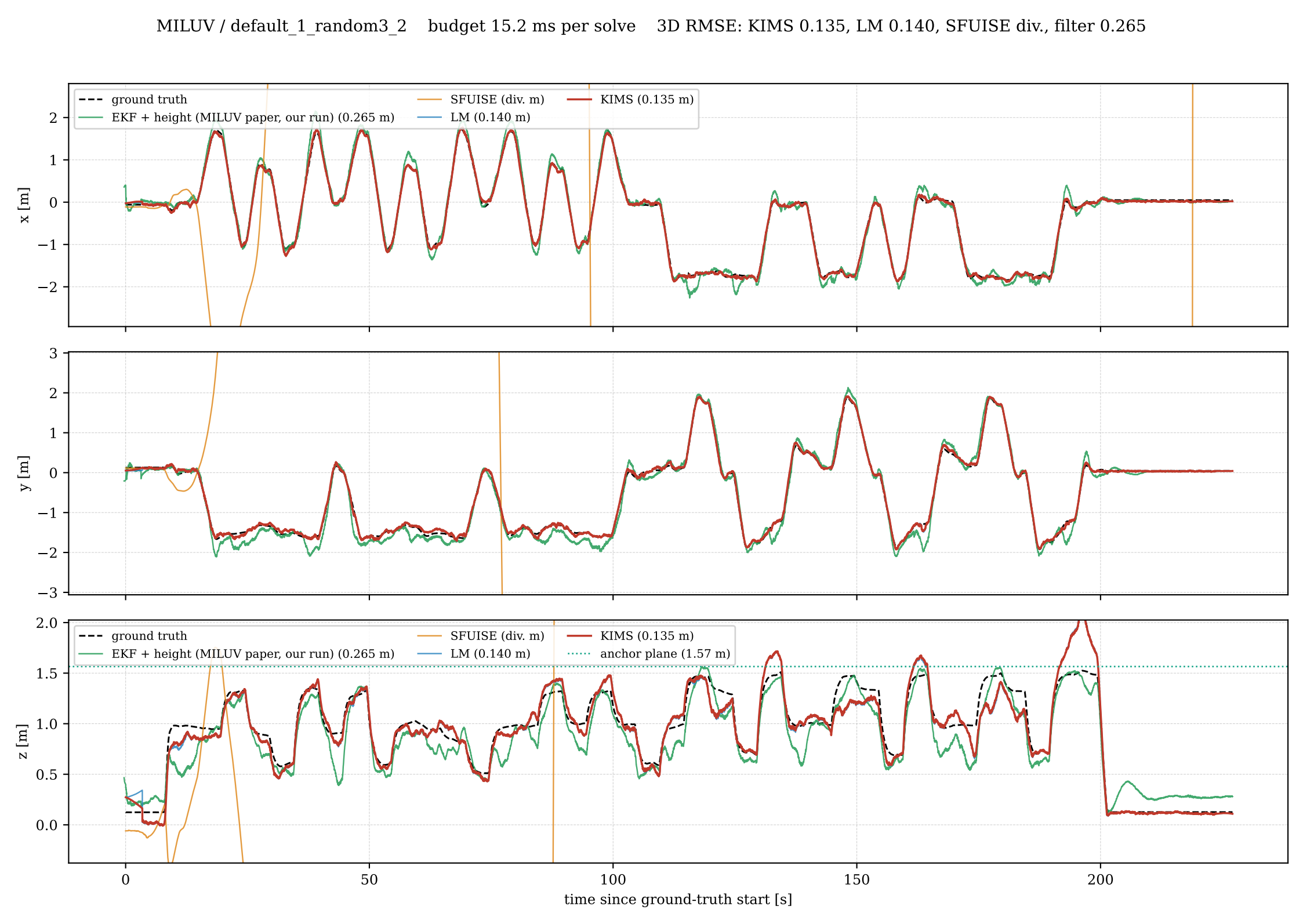

MILUV / default_1_remoteControlLowPace_0_v2 budget 14.4 ms per solve 3D RMSE: KIMS 0.086, LM 0.089, SFUISE 0.114, filter 0.219
ground truth
EKF + height (MILUV paper, our run) (0.219 m)
SFUISE (0.114 m)
LM (0.089 m)
KIMS (0.086 m)
x [m]
y [m]
z [m]
anchor plane (1.77 m)
time since ground-truth start [s]

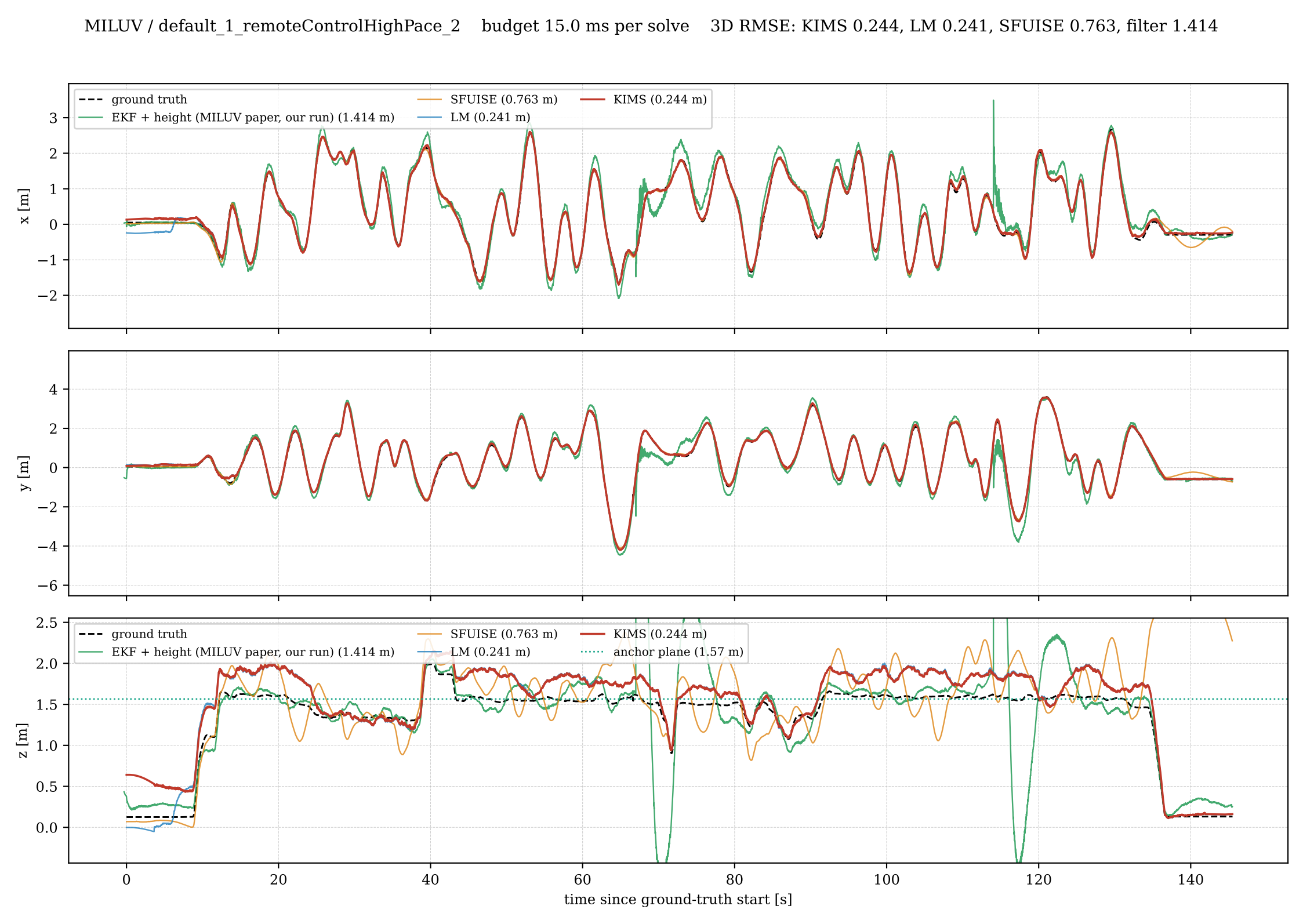

MILUV / default_1_remoteControlLowPace_0 budget 14.3 ms per solve 3D RMSE: KIMS 0.202, LM 0.219, SFUISE 0.955, filter 0.247
ground truth
EKF + height (MILUV paper, our run) (0.247 m)
SFUISE (0.955 m)
LM (0.219 m)
KIMS (0.202 m)
x [m]
y [m]
z [m]
anchor plane (1.77 m)
time since ground-truth start [s]

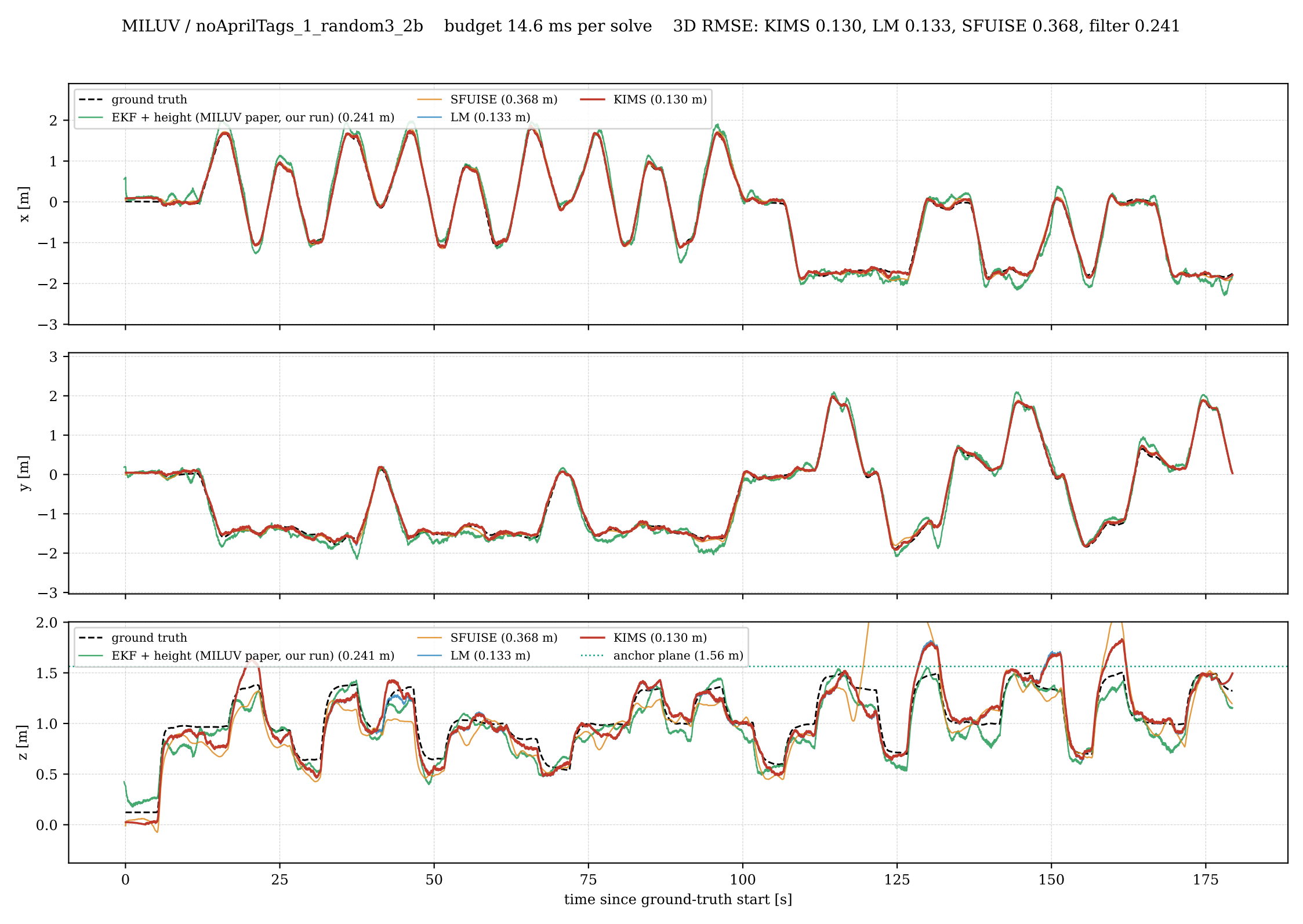

MILUV / default_1_circular2D_0 budget 14.7 ms per solve 3D RMSE: KIMS 0.134, LM 0.133, SFUISE 0.192, filter 0.335
ground truth
EKF + height (MILUV paper, our run) (0.335 m)
SFUISE (0.192 m)
LM (0.133 m)
KIMS (0.134 m)
x [m]
y [m]
z [m]
anchor plane (1.77 m)
time since ground-truth start [s]

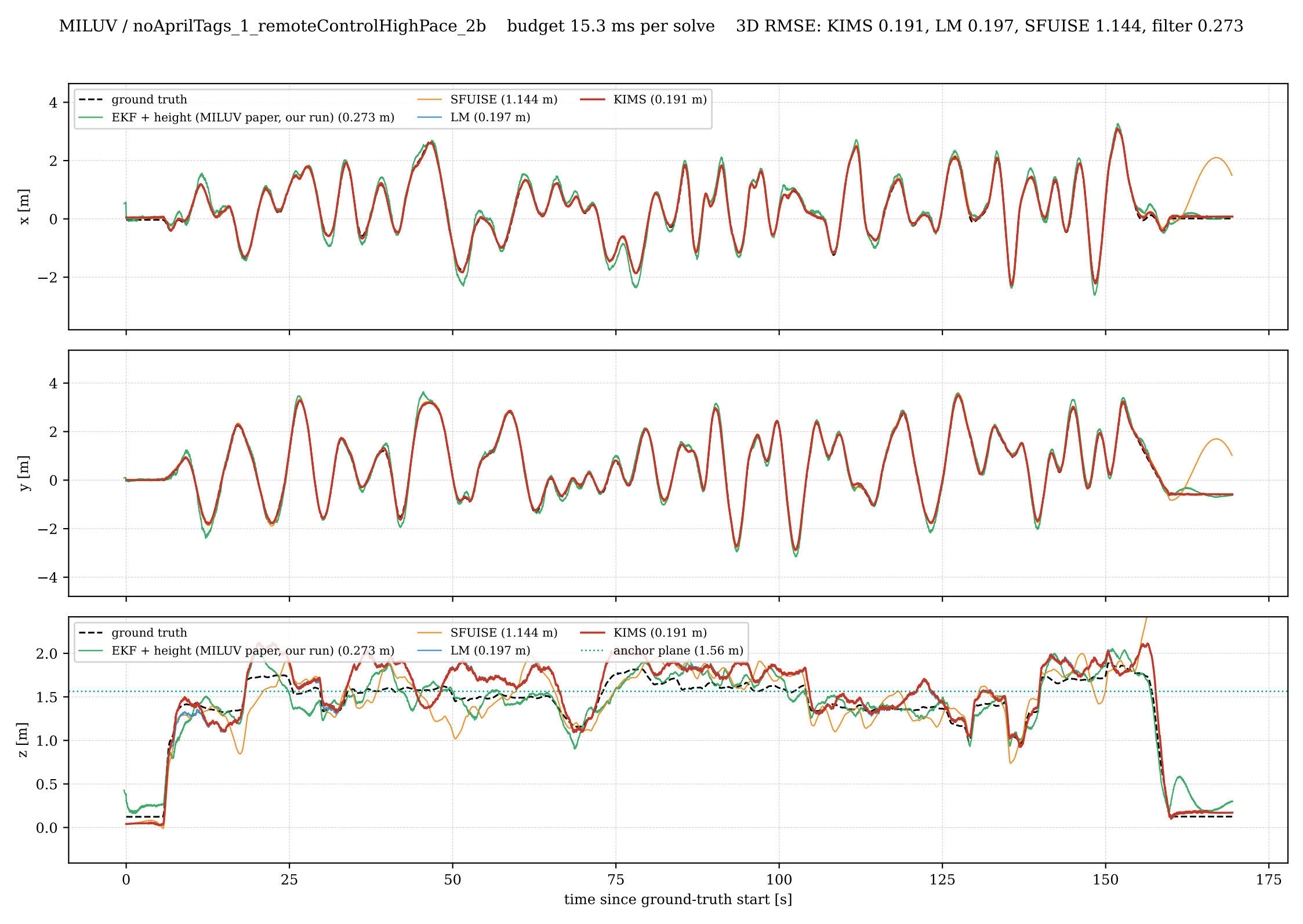

MILUV / default_1_remoteControlHighPace_0 budget 14.6 ms per solve 3D RMSE: KIMS 0.100, LM 0.097, SFUISE div., filter 1.358
ground truth
EKF + height (MILUV paper, our run) (1.358 m)
SFUISE (div. m)
LM (0.097 m)
KIMS (0.100 m)
x [m]
y [m]
z [m]
anchor plane (1.77 m)
time since ground-truth start [s]

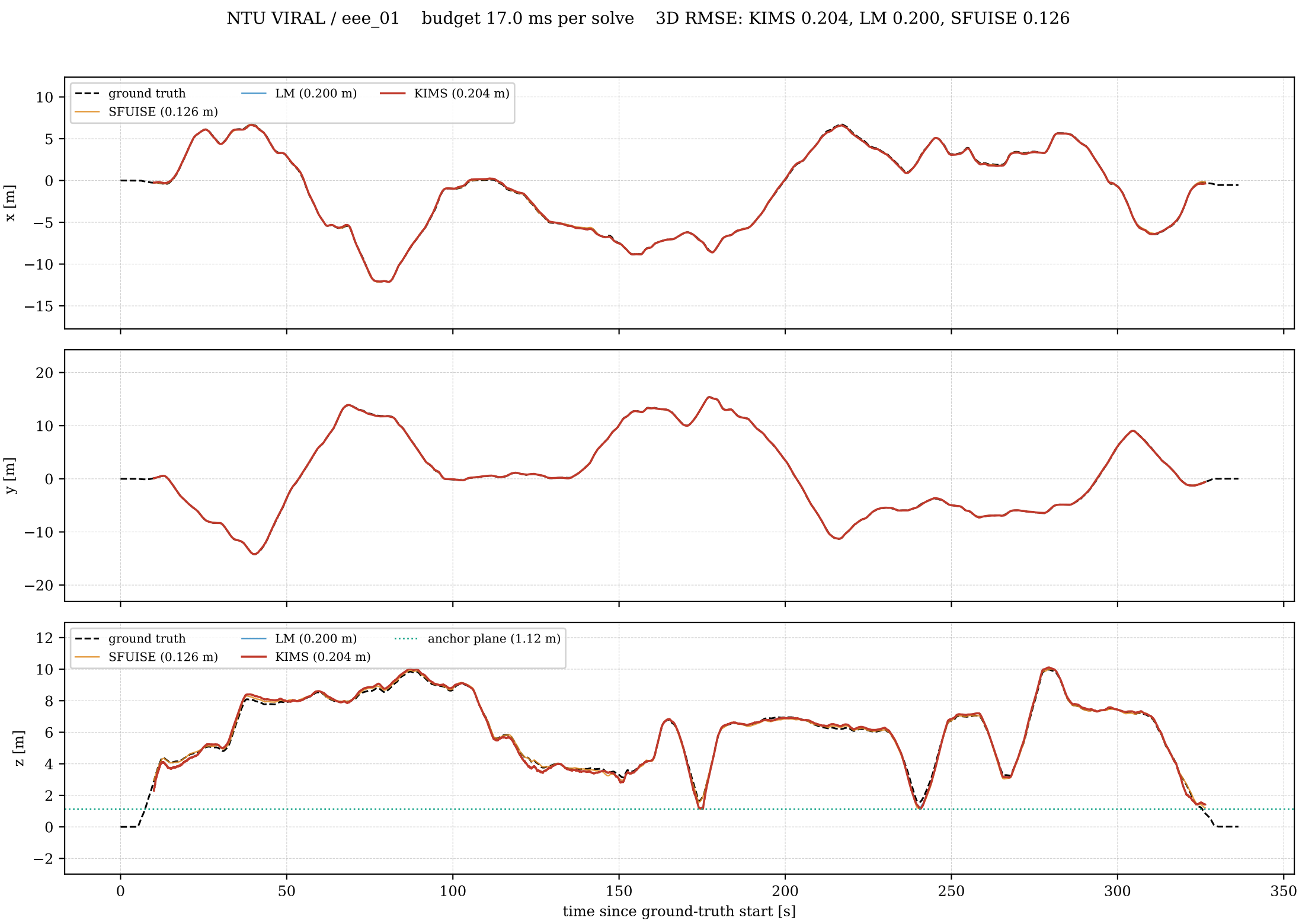

MILUV / obstacles_1_random3_0b budget 14.0 ms per solve 3D RMSE: KIMS 0.130, LM 0.595, SFUISE 0.639, filter 0.252
ground truth
EKF + height (MILUV paper, our run) (0.252 m)
SFUISE (0.639 m)
LM (0.595 m)
KIMS (0.130 m)
x [m]
y [m]
z [m]
anchor plane (1.77 m)
time since ground-truth start [s]

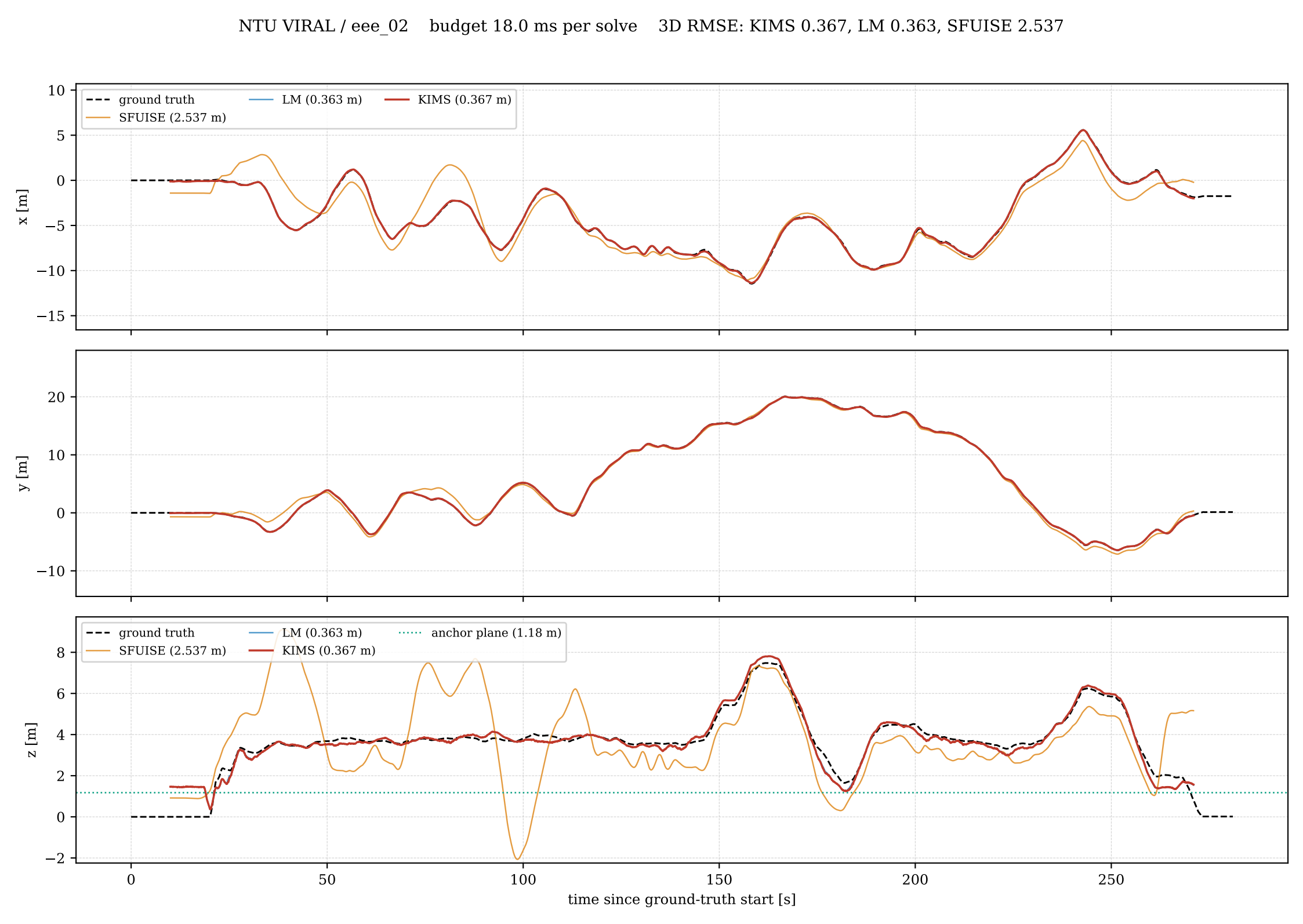

MILUV / default_1_random3_2 budget 15.2 ms per solve 3D RMSE: KIMS 0.135, LM 0.140, SFUISE div., filter 0.265
ground truth
EKF + height (MILUV paper, our run) (0.265 m)
SFUISE (div. m)
LM (0.140 m)
KIMS (0.135 m)
anchor plane (1.57 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

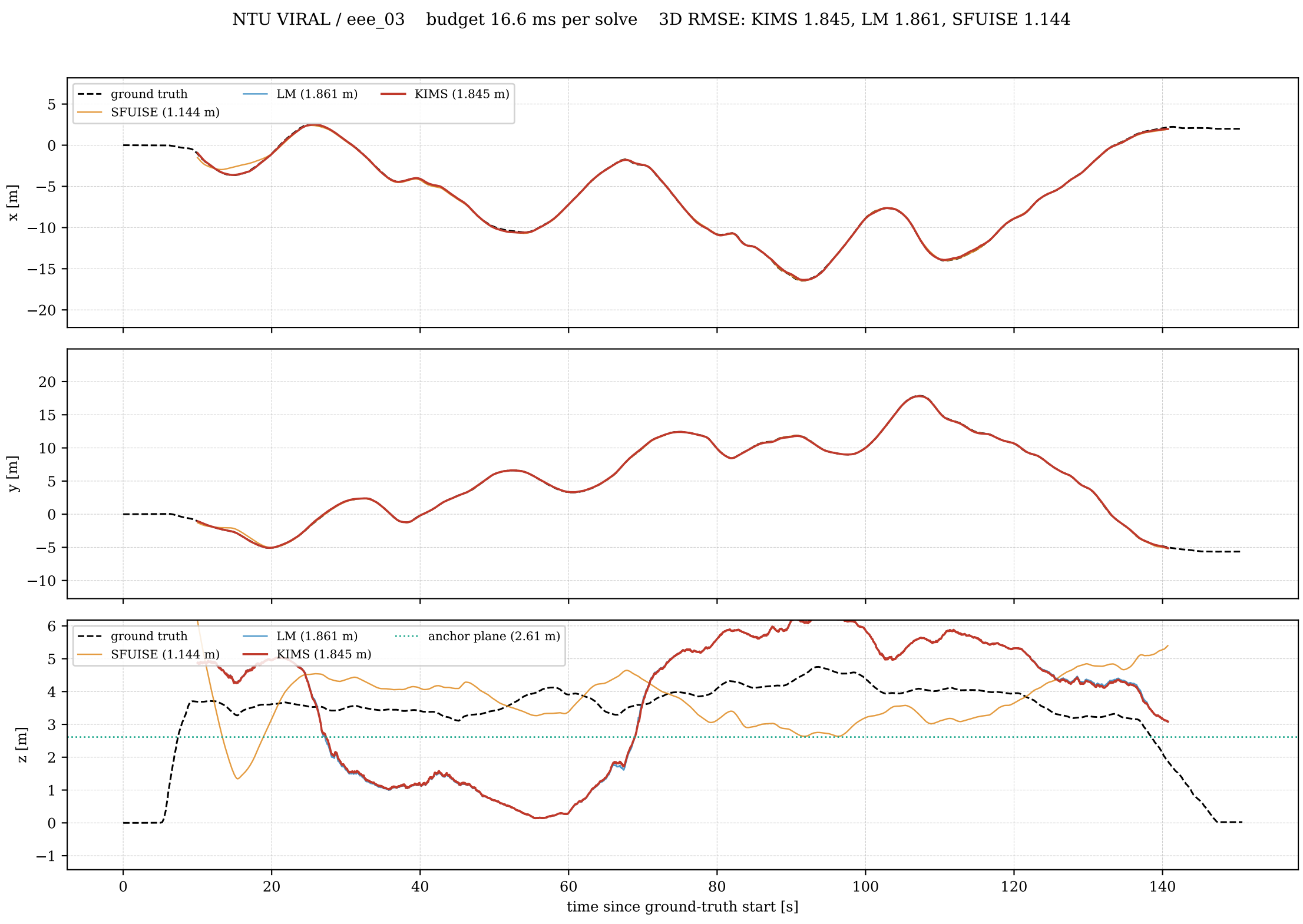

MILUV / default_1_remoteControlHighPace_2 budget 15.0 ms per solve 3D RMSE: KIMS 0.244, LM 0.241, SFUISE 0.763, filter 1.414
ground truth
EKF + height (MILUV paper, our run) (1.414 m)
SFUISE (0.763 m)
LM (0.241 m)
KIMS (0.244 m)
x [m]
y [m]
z [m]
anchor plane (1.57 m)
time since ground-truth start [s]

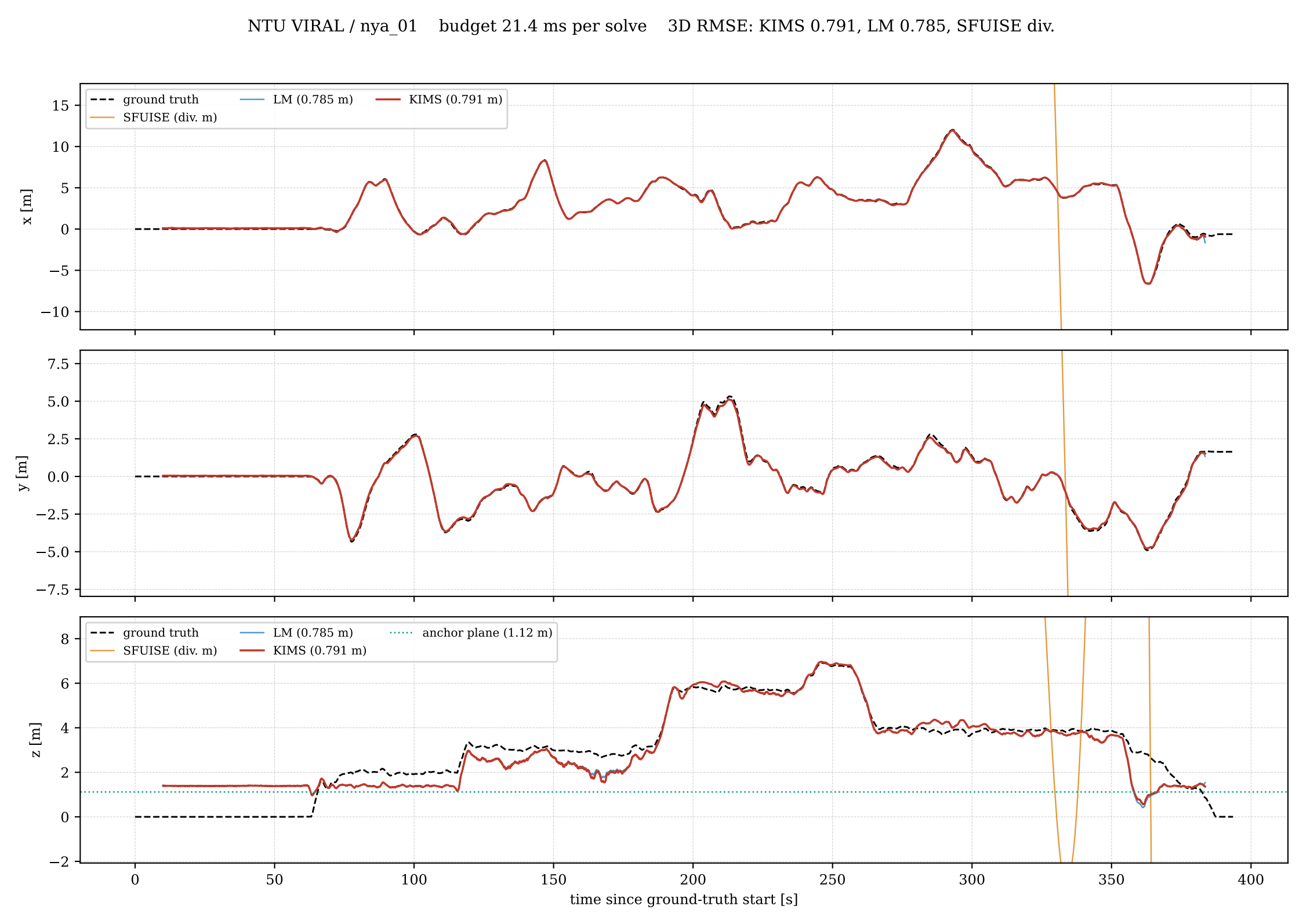

MILUV / noAprilTags_1_random3_2b budget 14.6 ms per solve 3D RMSE: KIMS 0.130, LM 0.133, SFUISE 0.368, filter 0.241
ground truth
EKF + height (MILUV paper, our run) (0.241 m)
SFUISE (0.368 m)
LM (0.133 m)
KIMS (0.130 m)
x [m]
y [m]
z [m]
anchor plane (1.56 m)
time since ground-truth start [s]

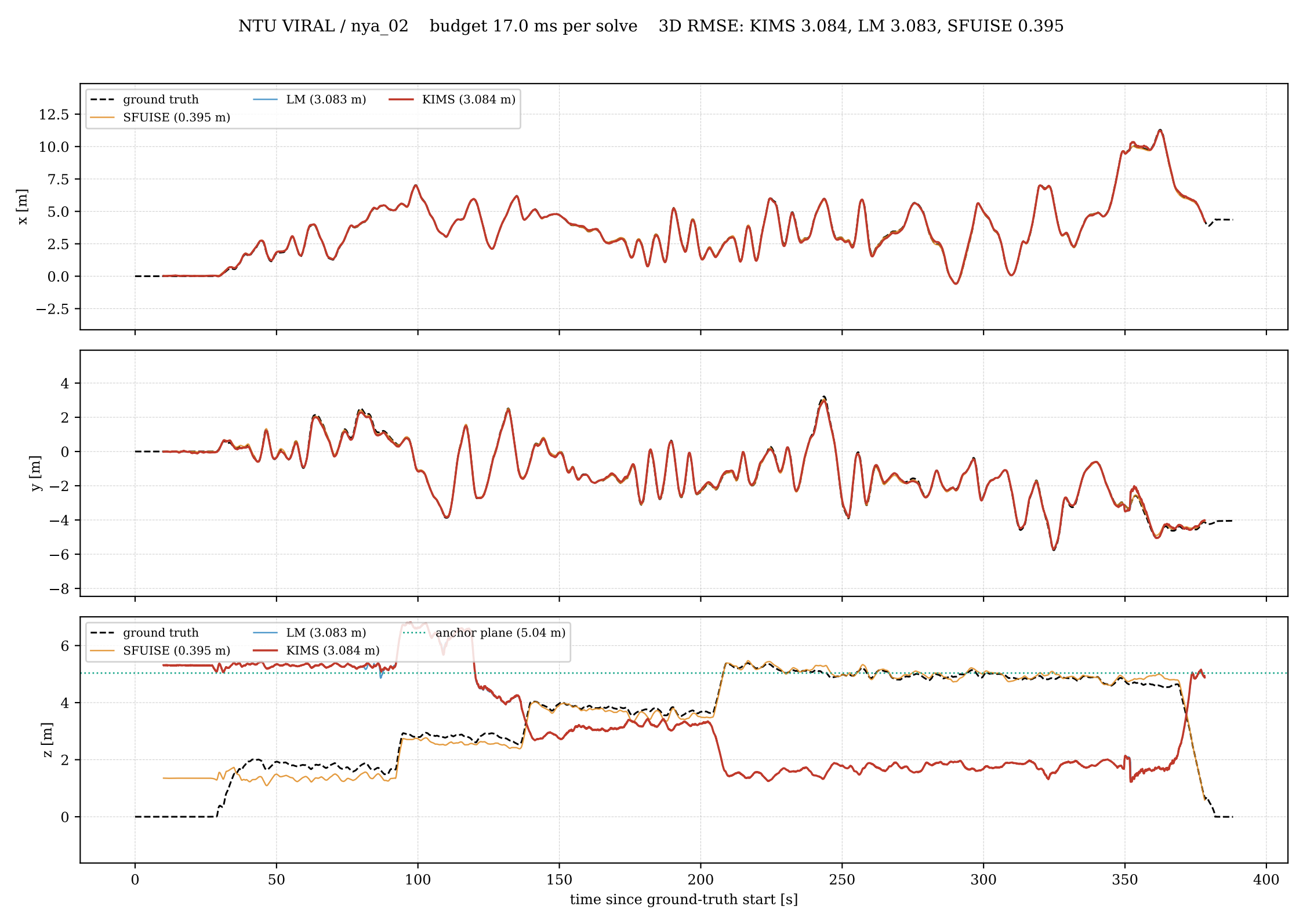

MILUV / noAprilTags_1_remoteControlHighPace_2b budget 15.3 ms per solve 3D RMSE: KIMS 0.191, LM 0.197, SFUISE 1.144, filter 0.273
ground truth
EKF + height (MILUV paper, our run) (0.273 m)
SFUISE (1.144 m)
LM (0.197 m)
KIMS (0.191 m)
x [m]
y [m]
z [m]
anchor plane (1.56 m)
time since ground-truth start [s]

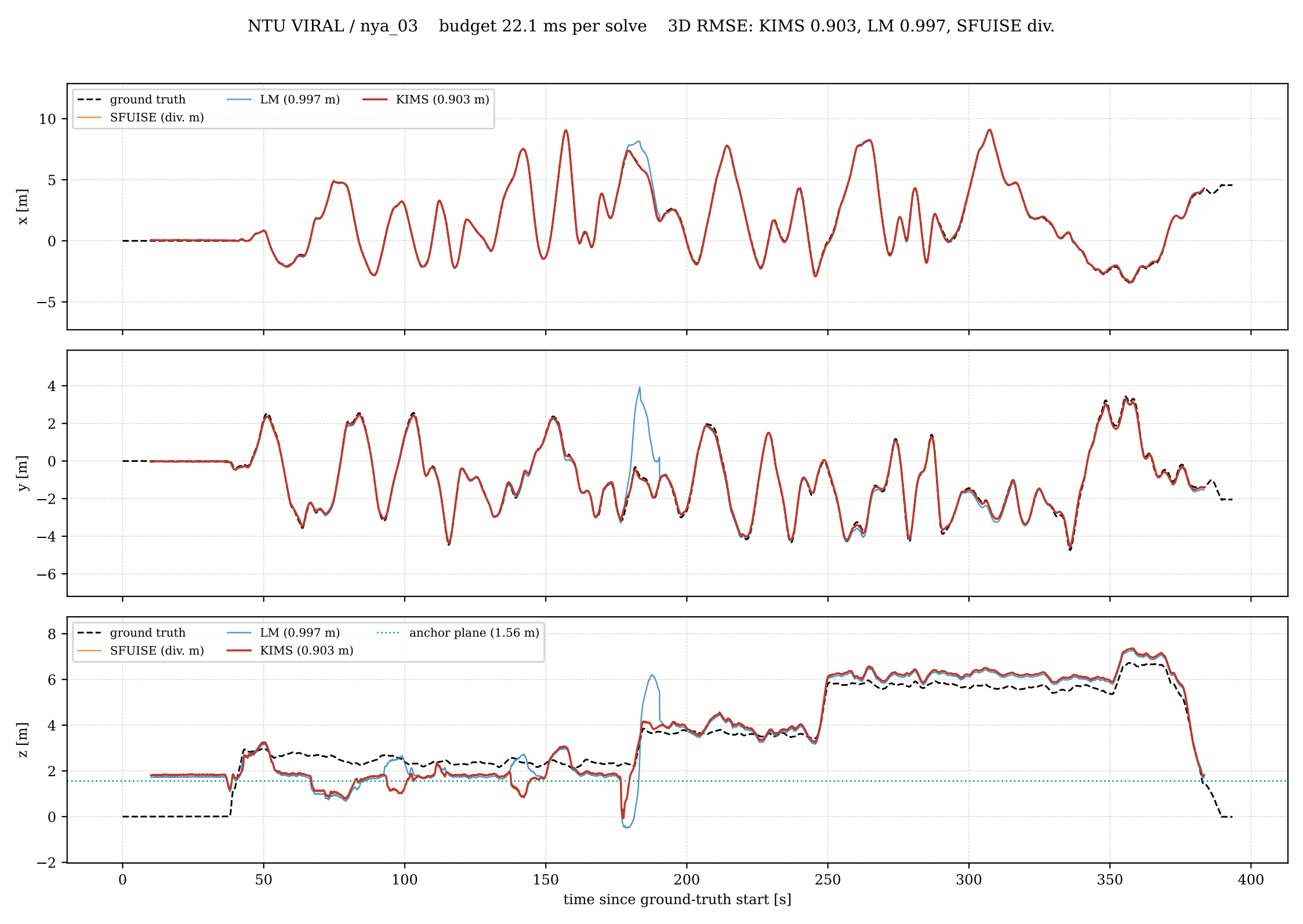

NTU VIRAL / eee_01 budget 17.0 ms per solve 3D RMSE: KIMS 0.204, LM 0.200, SFUISE 0.126
ground truth
SFUISE (0.126 m)
LM (0.200 m)
KIMS (0.204 m)
x [m]
y [m]
z [m]
anchor plane (1.12 m)
time since ground-truth start [s]

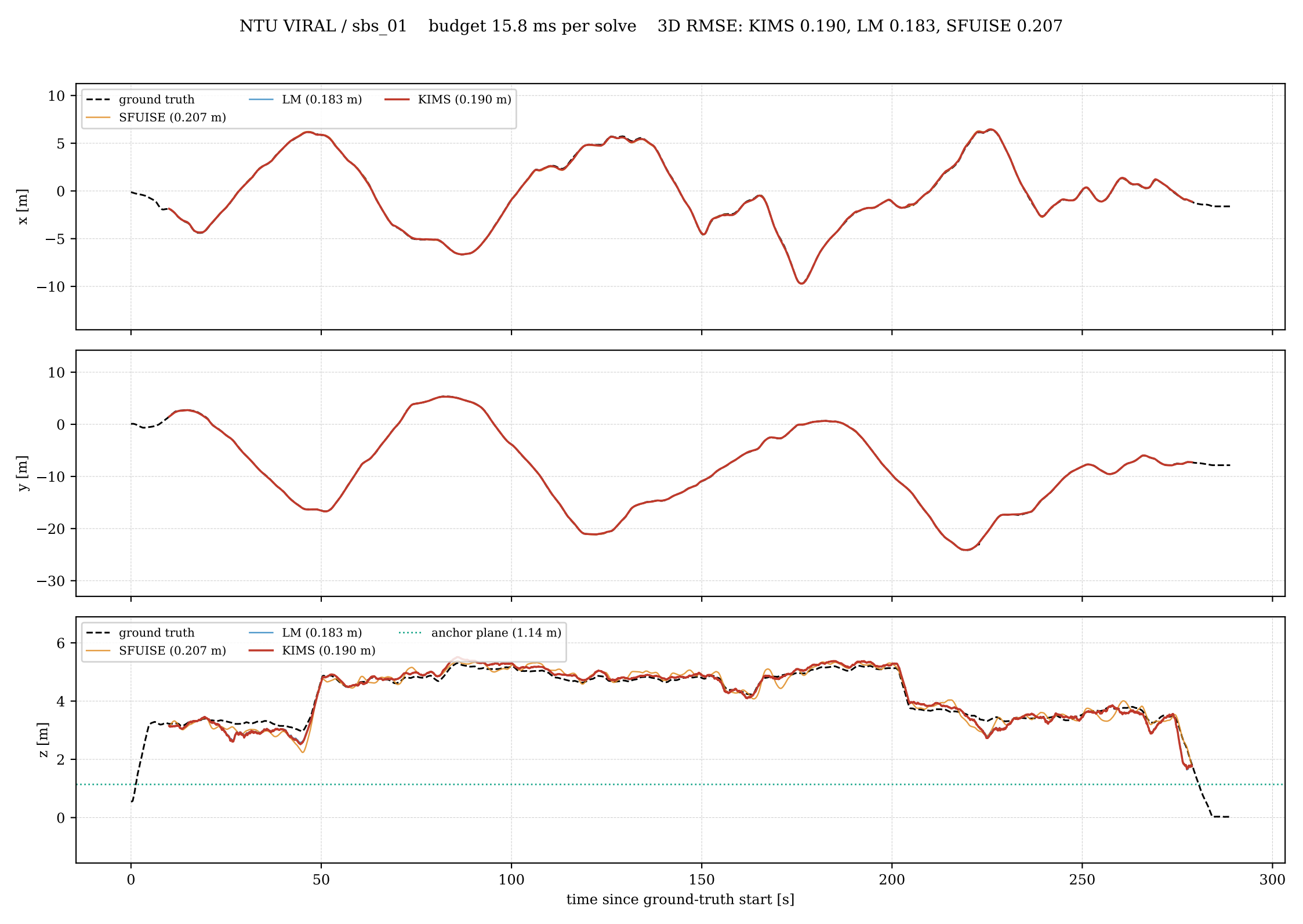

NTU VIRAL / eee_02 budget 18.0 ms per solve 3D RMSE: KIMS 0.367, LM 0.363, SFUISE 2.537
ground truth
LM (0.363 m)
KIMS (0.367 m)
SFUISE (2.537 m)
x [m]
y [m]
z [m]
anchor plane (1.18 m)
time since ground-truth start [s]

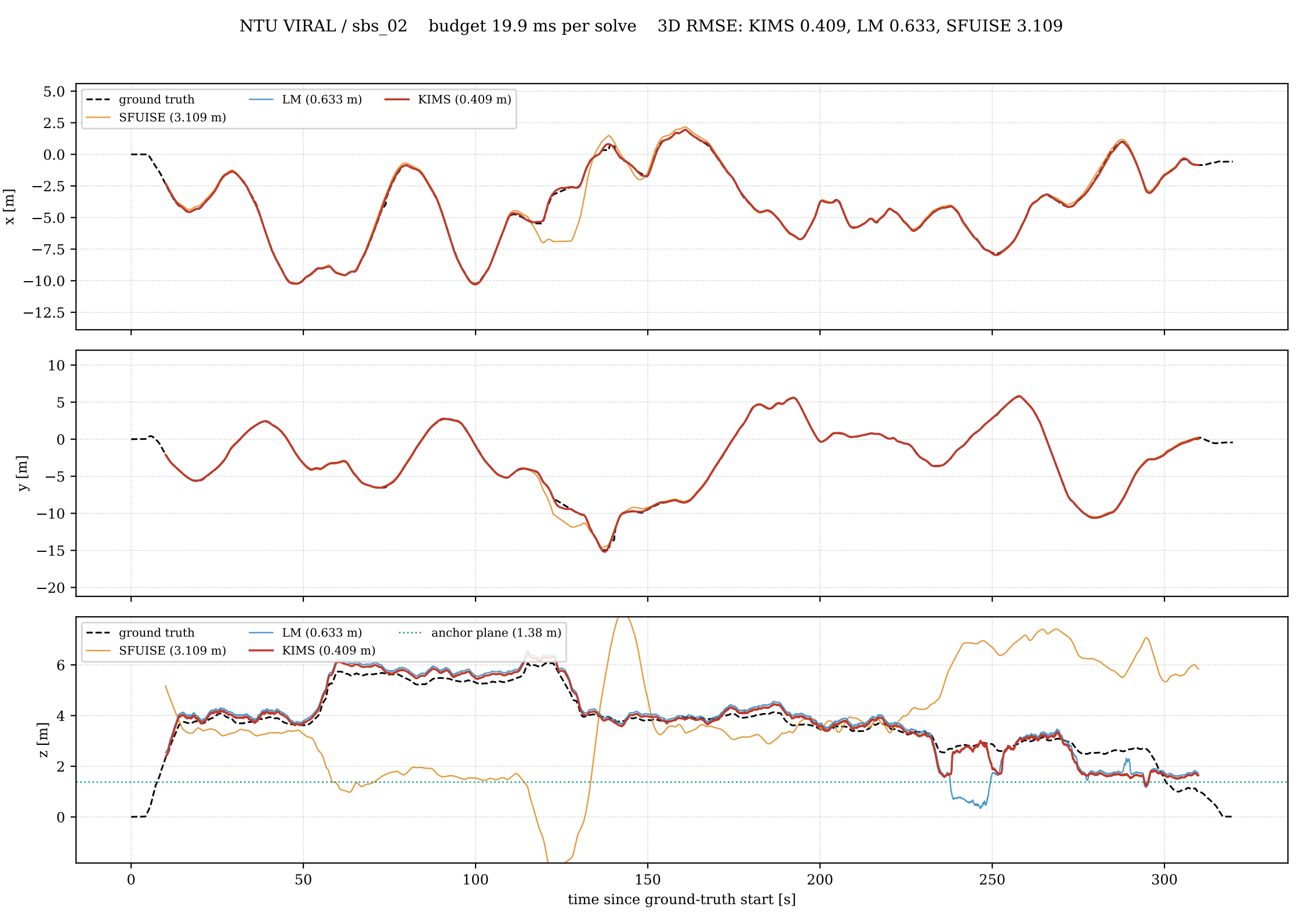

NTU VIRAL / eee_03 budget 16.6 ms per solve 3D RMSE: KIMS 1.845, LM 1.861, SFUISE 1.144
ground truth
LM (1.861 m)
KIMS (1.845 m)
SFUISE (1.144 m)
x [m]
y [m]
z [m]
anchor plane (2.61 m)
time since ground-truth start [s]

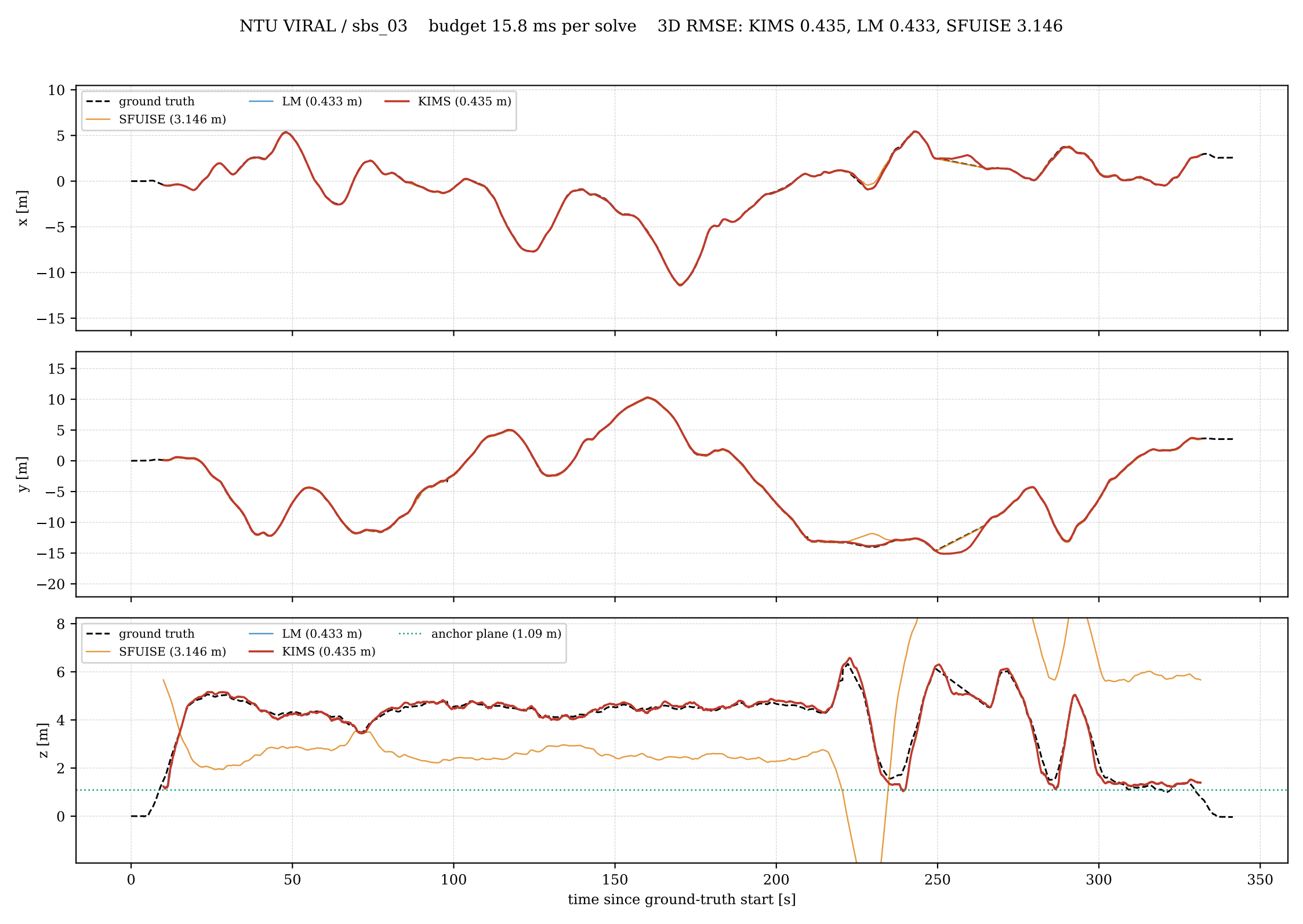

NTU VIRAL / nya_01 budget 21.4 ms per solve 3D RMSE: KIMS 0.791, LM 0.785, SFUISE div.
ground truth
LM (0.785 m)
KIMS (0.791 m)
SFUISE (div. m)
x [m]
y [m]
z [m]
anchor plane (1.12 m)
time since ground-truth start [s]

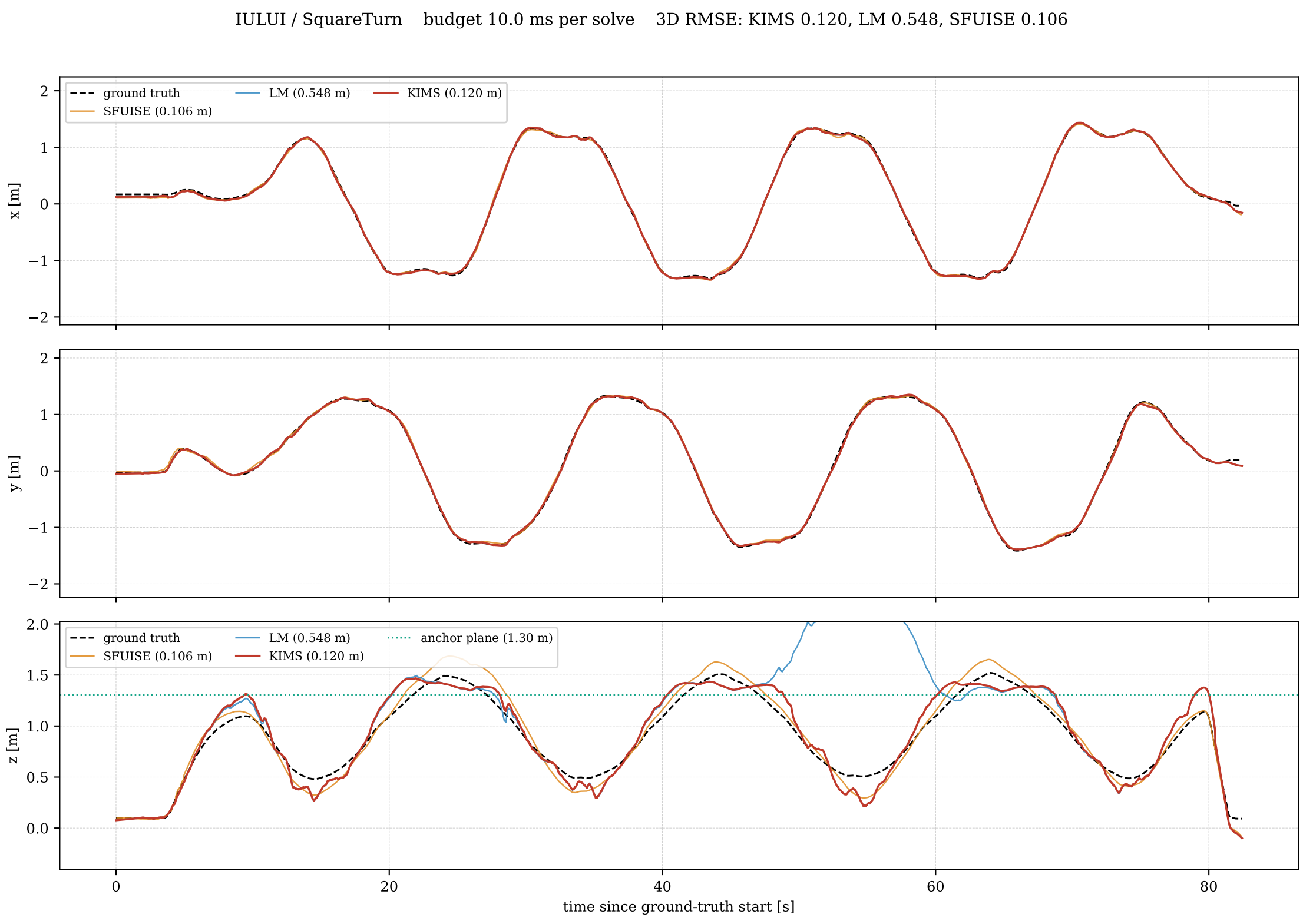

NTU VIRAL / nya_02 budget 17.0 ms per solve 3D RMSE: KIMS 3.084, LM 3.083, SFUISE 0.395
ground truth
LM (3.083 m)
KIMS (3.084 m)
SFUISE (0.395 m)
x [m]
y [m]
z [m]
anchor plane (5.04 m)
time since ground-truth start [s]

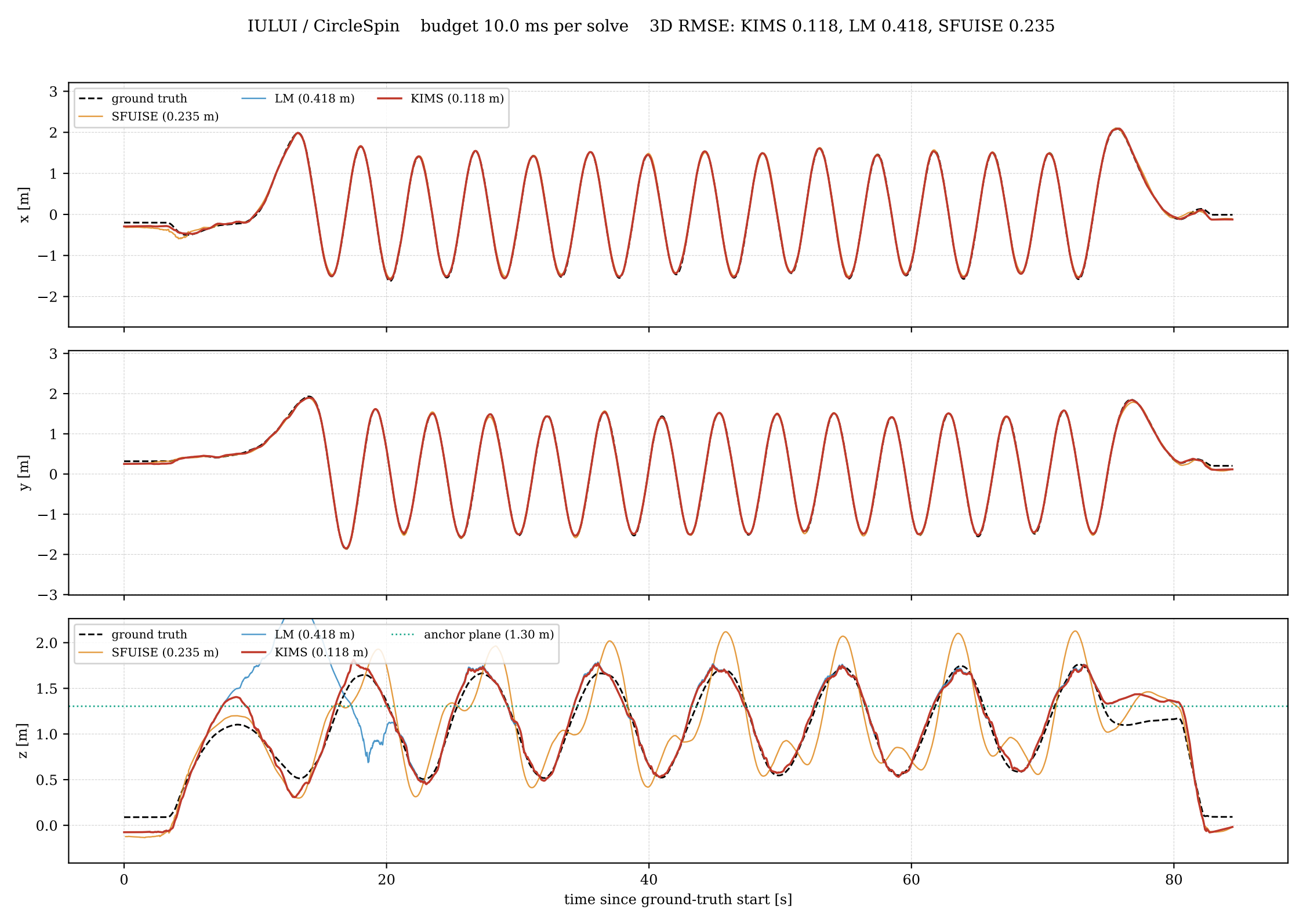

NTU VIRAL / nya_03 budget 22.1 ms per solve 3D RMSE: KIMS 0.903, LM 0.997, SFUISE div.
ground truth
LM (0.997 m)
KIMS (0.903 m)
SFUISE (div. m)
x [m]
y [m]
z [m]
anchor plane (1.56 m)
time since ground-truth start [s]

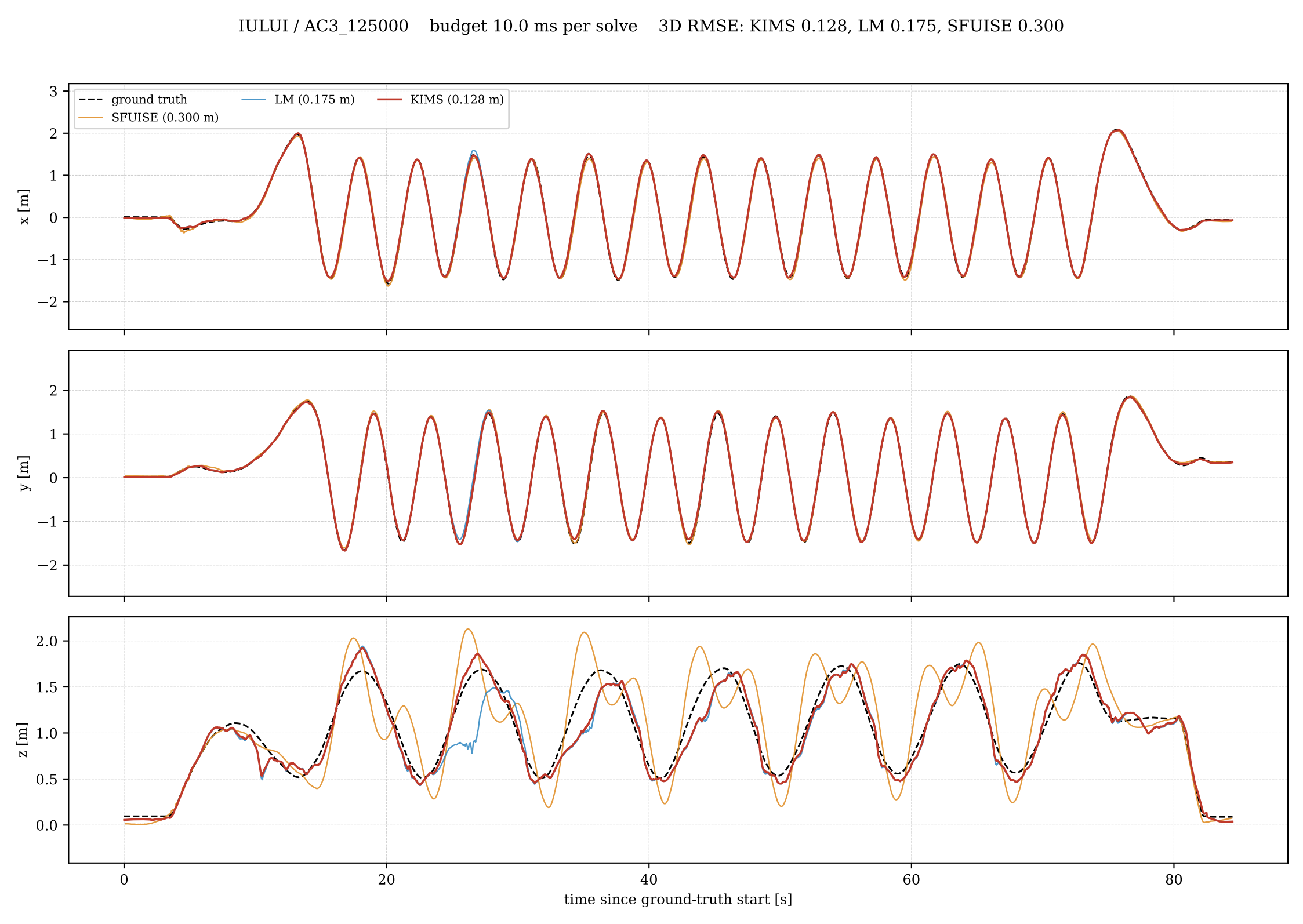

NTU VIRAL / sbs_01 budget 15.8 ms per solve 3D RMSE: KIMS 0.190, LM 0.183, SFUISE 0.207
ground truth
LM (0.183 m)
KIMS (0.190 m)
SFUISE (0.207 m)
x [m]
y [m]
z [m]
anchor plane (1.14 m)
time since ground-truth start [s]

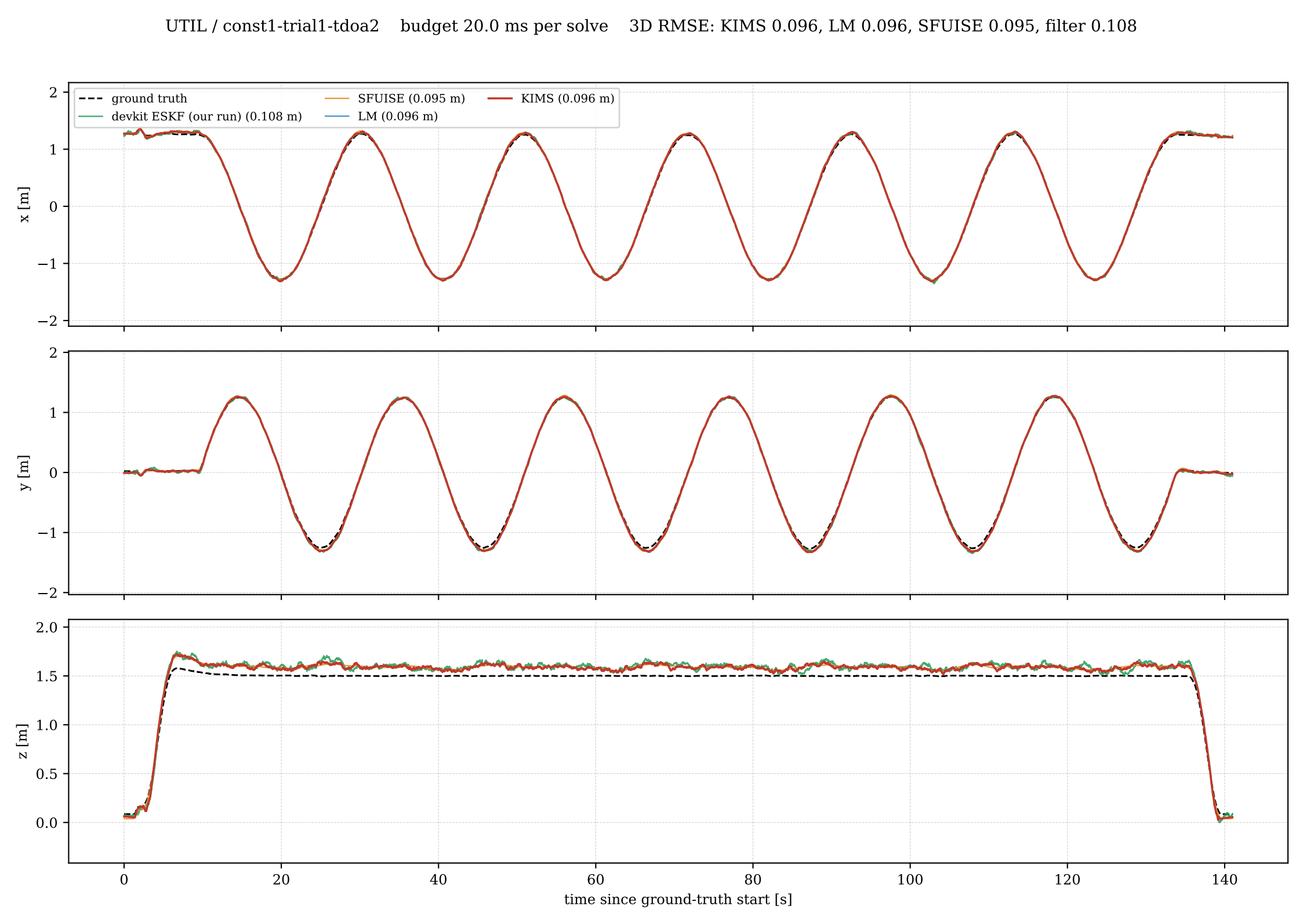

NTU VIRAL / sbs_02 budget 19.9 ms per solve 3D RMSE: KIMS 0.409, LM 0.633, SFUISE 3.109
ground truth
LM (0.633 m)
KIMS (0.409 m)
SFUISE (3.109 m)
x [m]
y [m]
z [m]
anchor plane (1.38 m)
time since ground-truth start [s]

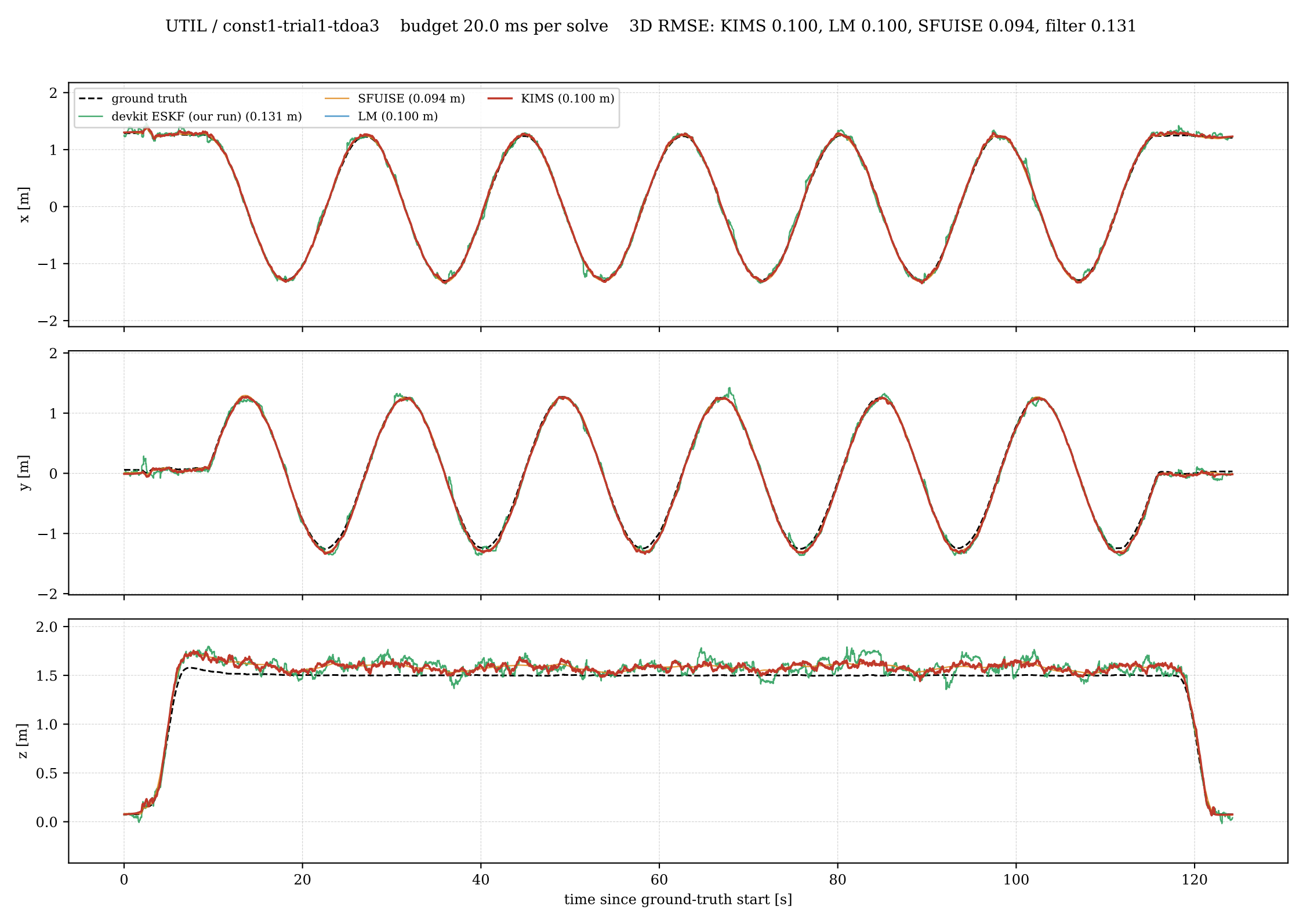

NTU VIRAL / sbs_03 budget 15.8 ms per solve 3D RMSE: KIMS 0.435, LM 0.433, SFUISE 3.146
ground truth
LM (0.433 m)
KIMS (0.435 m)
SFUISE (3.146 m)
x [m]
y [m]
anchor plane (1.09 m)
z [m]
time since ground-truth start [s]

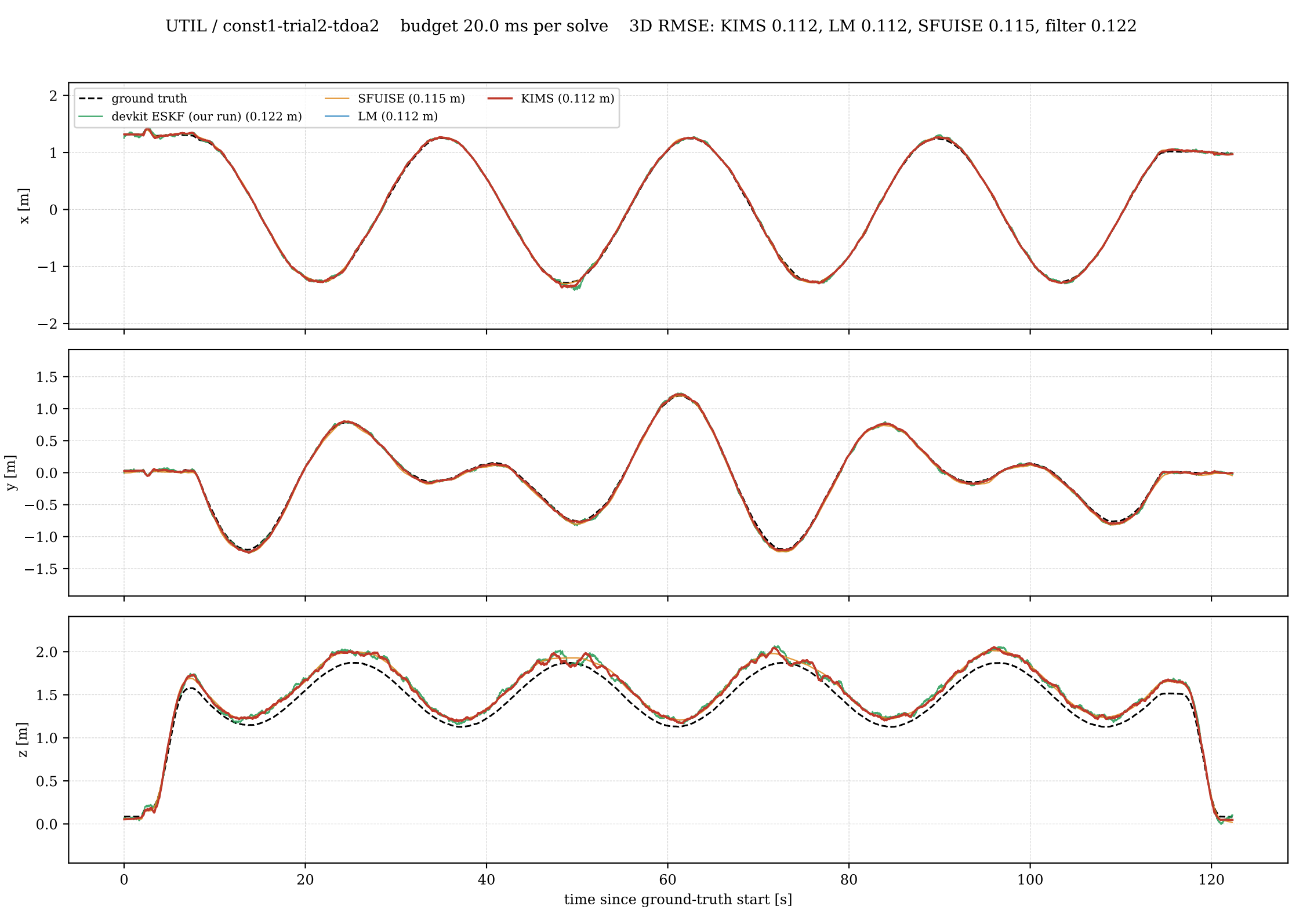

IULUI / SquareTurn budget 10.0 ms per solve 3D RMSE: KIMS 0.120, LM 0.548, SFUISE 0.106
ground truth
SFUISE (0.106 m)
LM (0.548 m)
KIMS (0.120 m)
x [m]
y [m]
z [m]
anchor plane (1.30 m)
time since ground-truth start [s]

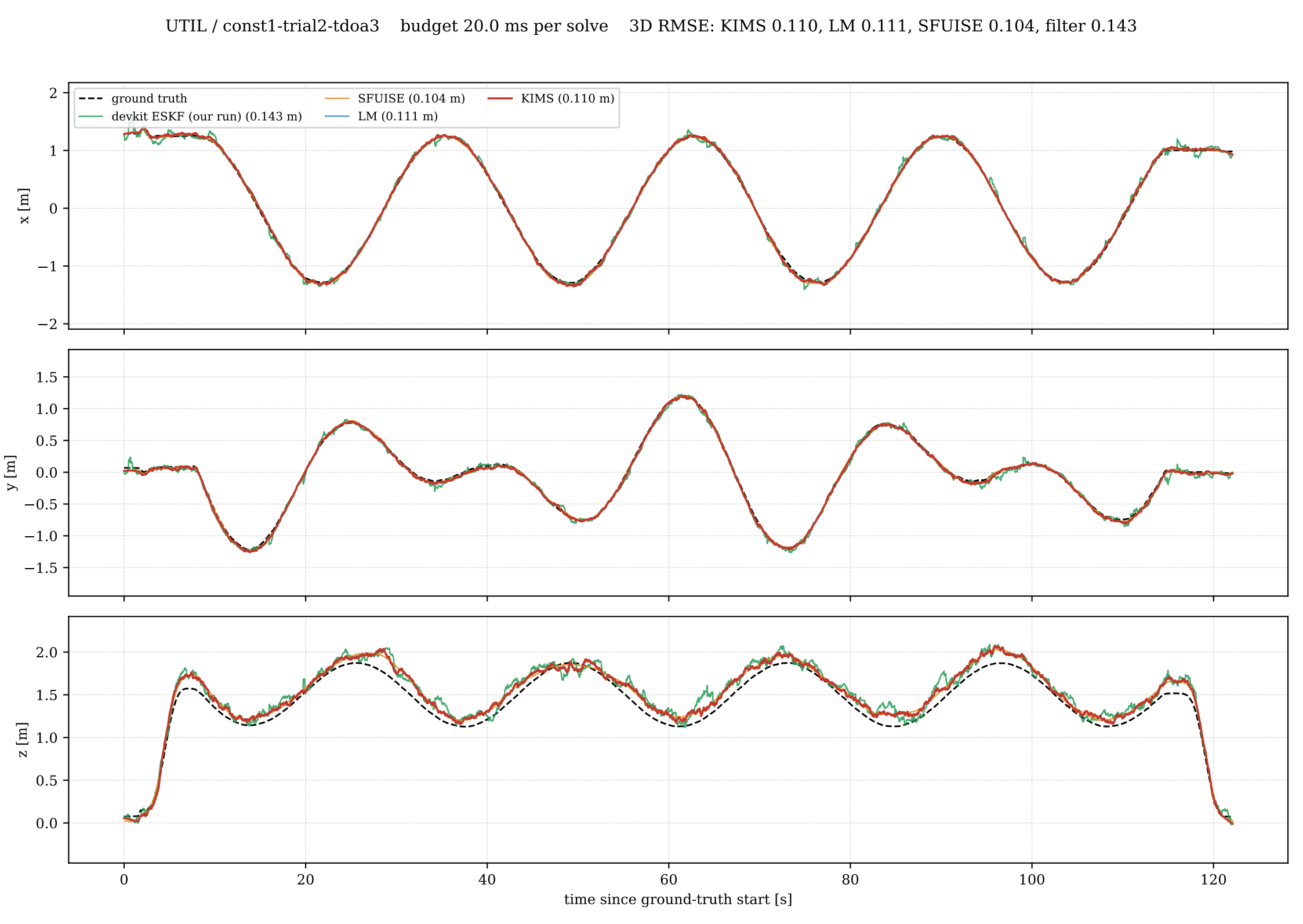

IULUI / CircleSpin budget 10.0 ms per solve 3D RMSE: KIMS 0.118, LM 0.418, SFUISE 0.235
ground truth
SFUISE (0.235 m)
LM (0.418 m)
KIMS (0.118 m)
x [m]
y [m]
z [m]
anchor plane (1.30 m)
time since ground-truth start [s]

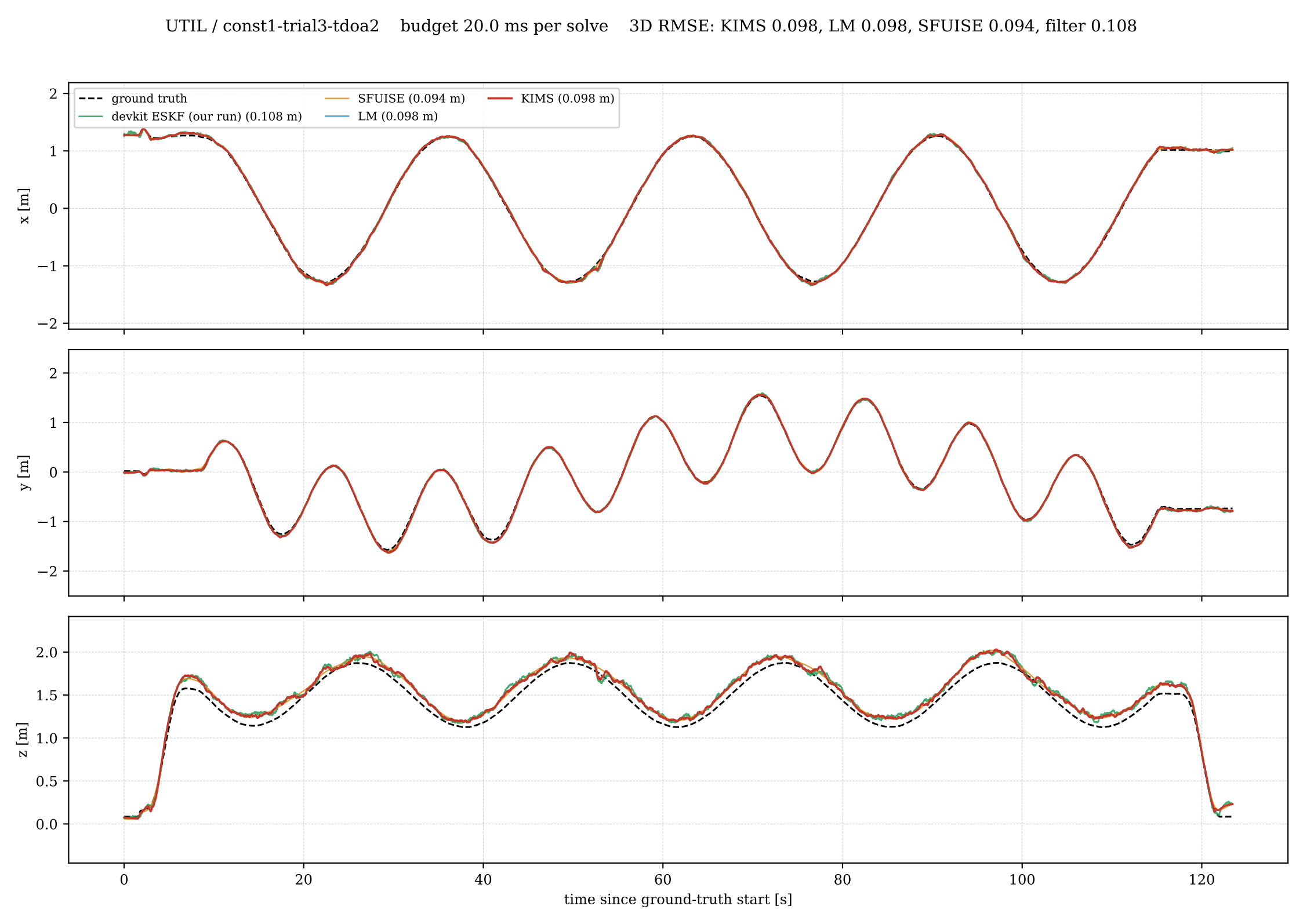

IULUI / AC3_125000 budget 10.0 ms per solve 3D RMSE: KIMS 0.128, LM 0.175, SFUISE 0.300
ground truth
LM (0.175 m)
KIMS (0.128 m)
SFUISE (0.300 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

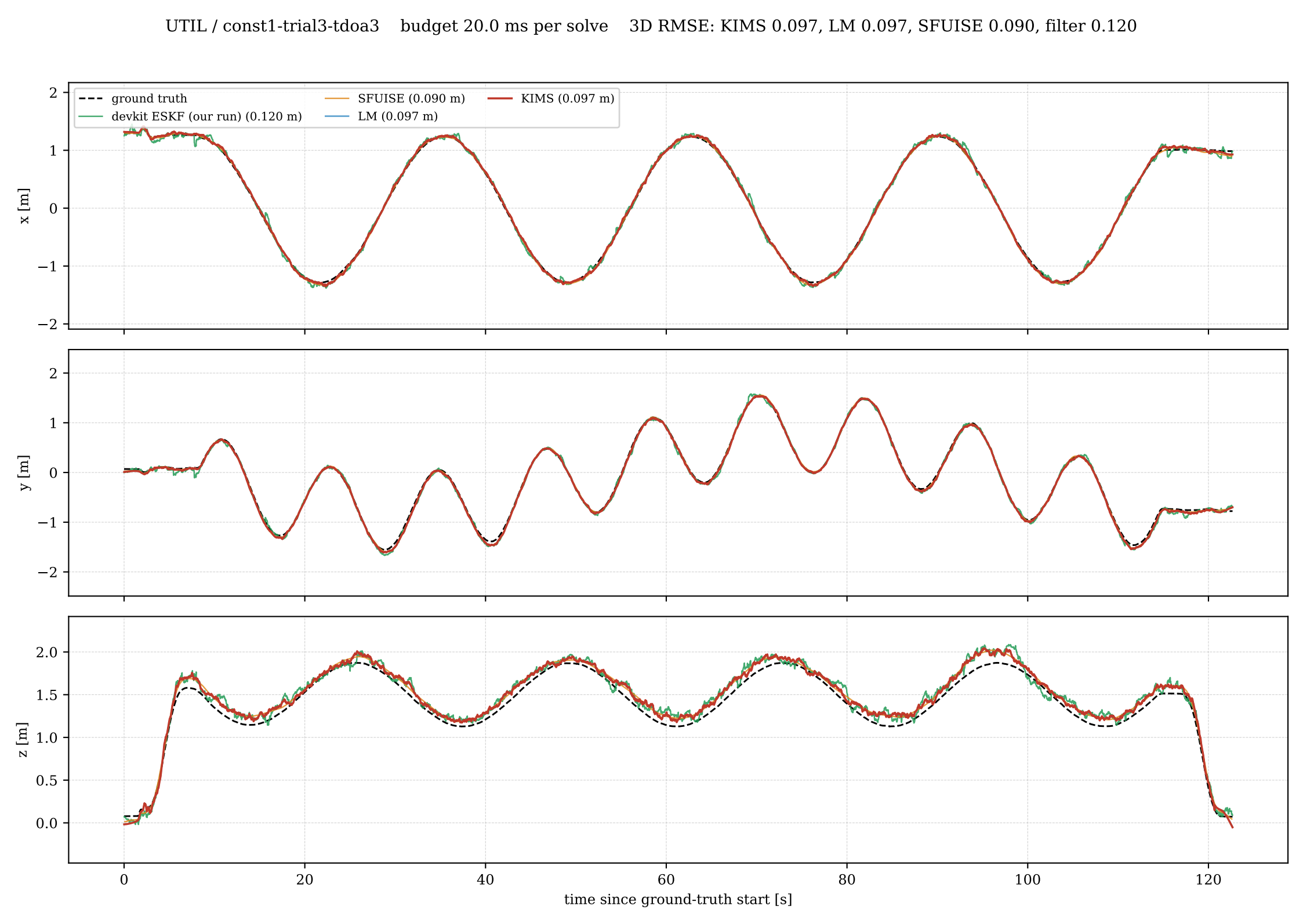

UTIL / const1-trial1-tdoa2 budget 20.0 ms per solve 3D RMSE: KIMS 0.096, LM 0.096, SFUISE 0.095, filter 0.108
ground truth
devkit ESKF (our run) (0.108 m)
SFUISE (0.095 m)
LM (0.096 m)
KIMS (0.096 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

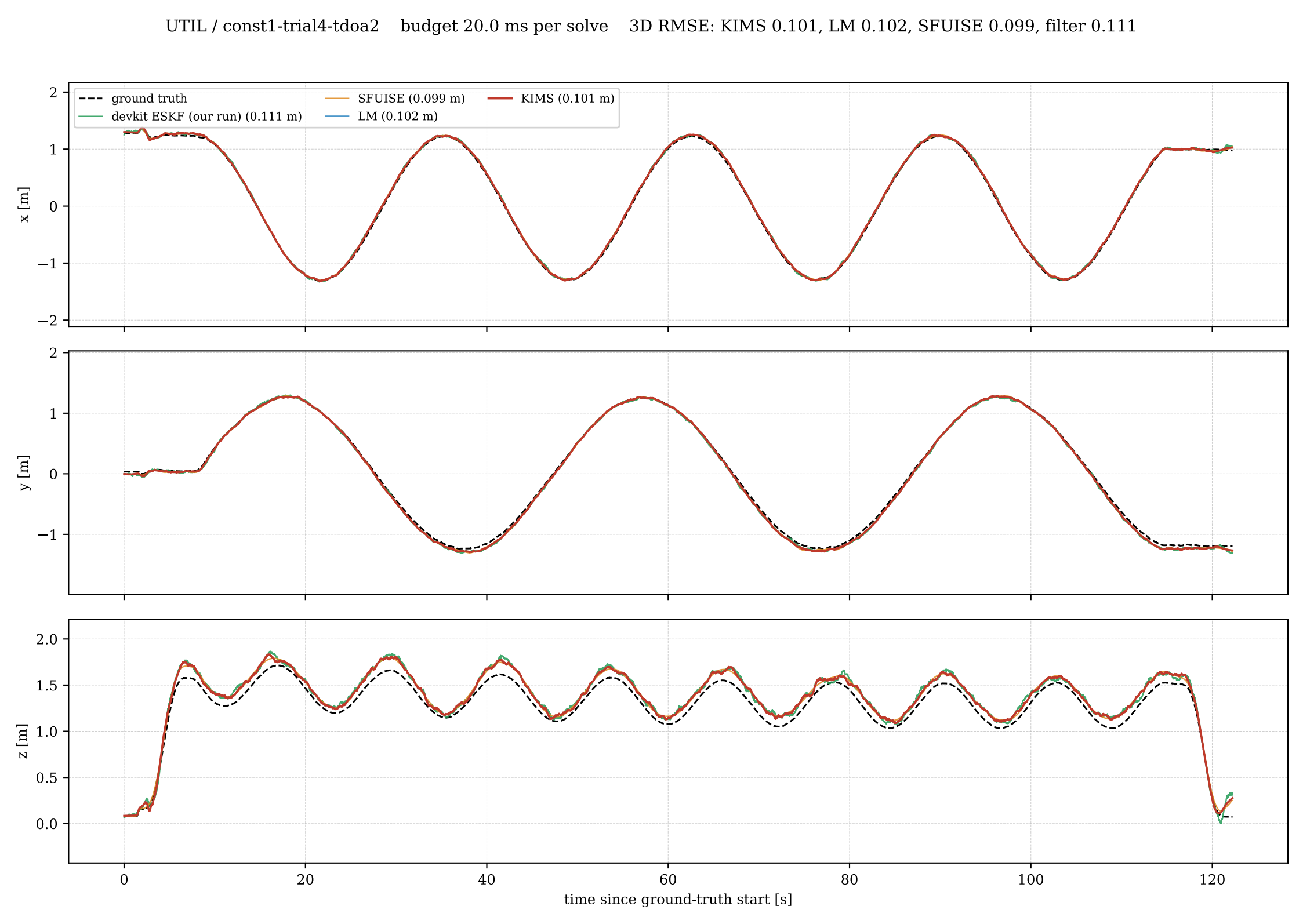

UTIL / const1-trial1-tdoa3 budget 20.0 ms per solve 3D RMSE: KIMS 0.100, LM 0.100, SFUISE 0.094, filter 0.131
ground truth
devkit ESKF (our run) (0.131 m)
SFUISE (0.094 m)
LM (0.100 m)
KIMS (0.100 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

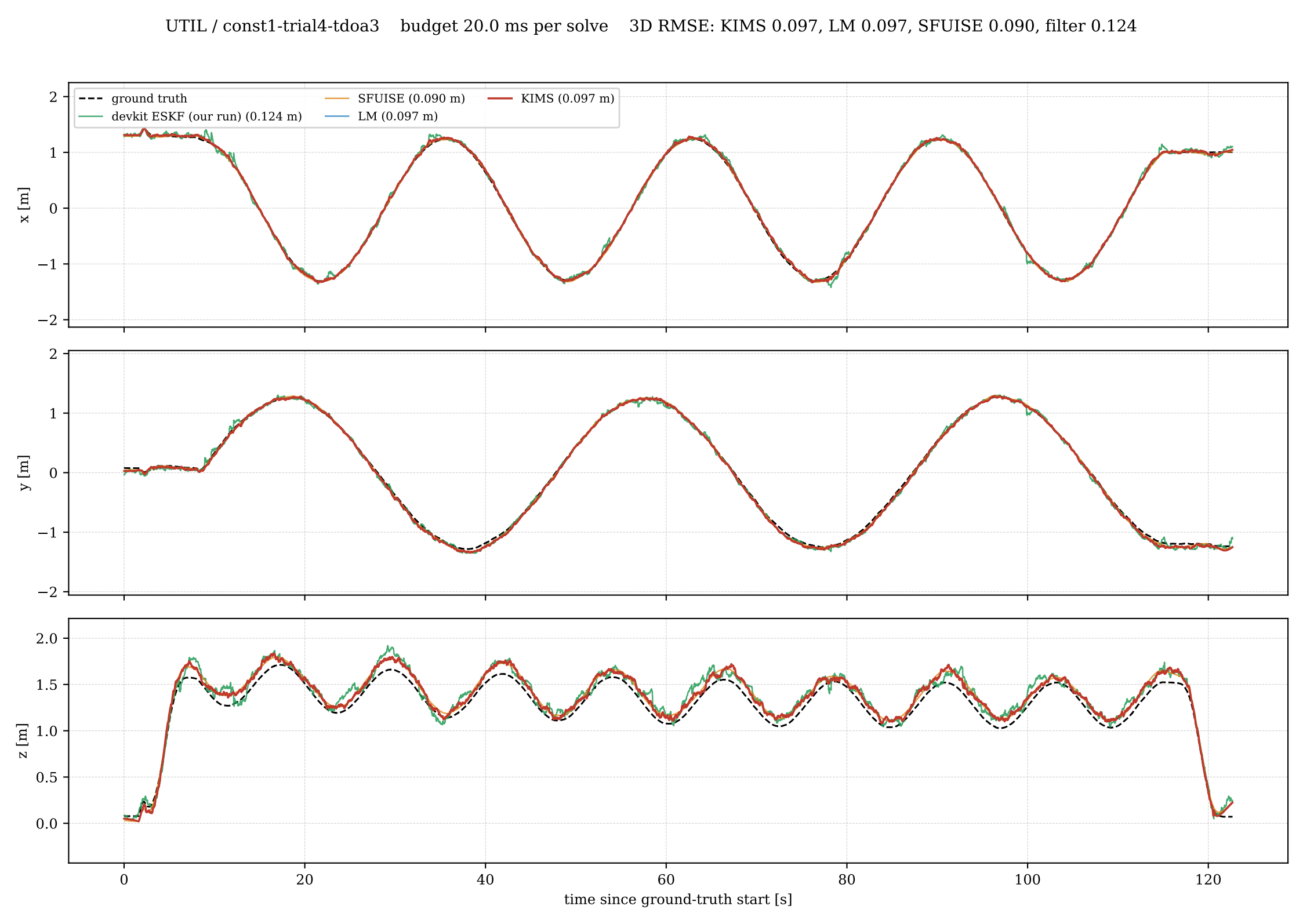

UTIL / const1-trial2-tdoa2 budget 20.0 ms per solve 3D RMSE: KIMS 0.112, LM 0.112, SFUISE 0.115, filter 0.122
ground truth
devkit ESKF (our run) (0.122 m)
SFUISE (0.115 m)
LM (0.112 m)
KIMS (0.112 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

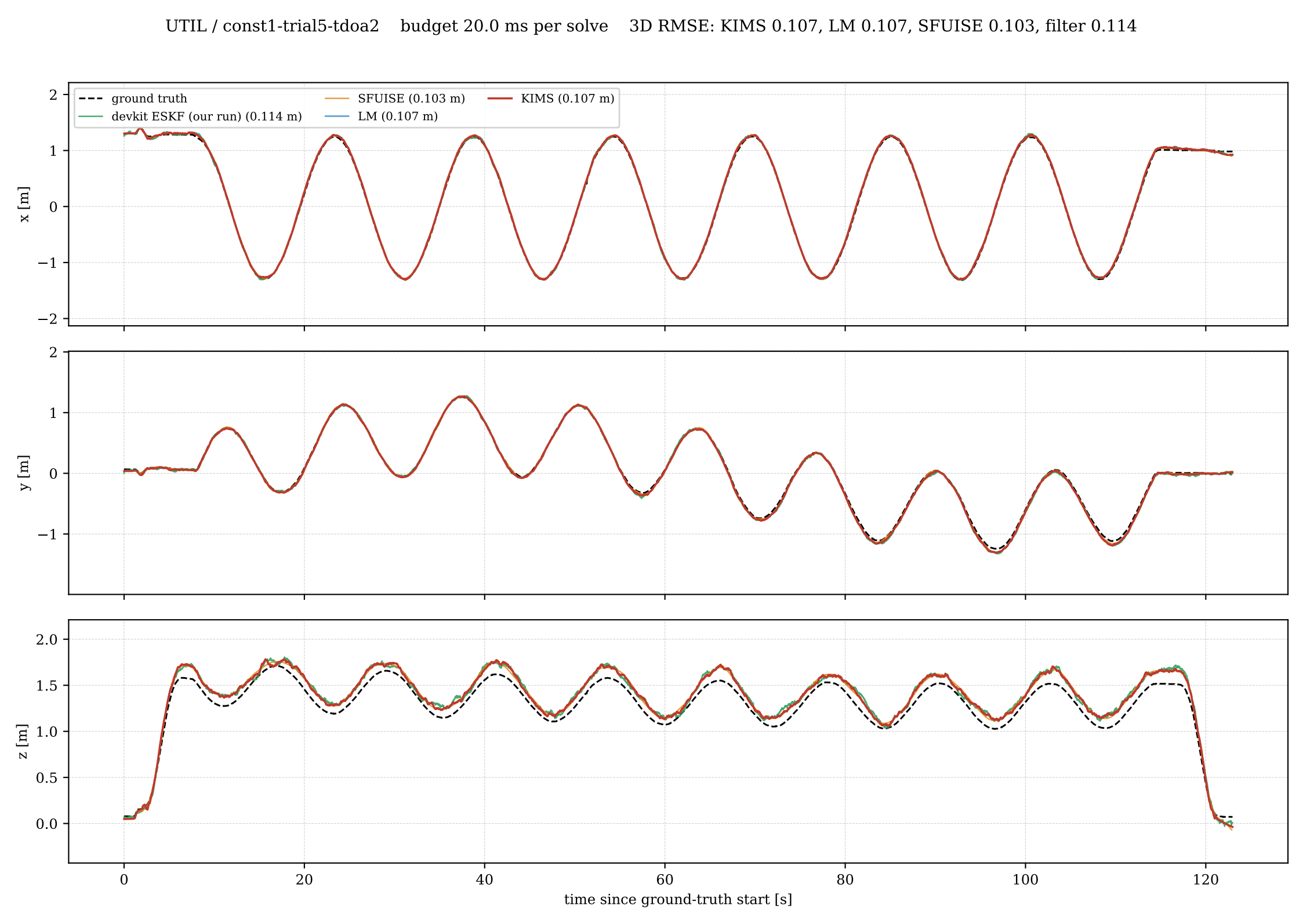

UTIL / const1-trial2-tdoa3 budget 20.0 ms per solve 3D RMSE: KIMS 0.110, LM 0.111, SFUISE 0.104, filter 0.143
ground truth
devkit ESKF (our run) (0.143 m)
SFUISE (0.104 m)
LM (0.111 m)
KIMS (0.110 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

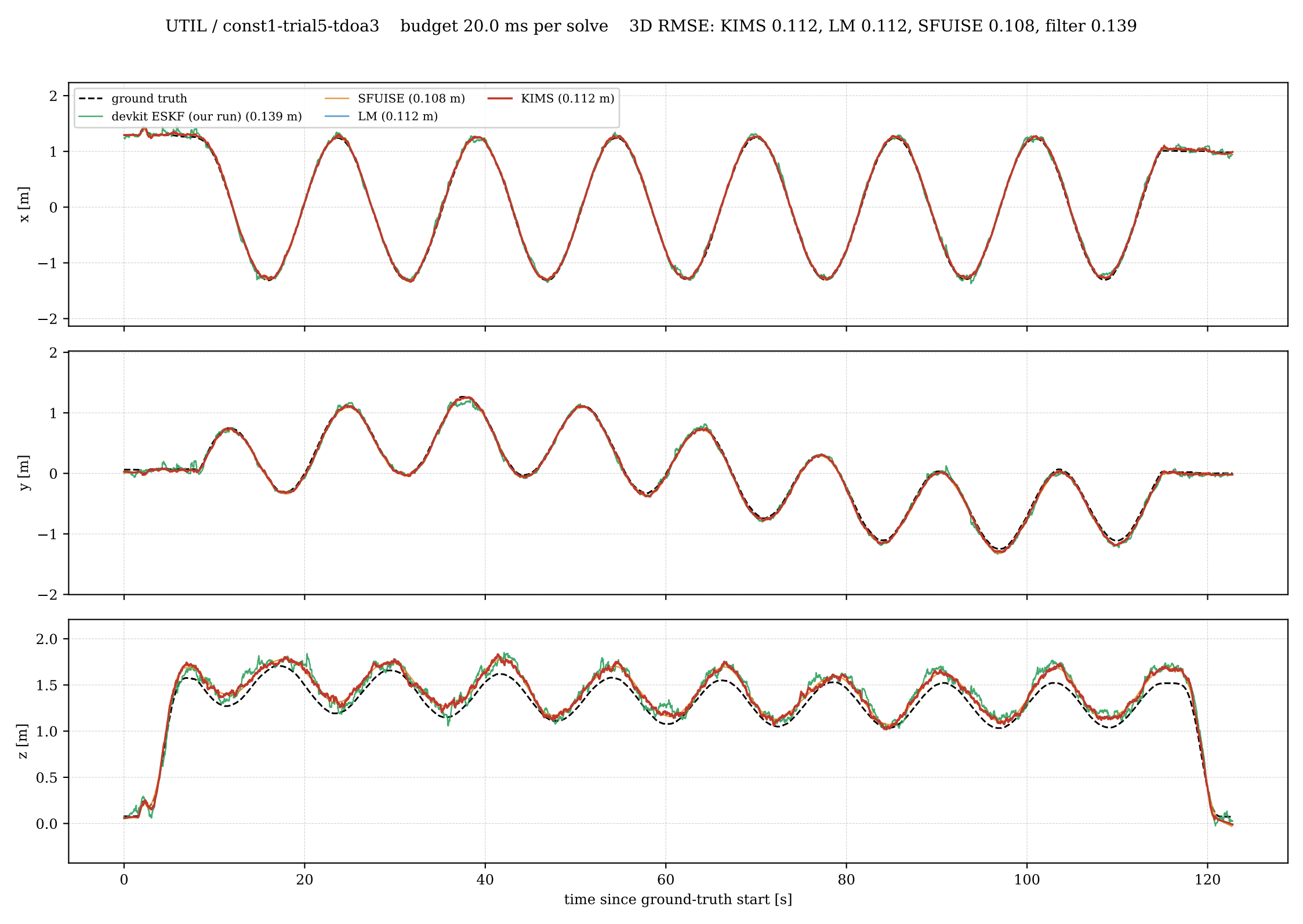

UTIL / const1-trial3-tdoa2 budget 20.0 ms per solve 3D RMSE: KIMS 0.098, LM 0.098, SFUISE 0.094, filter 0.108
ground truth
devkit ESKF (our run) (0.108 m)
SFUISE (0.094 m)
LM (0.098 m)
KIMS (0.098 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

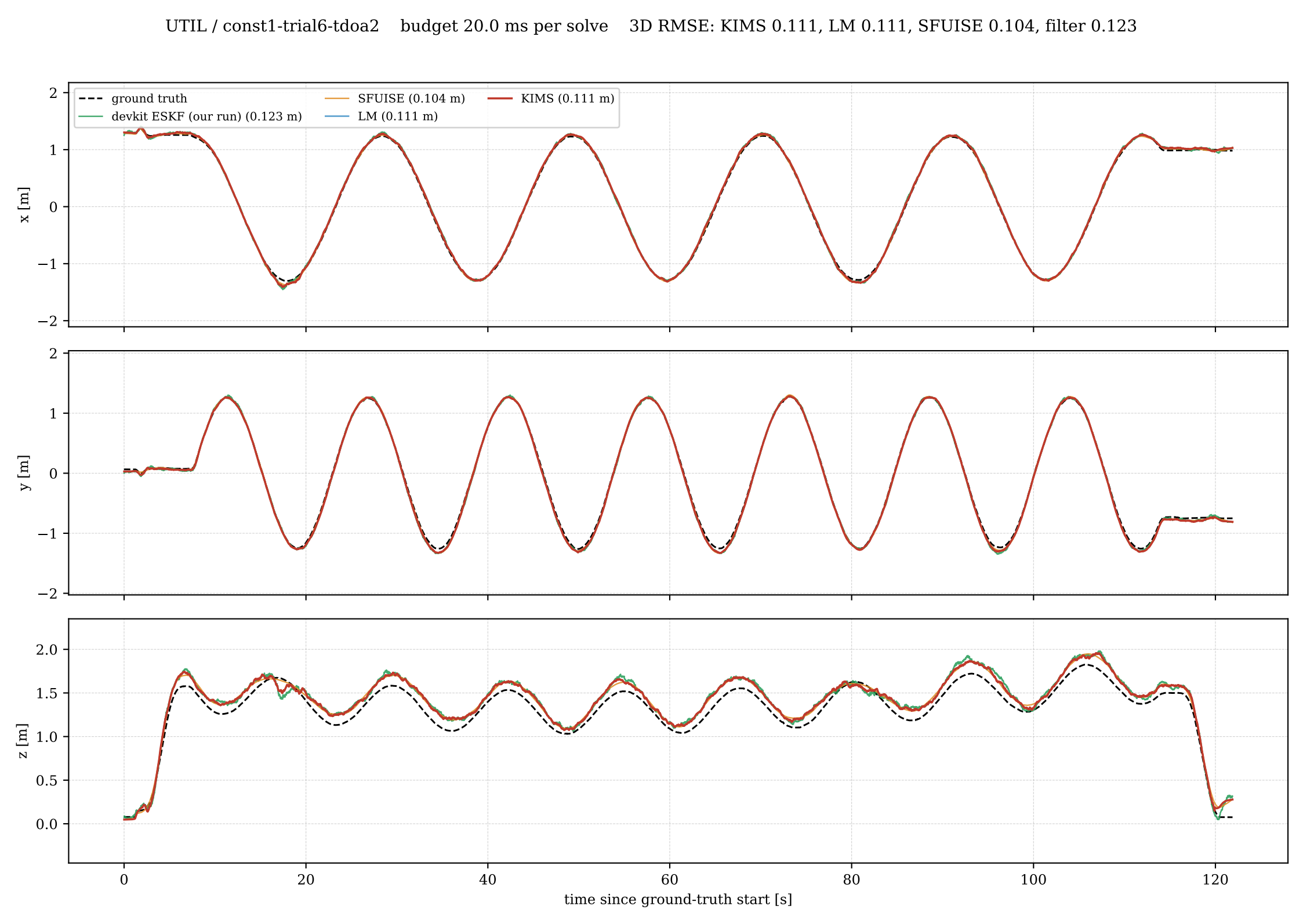

UTIL / const1-trial3-tdoa3 budget 20.0 ms per solve 3D RMSE: KIMS 0.097, LM 0.097, SFUISE 0.090, filter 0.120
ground truth
devkit ESKF (our run) (0.120 m)
SFUISE (0.090 m)
LM (0.097 m)
KIMS (0.097 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

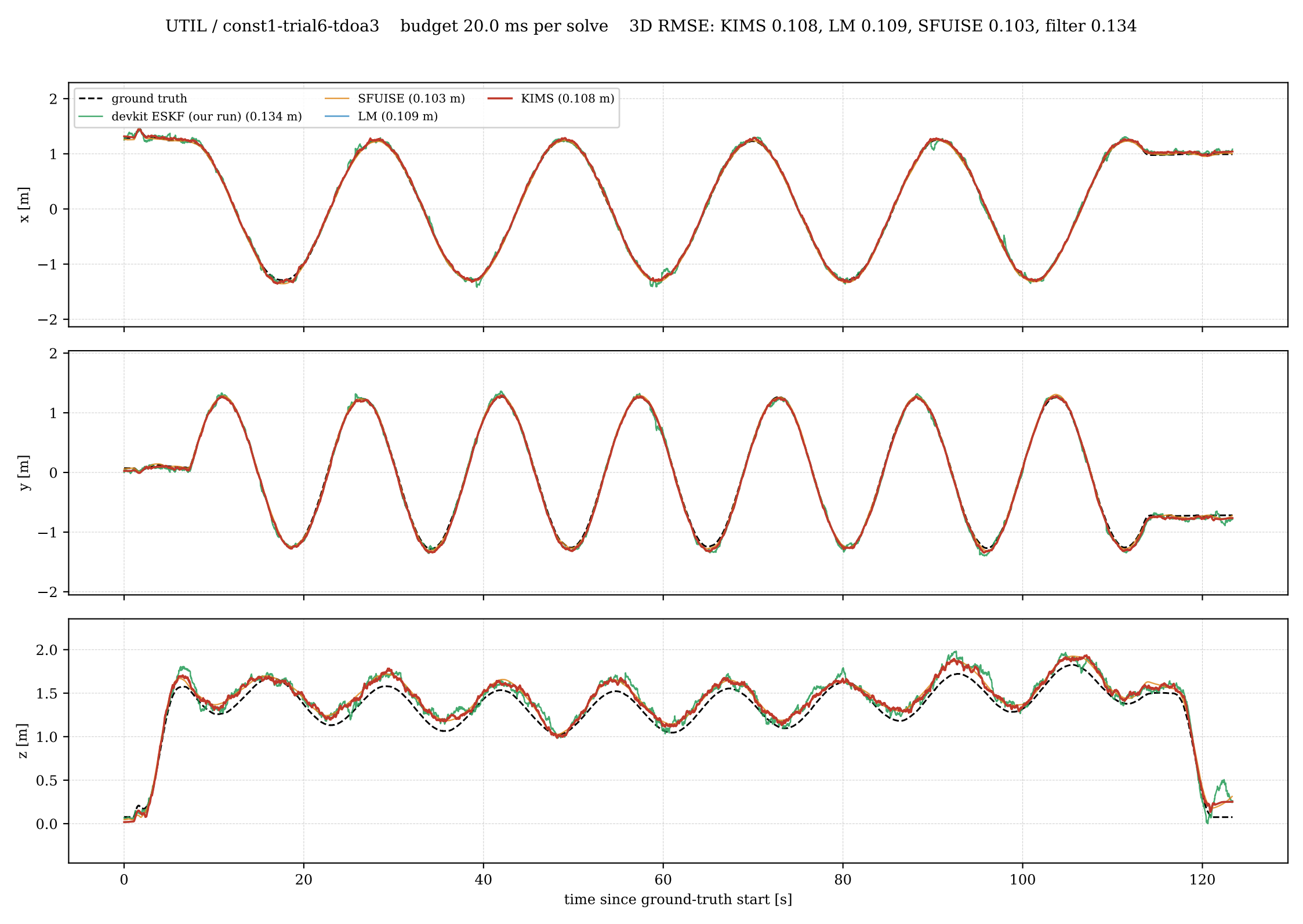

UTIL / const1-trial4-tdoa2 budget 20.0 ms per solve 3D RMSE: KIMS 0.101, LM 0.102, SFUISE 0.099, filter 0.111
ground truth
devkit ESKF (our run) (0.111 m)
SFUISE (0.099 m)
LM (0.102 m)
KIMS (0.101 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

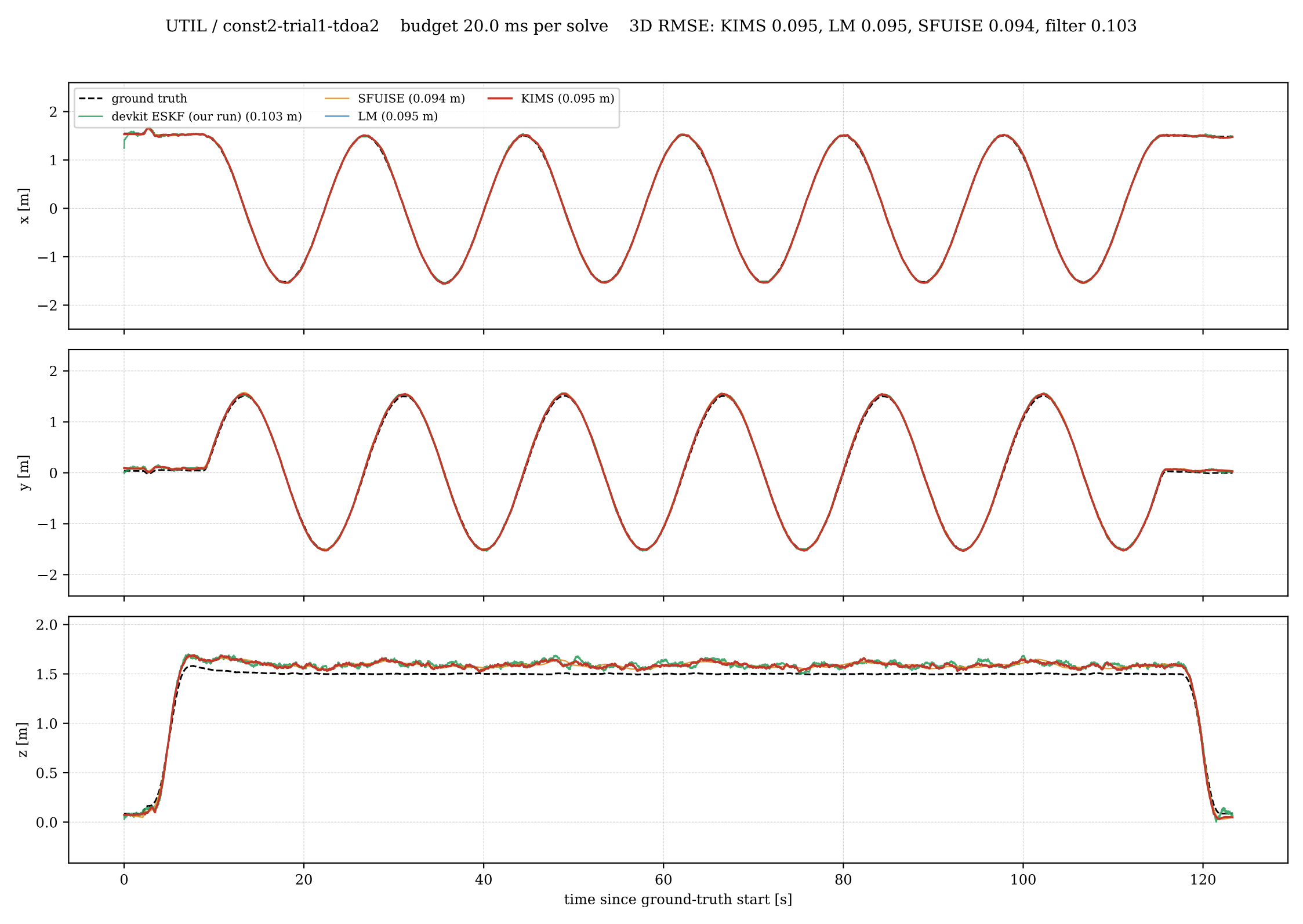

UTIL / const1-trial4-tdoa3 budget 20.0 ms per solve 3D RMSE: KIMS 0.097, LM 0.097, SFUISE 0.090, filter 0.124
ground truth
devkit ESKF (our run) (0.124 m)
SFUISE (0.090 m)
LM (0.097 m)
KIMS (0.097 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

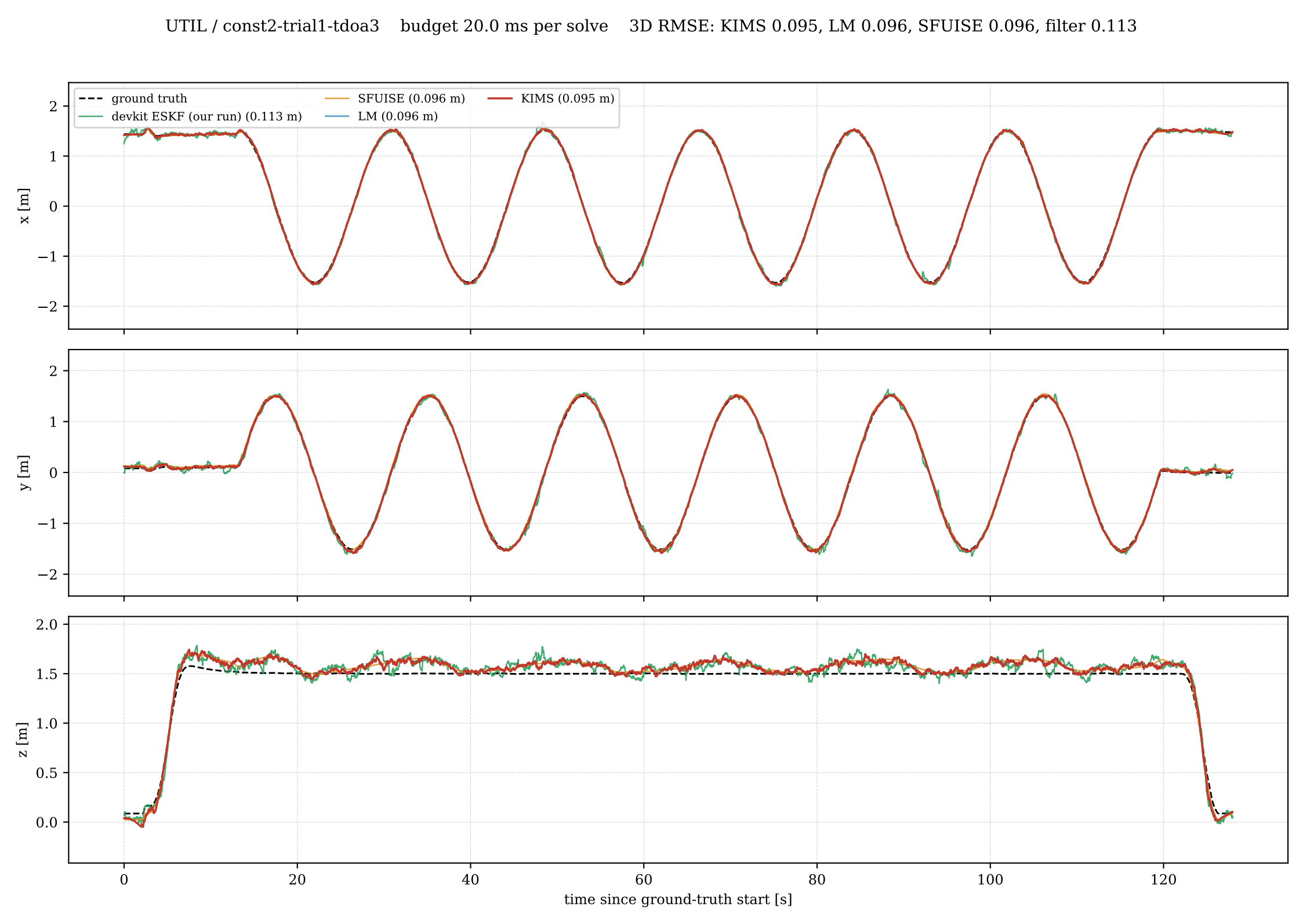

UTIL / const1-trial5-tdoa2 budget 20.0 ms per solve 3D RMSE: KIMS 0.107, LM 0.107, SFUISE 0.103, filter 0.114
ground truth
devkit ESKF (our run) (0.114 m)
SFUISE (0.103 m)
LM (0.107 m)
KIMS (0.107 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

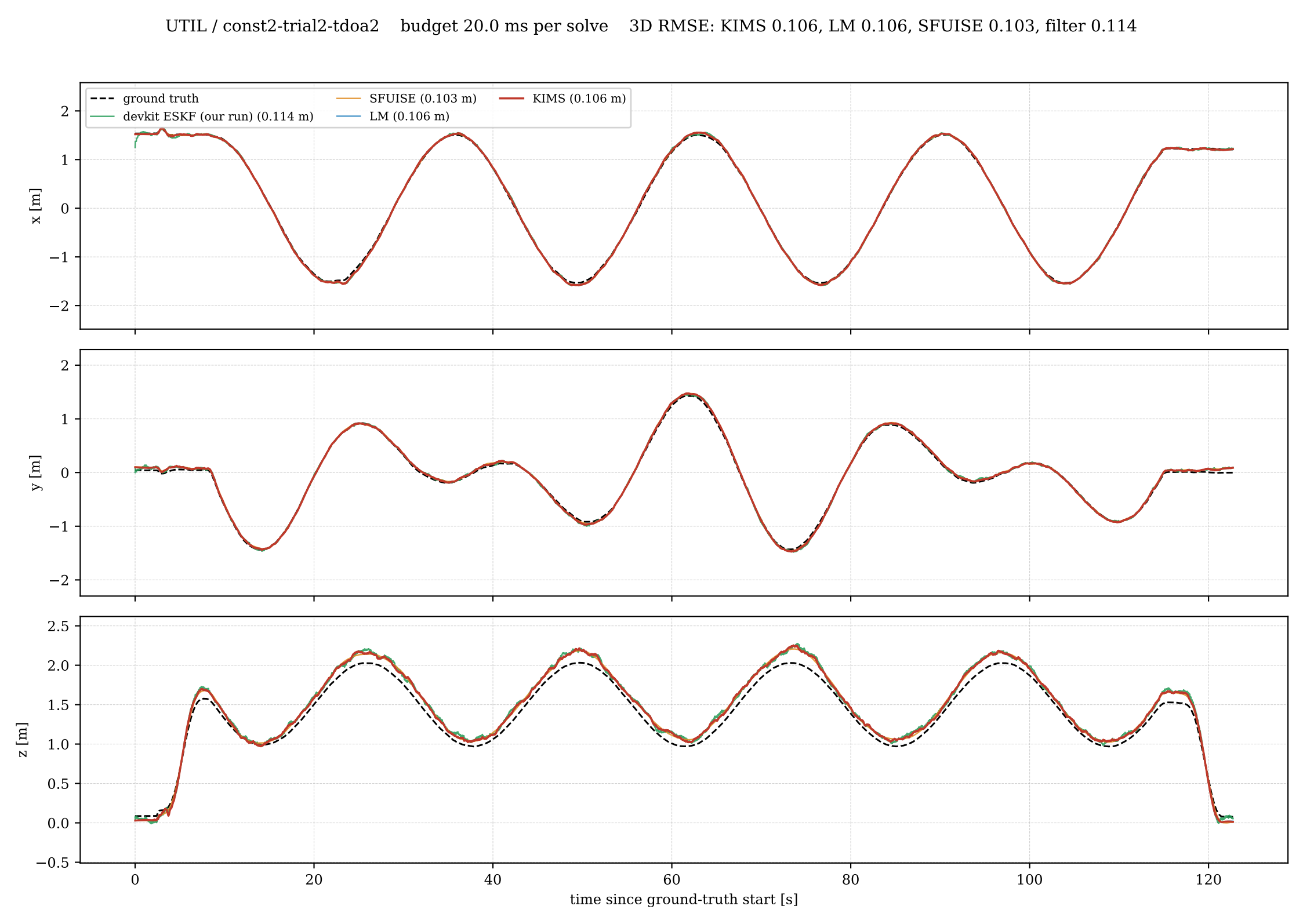

UTIL / const1-trial5-tdoa3 budget 20.0 ms per solve 3D RMSE: KIMS 0.112, LM 0.112, SFUISE 0.108, filter 0.139
ground truth
devkit ESKF (our run) (0.139 m)
SFUISE (0.108 m)
LM (0.112 m)
KIMS (0.112 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

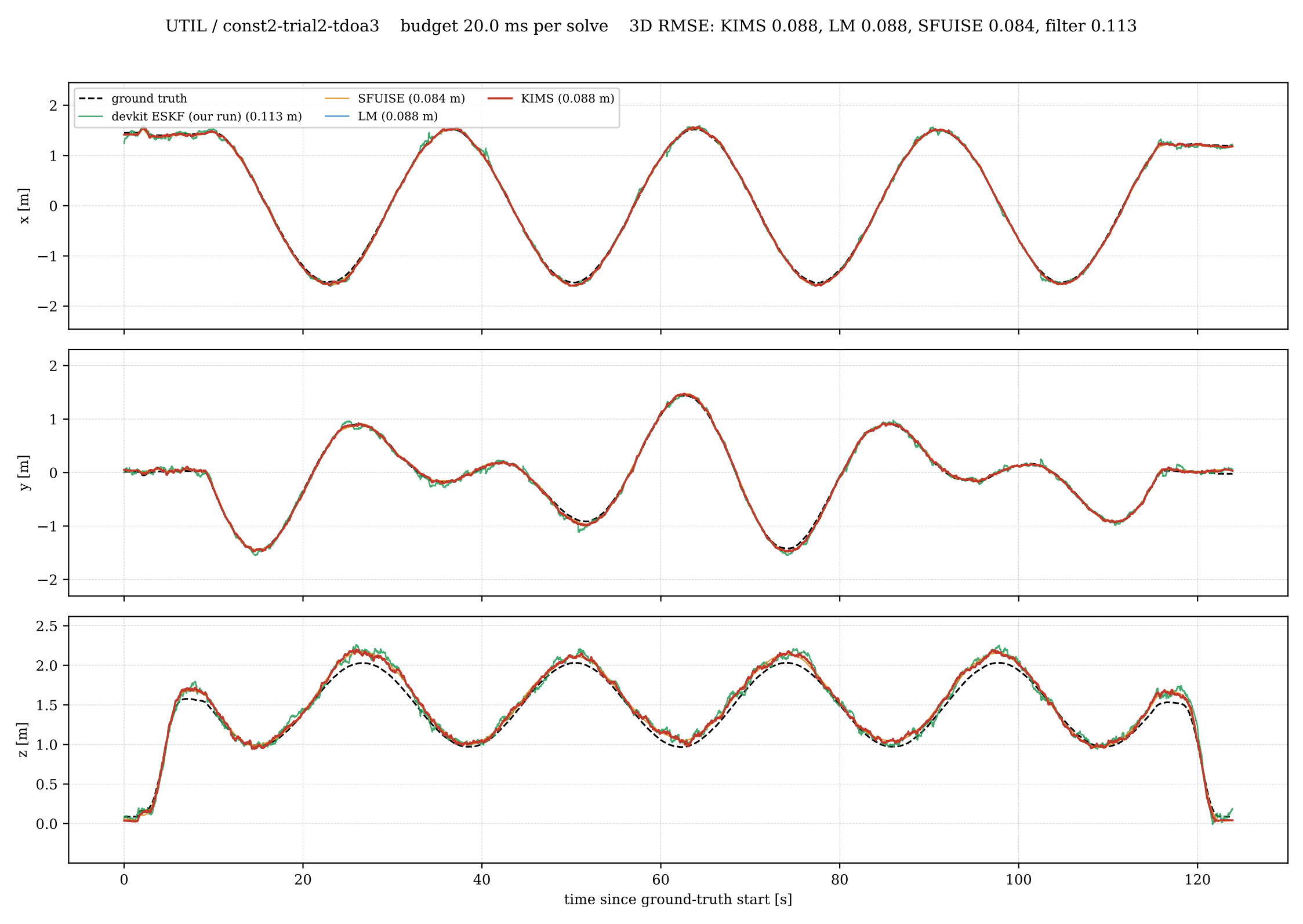

UTIL / const1-trial6-tdoa2 budget 20.0 ms per solve 3D RMSE: KIMS 0.111, LM 0.111, SFUISE 0.104, filter 0.123
ground truth
devkit ESKF (our run) (0.123 m)
SFUISE (0.104 m)
LM (0.111 m)
KIMS (0.111 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

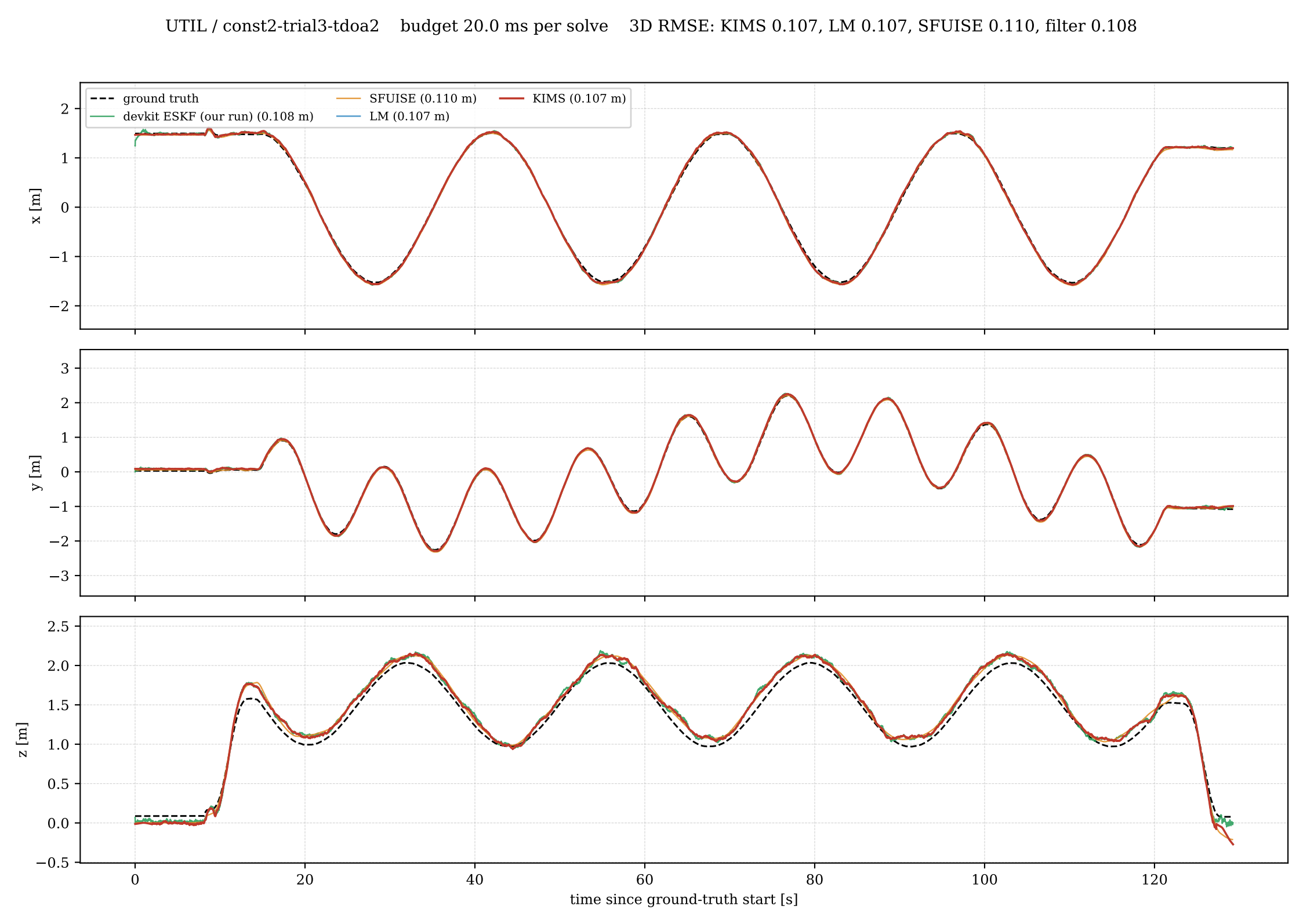

UTIL / const1-trial6-tdoa3 budget 20.0 ms per solve 3D RMSE: KIMS 0.108, LM 0.109, SFUISE 0.103, filter 0.134
ground truth
devkit ESKF (our run) (0.134 m)
SFUISE (0.103 m)
LM (0.109 m)
KIMS (0.108 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

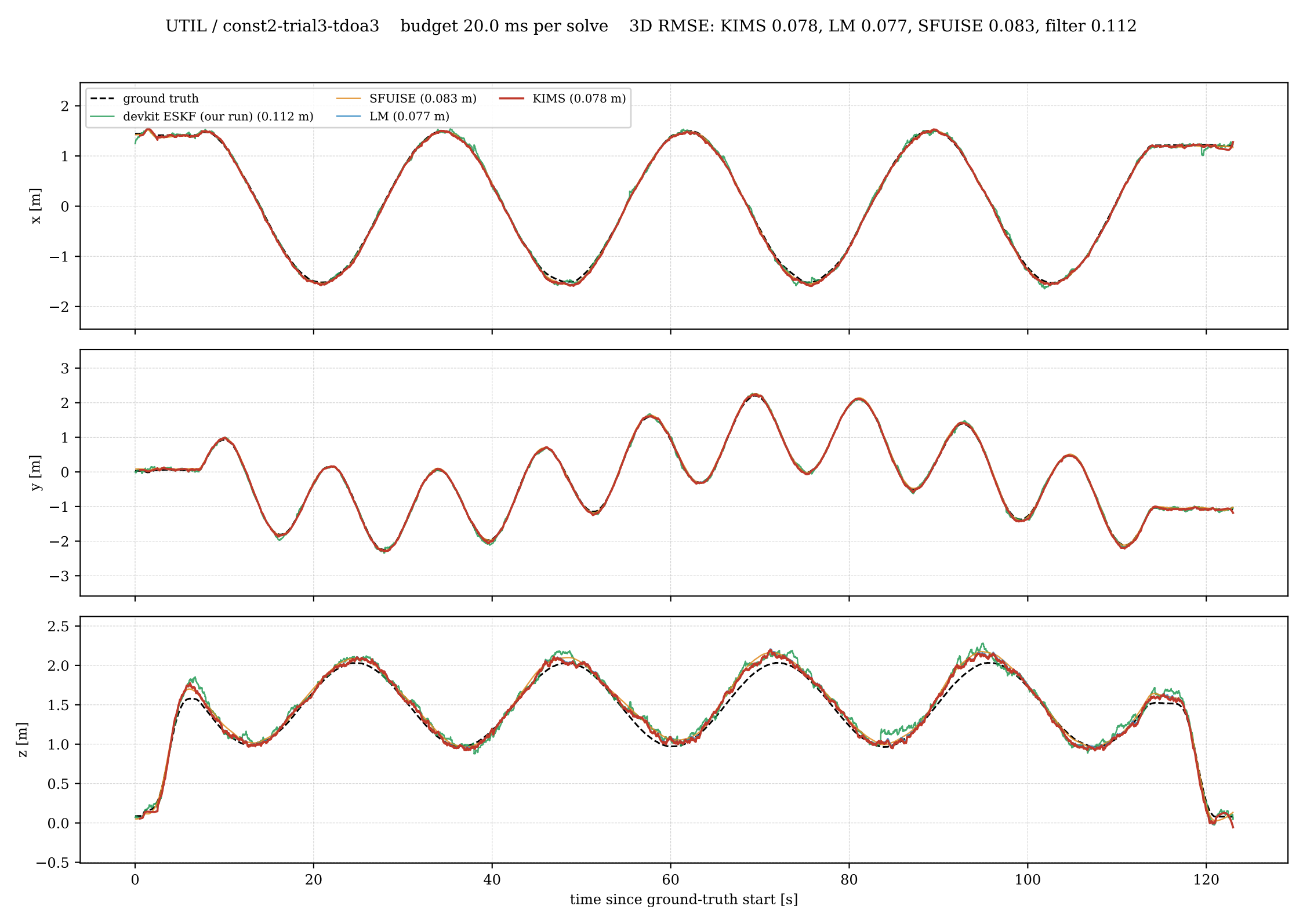

UTIL / const2-trial1-tdoa2 budget 20.0 ms per solve 3D RMSE: KIMS 0.095, LM 0.095, SFUISE 0.094, filter 0.103
ground truth
devkit ESKF (our run) (0.103 m)
SFUISE (0.094 m)
LM (0.095 m)
KIMS (0.095 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

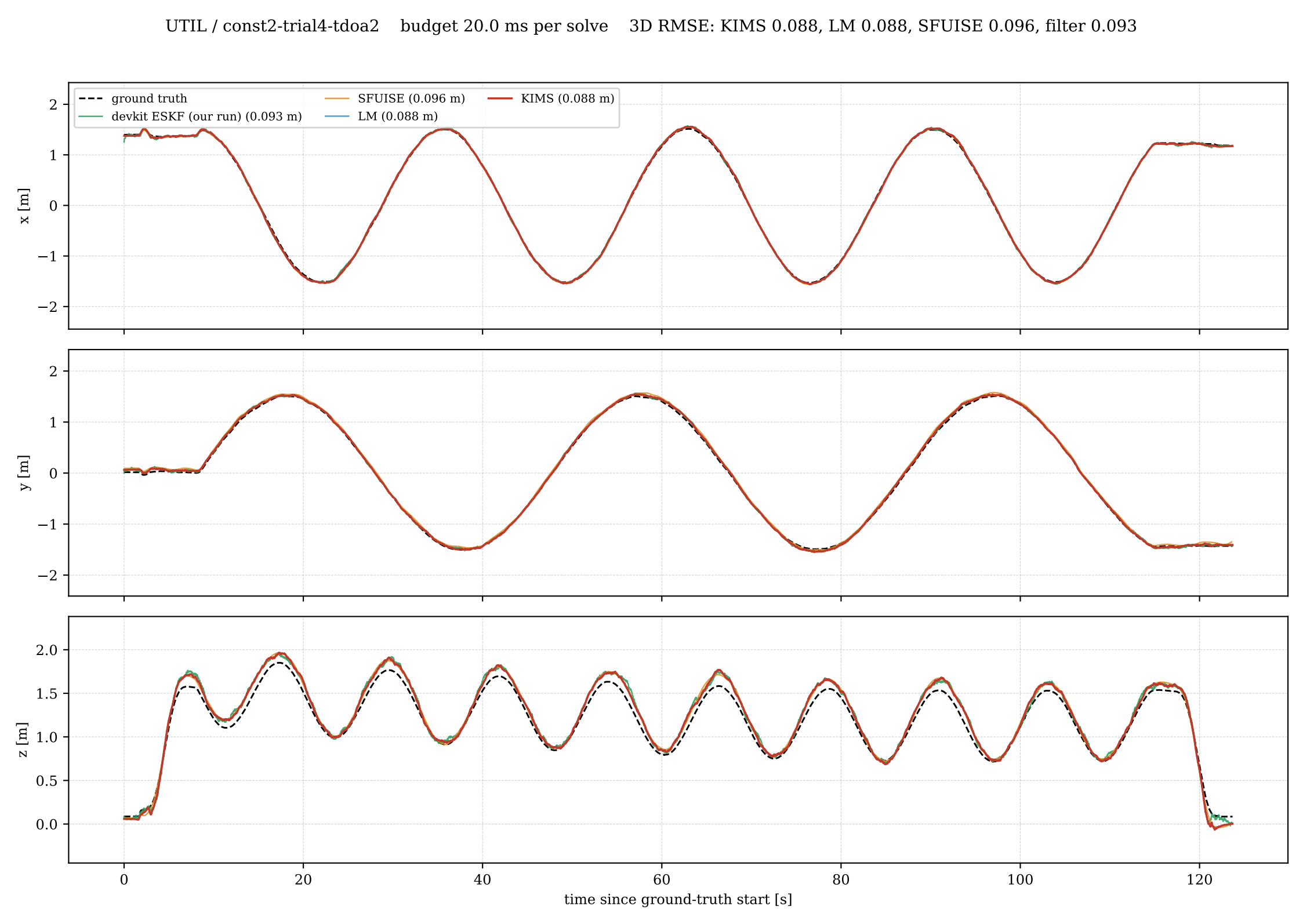

UTIL / const2-trial1-tdoa3 budget 20.0 ms per solve 3D RMSE: KIMS 0.095, LM 0.096, SFUISE 0.096, filter 0.113
ground truth
devkit ESKF (our run) (0.113 m)
SFUISE (0.096 m)
LM (0.096 m)
KIMS (0.095 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

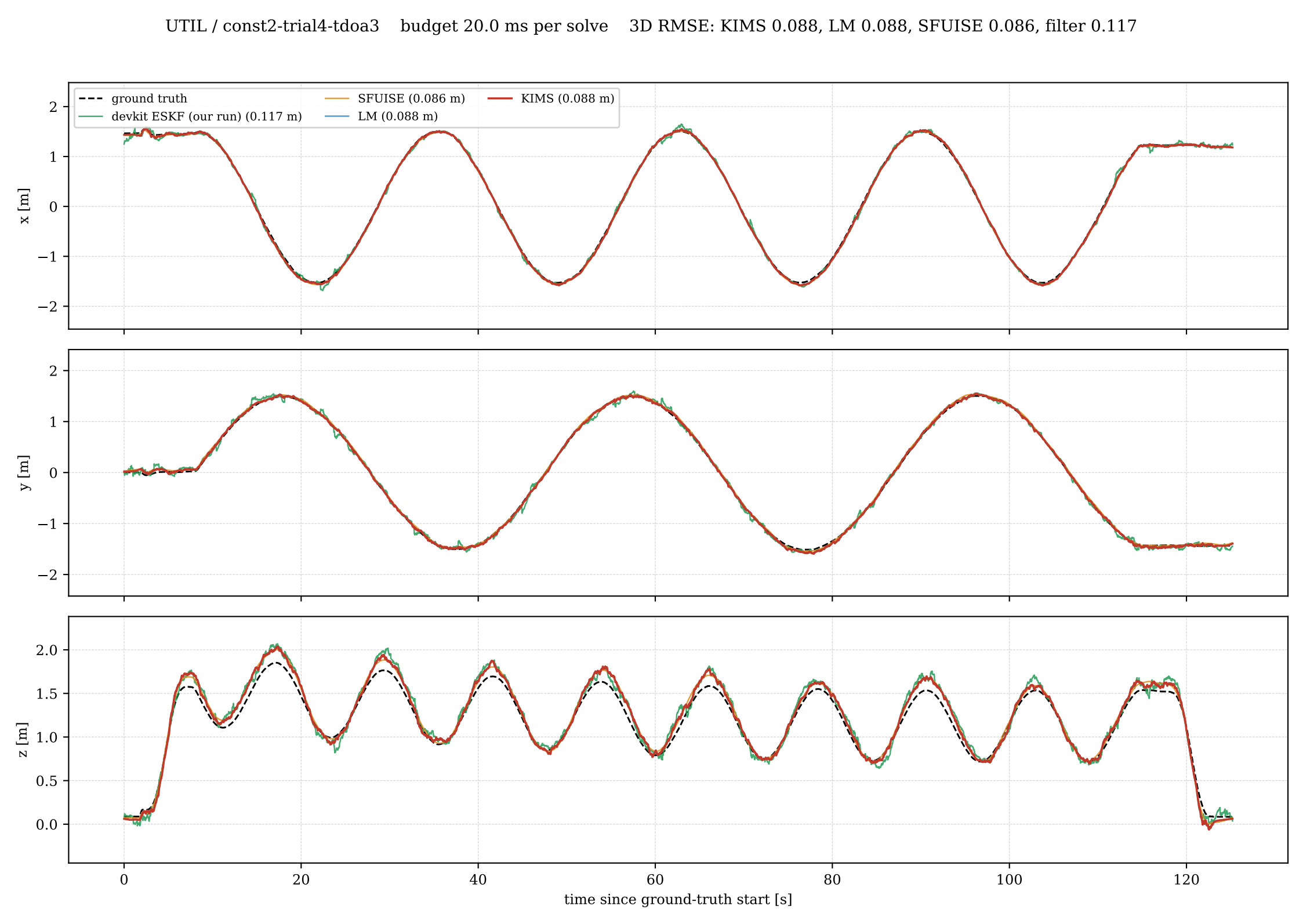

UTIL / const2-trial2-tdoa2 budget 20.0 ms per solve 3D RMSE: KIMS 0.106, LM 0.106, SFUISE 0.103, filter 0.114
ground truth
devkit ESKF (our run) (0.114 m)
SFUISE (0.103 m)
LM (0.106 m)
KIMS (0.106 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

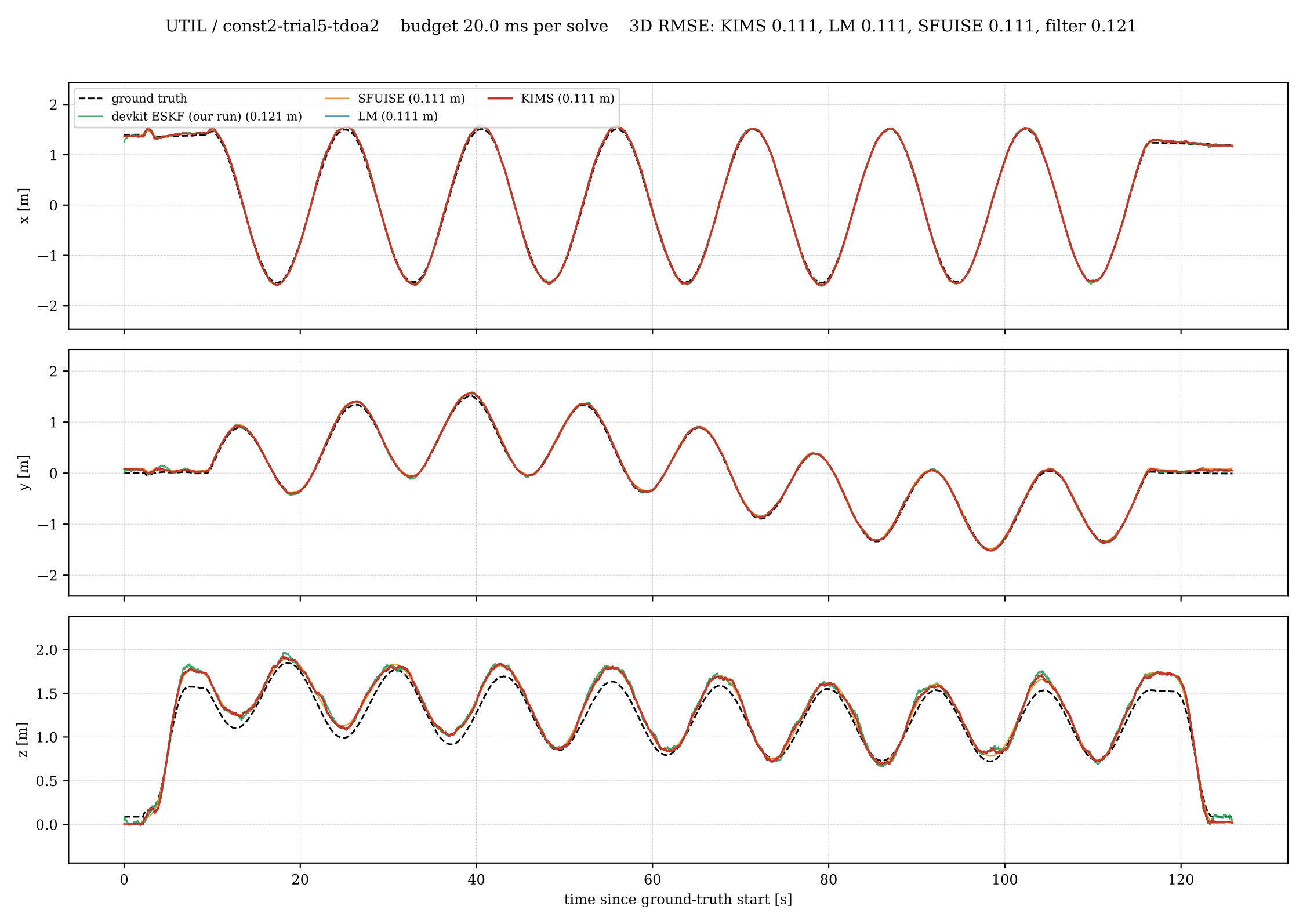

UTIL / const2-trial2-tdoa3 budget 20.0 ms per solve 3D RMSE: KIMS 0.088, LM 0.088, SFUISE 0.084, filter 0.113
ground truth
devkit ESKF (our run) (0.113 m)
SFUISE (0.084 m)
LM (0.088 m)
KIMS (0.088 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

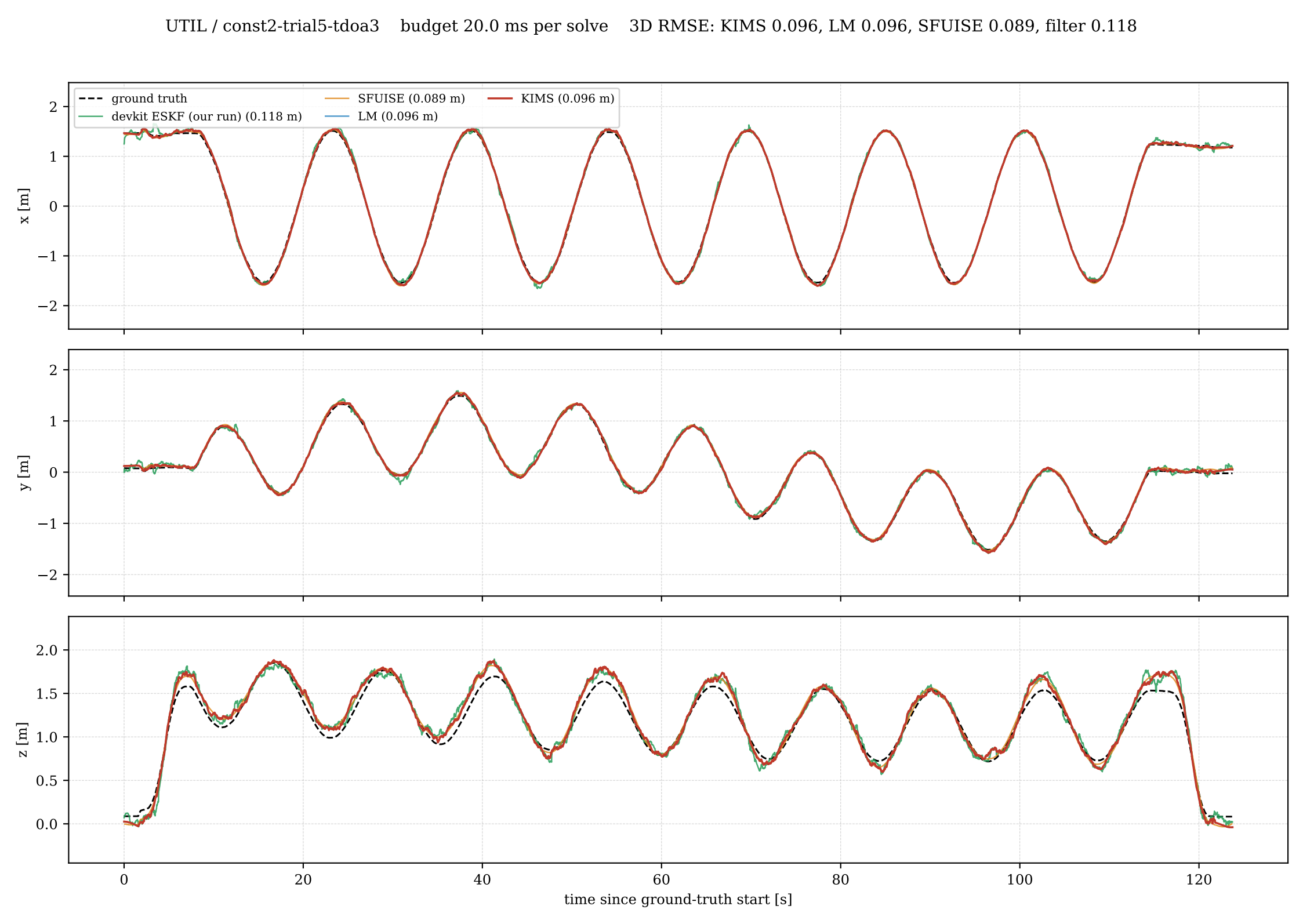

UTIL / const2-trial3-tdoa2 budget 20.0 ms per solve 3D RMSE: KIMS 0.107, LM 0.107, SFUISE 0.110, filter 0.108
ground truth
devkit ESKF (our run) (0.108 m)
SFUISE (0.110 m)
LM (0.107 m)
KIMS (0.107 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

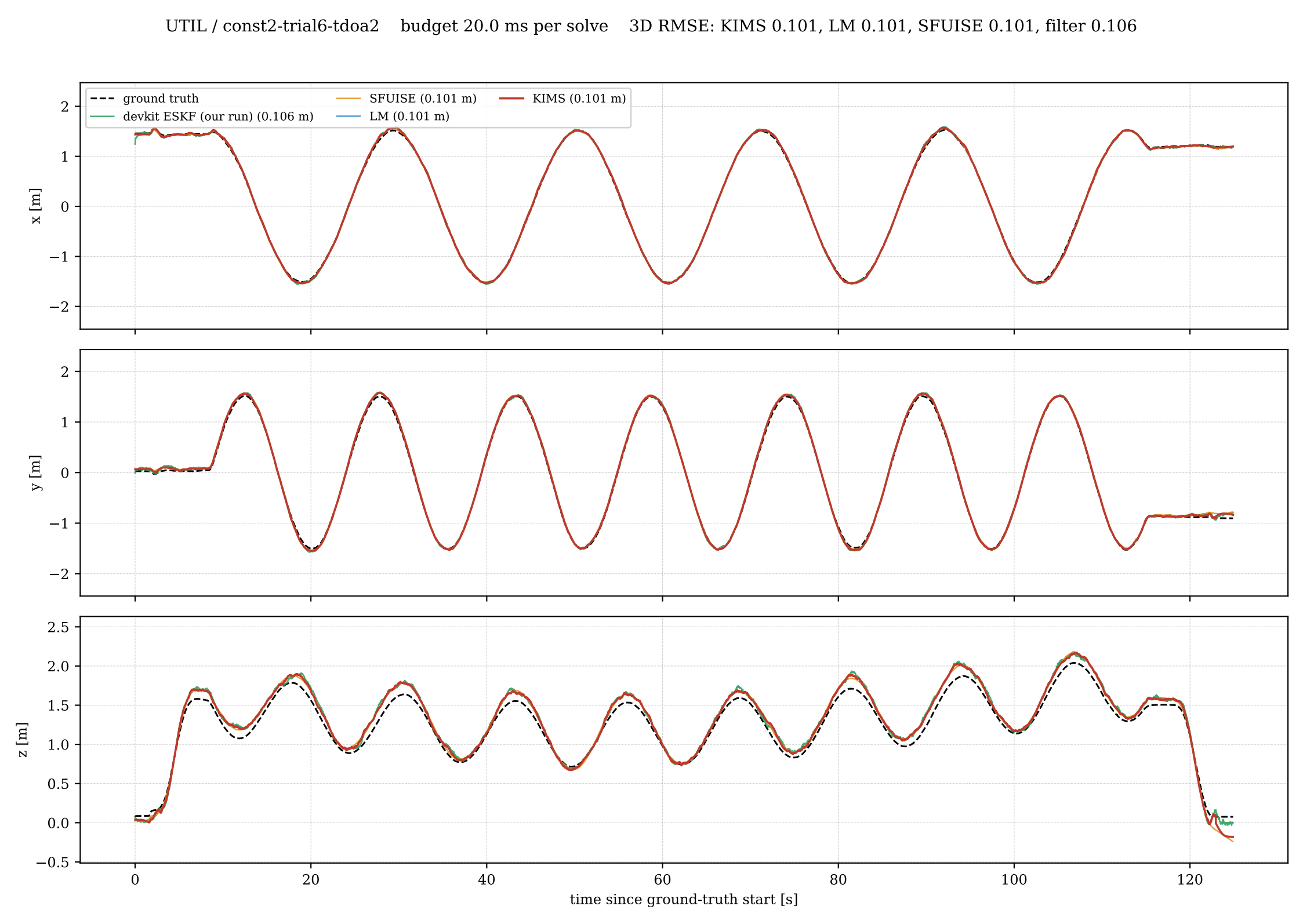

UTIL / const2-trial3-tdoa3 budget 20.0 ms per solve 3D RMSE: KIMS 0.078, LM 0.077, SFUISE 0.083, filter 0.112
ground truth
devkit ESKF (our run) (0.112 m)
SFUISE (0.083 m)
LM (0.077 m)
KIMS (0.078 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

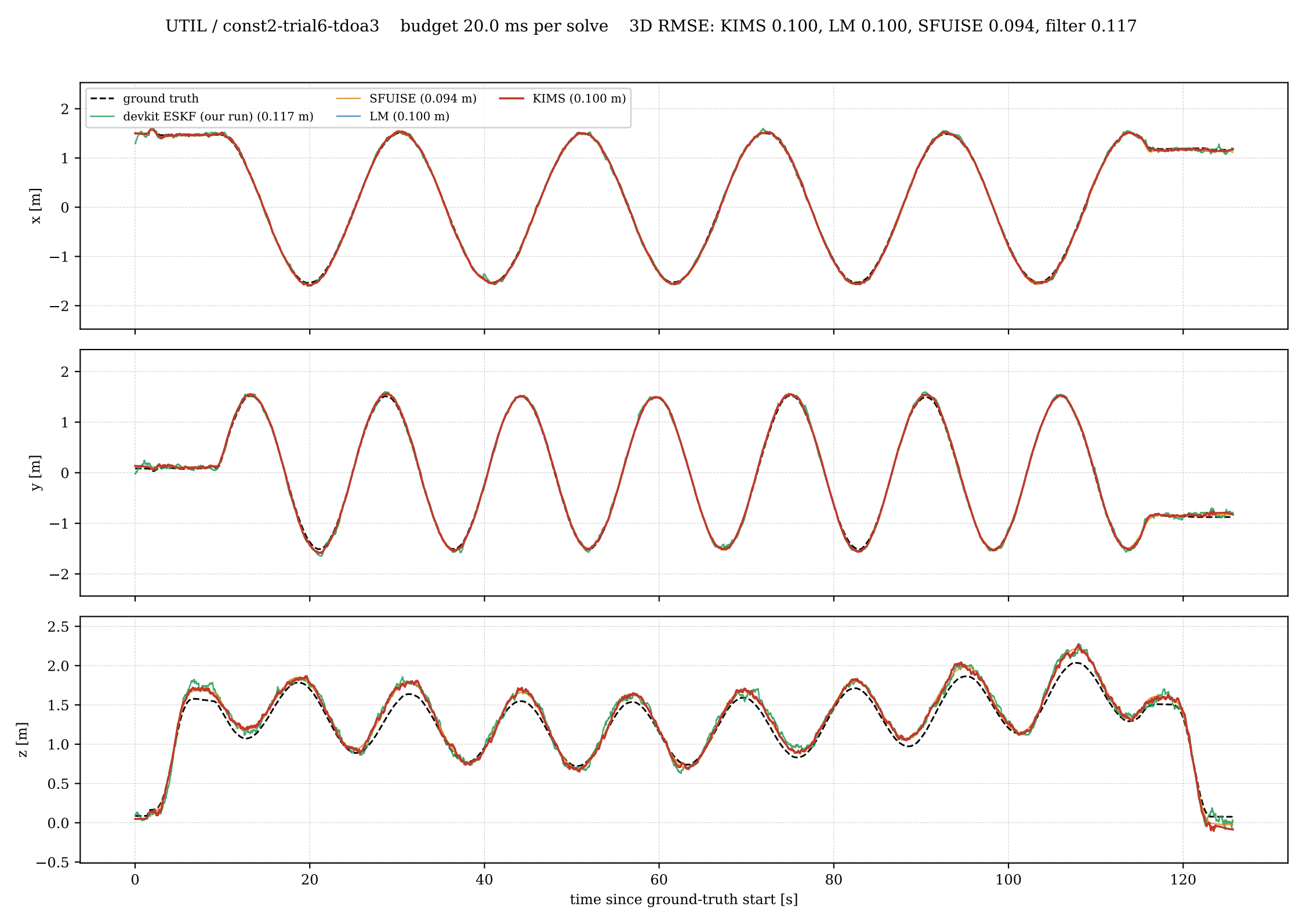

UTIL / const2-trial4-tdoa2 budget 20.0 ms per solve 3D RMSE: KIMS 0.088, LM 0.088, SFUISE 0.096, filter 0.093
ground truth
devkit ESKF (our run) (0.093 m)
SFUISE (0.096 m)
LM (0.088 m)
KIMS (0.088 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

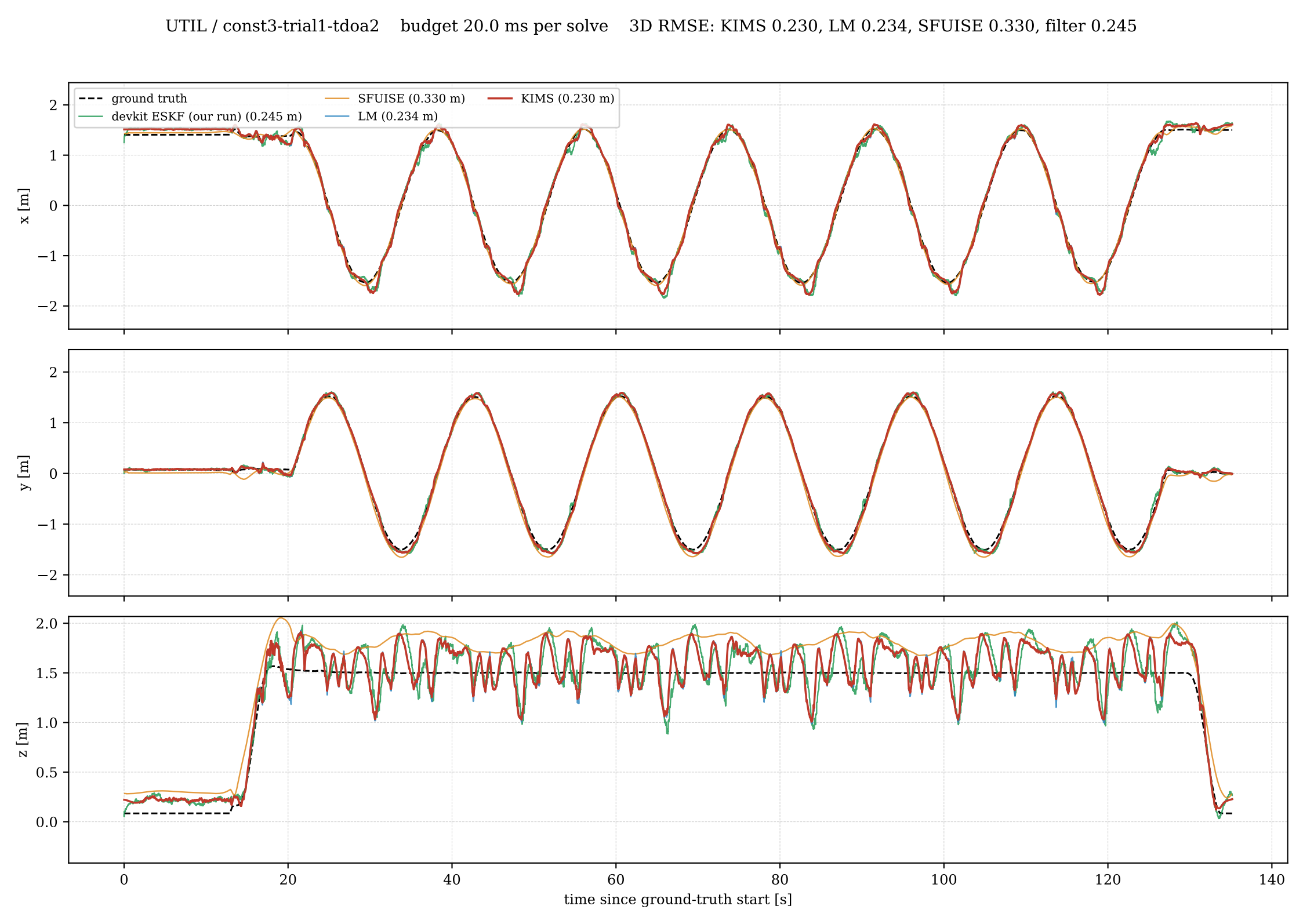

UTIL / const2-trial4-tdoa3 budget 20.0 ms per solve 3D RMSE: KIMS 0.088, LM 0.088, SFUISE 0.086, filter 0.117
ground truth
devkit ESKF (our run) (0.117 m)
SFUISE (0.086 m)
LM (0.088 m)
KIMS (0.088 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

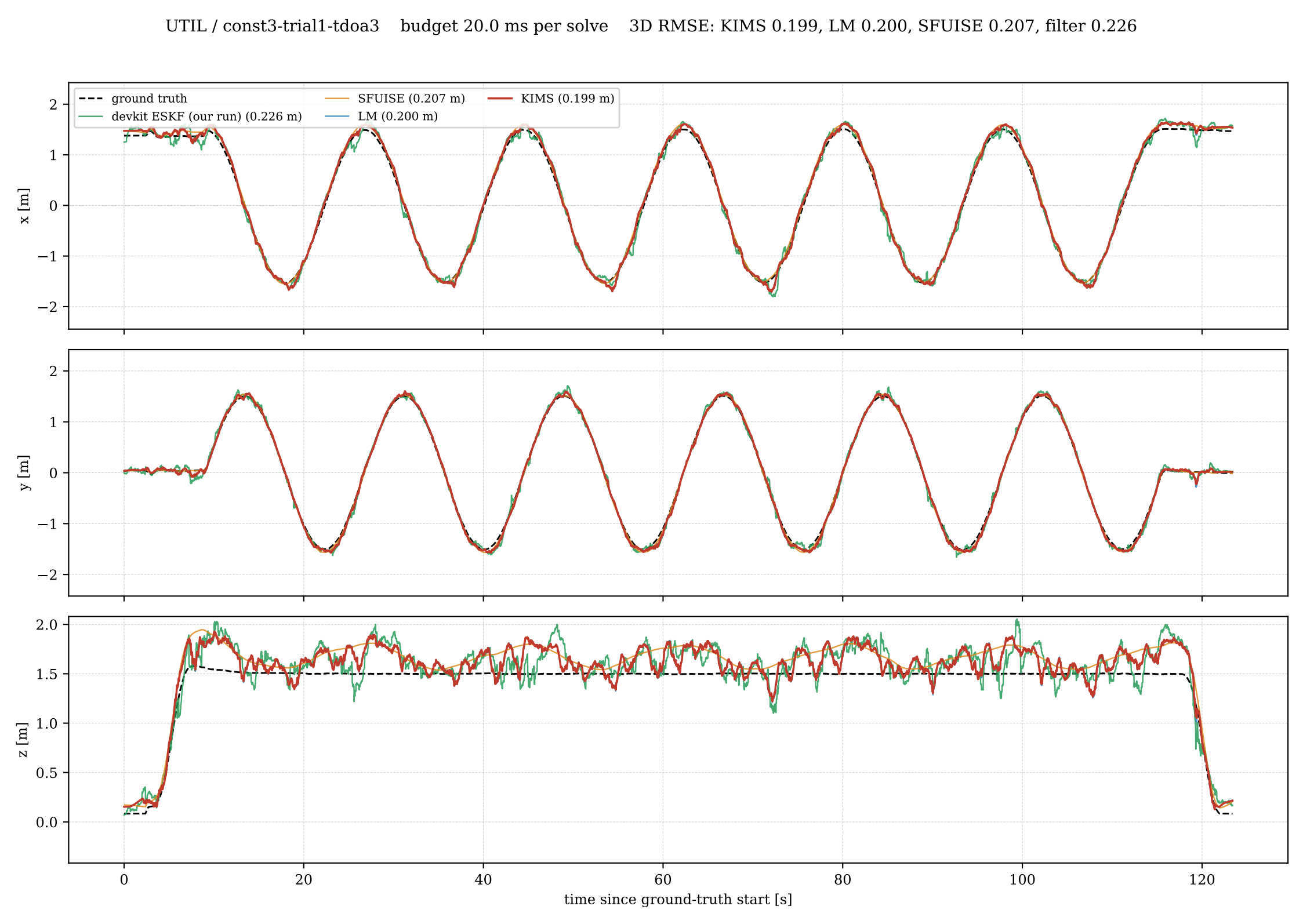

UTIL / const2-trial5-tdoa2 budget 20.0 ms per solve 3D RMSE: KIMS 0.111, LM 0.111, SFUISE 0.111, filter 0.121
ground truth
devkit ESKF (our run) (0.121 m)
SFUISE (0.111 m)
LM (0.111 m)
KIMS (0.111 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

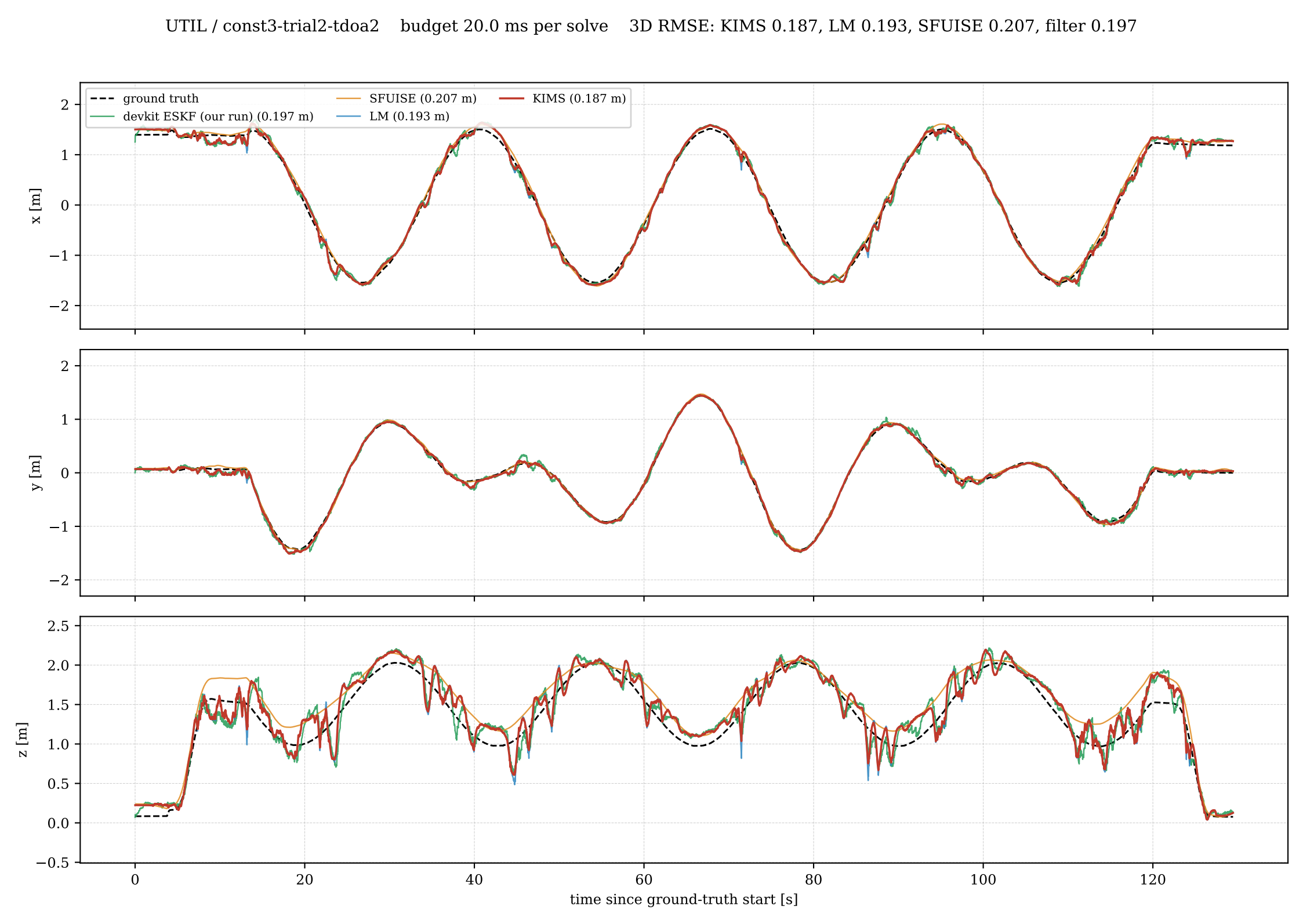

UTIL / const2-trial5-tdoa3 budget 20.0 ms per solve 3D RMSE: KIMS 0.096, LM 0.096, SFUISE 0.089, filter 0.118
ground truth
devkit ESKF (our run) (0.118 m)
SFUISE (0.089 m)
LM (0.096 m)
KIMS (0.096 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

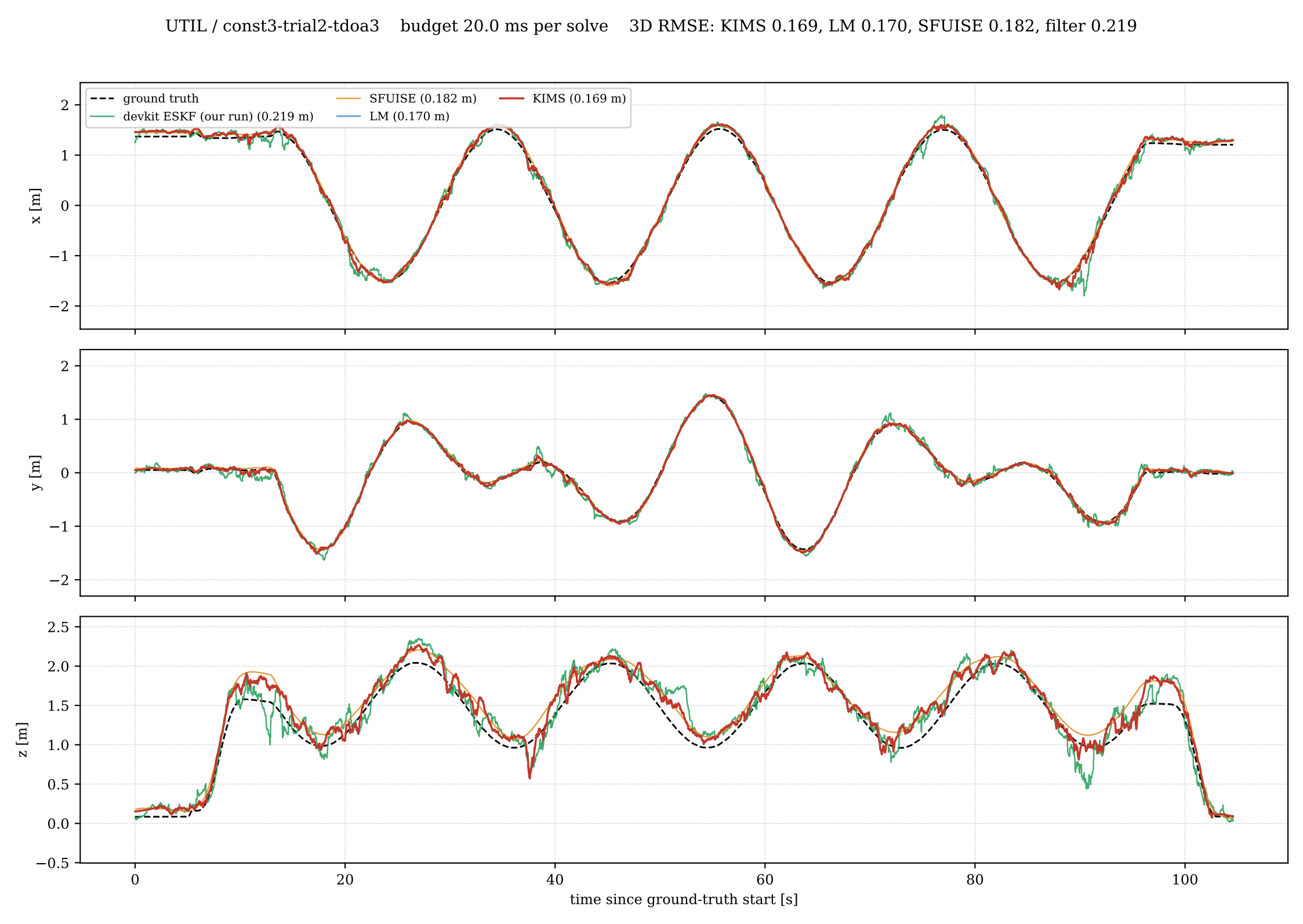

UTIL / const2-trial6-tdoa2 budget 20.0 ms per solve 3D RMSE: KIMS 0.101, LM 0.101, SFUISE 0.101, filter 0.106
ground truth
devkit ESKF (our run) (0.106 m)
SFUISE (0.101 m)
LM (0.101 m)
KIMS (0.101 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

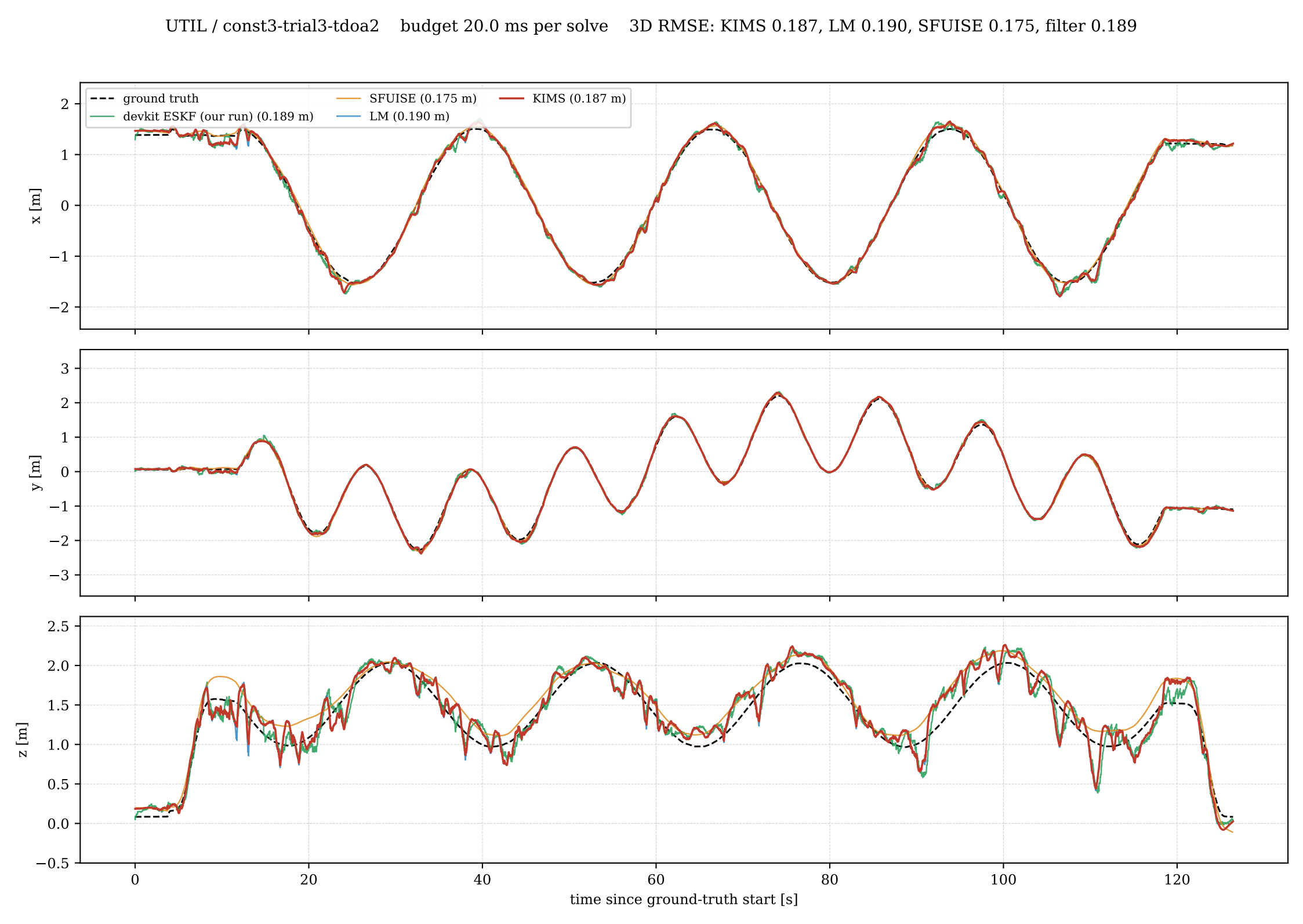

UTIL / const2-trial6-tdoa3 budget 20.0 ms per solve 3D RMSE: KIMS 0.100, LM 0.100, SFUISE 0.094, filter 0.117
ground truth
devkit ESKF (our run) (0.117 m)
SFUISE (0.094 m)
LM (0.100 m)
KIMS (0.100 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

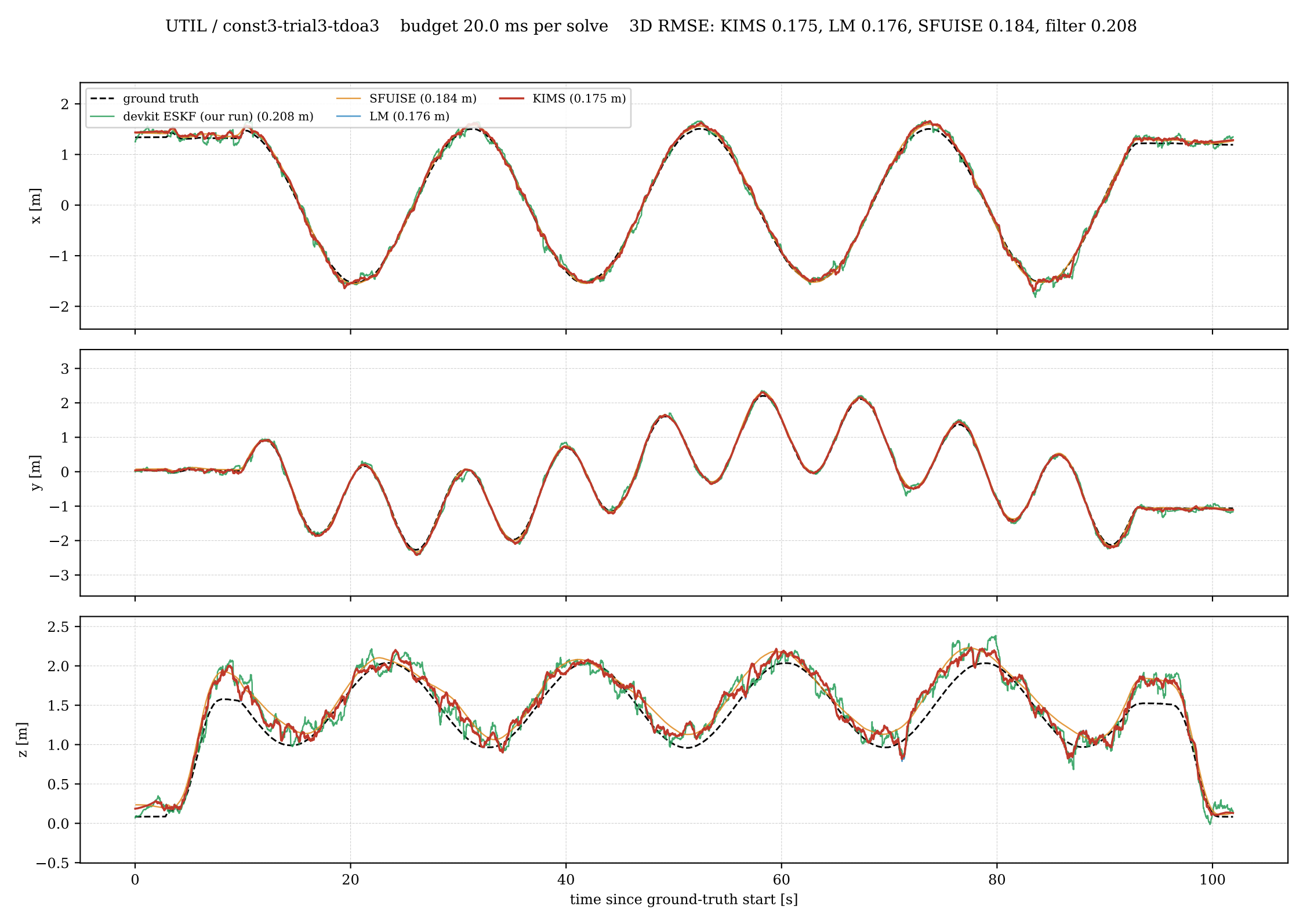

UTIL / const3-trial1-tdoa2 budget 20.0 ms per solve 3D RMSE: KIMS 0.230, LM 0.234, SFUISE 0.330, filter 0.245
ground truth
devkit ESKF (our run) (0.245 m)
SFUISE (0.330 m)
LM (0.234 m)
KIMS (0.230 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

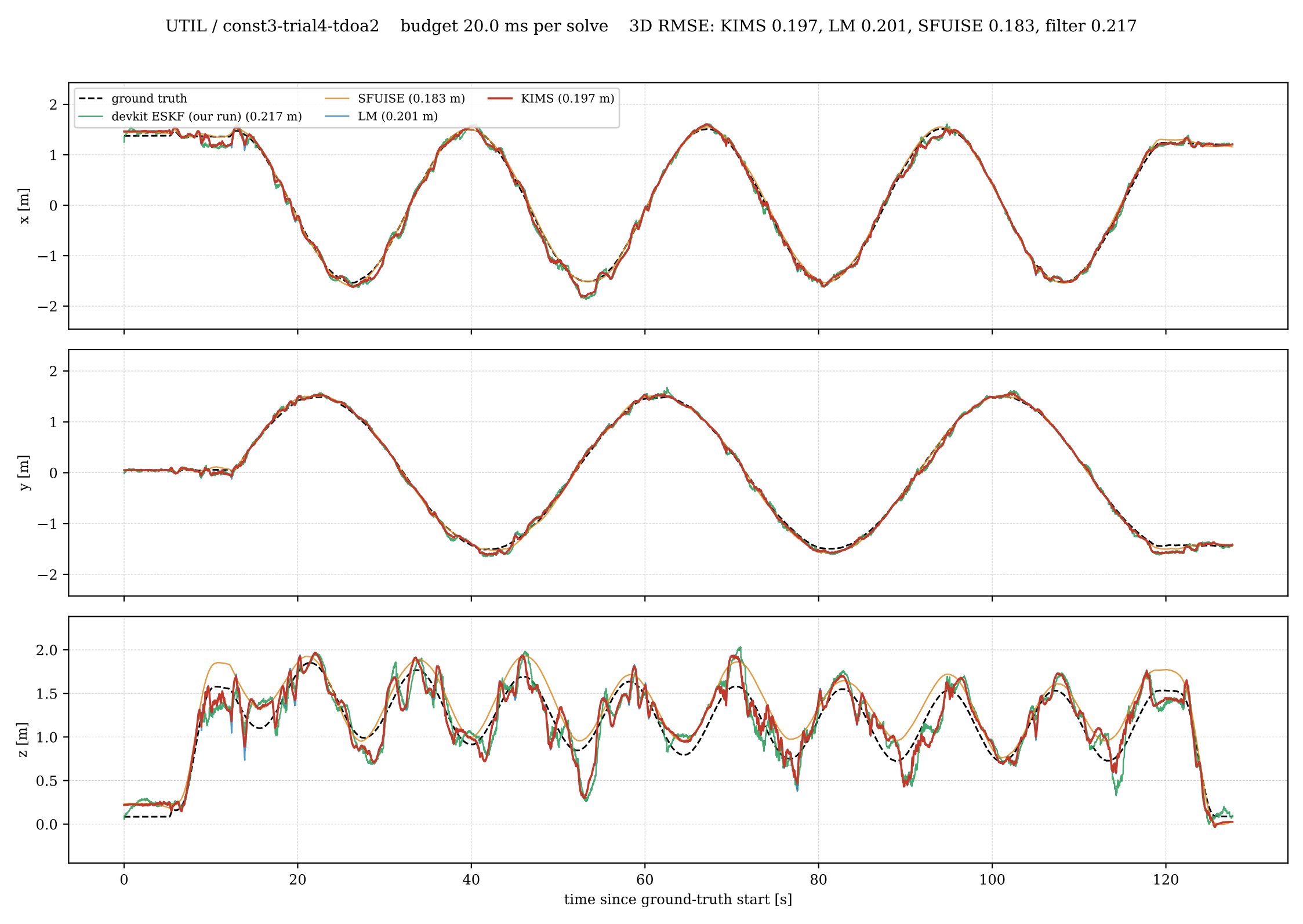

UTIL / const3-trial1-tdoa3 budget 20.0 ms per solve 3D RMSE: KIMS 0.199, LM 0.200, SFUISE 0.207, filter 0.226
ground truth
devkit ESKF (our run) (0.226 m)
SFUISE (0.207 m)
LM (0.200 m)
KIMS (0.199 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

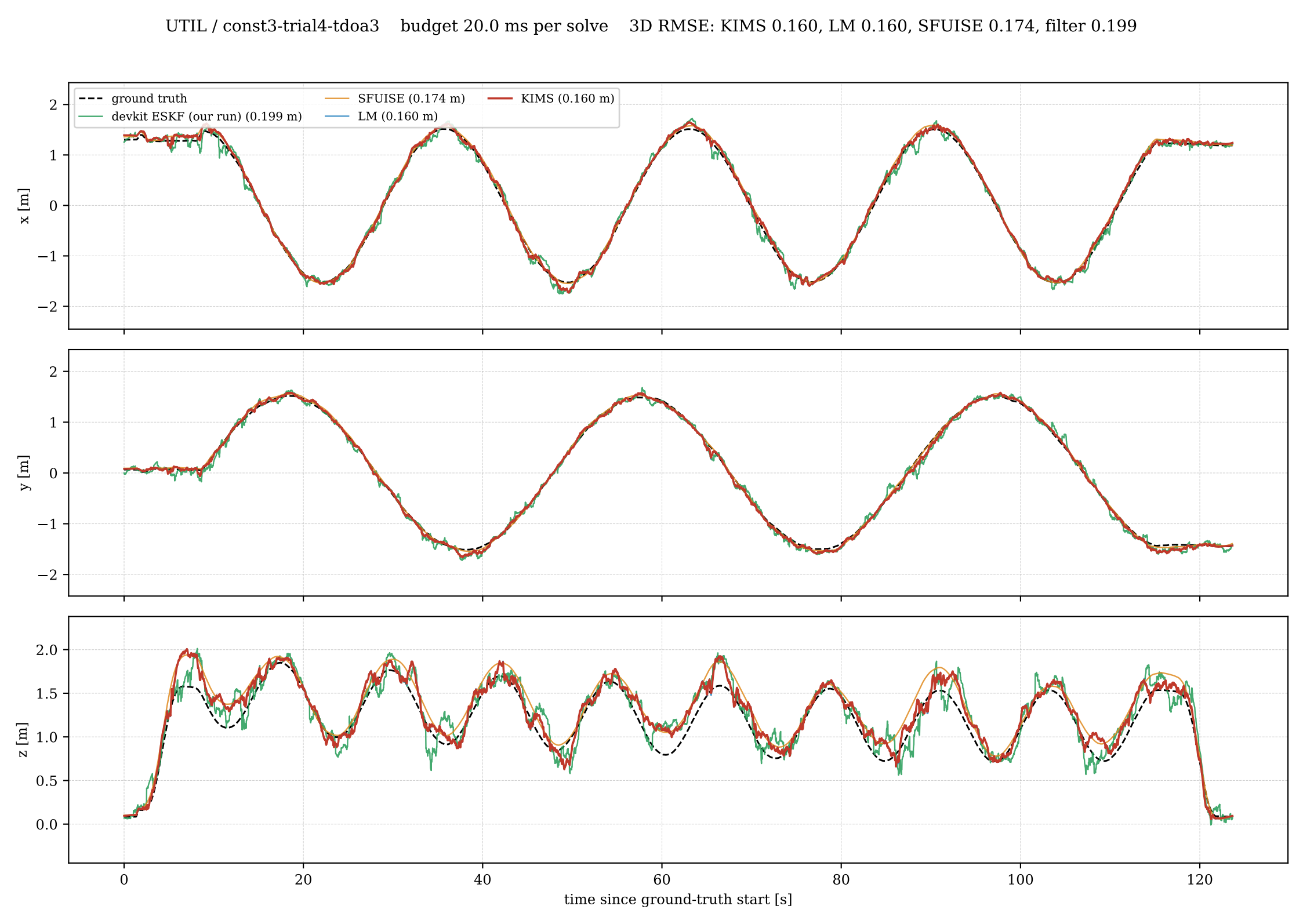

UTIL / const3-trial2-tdoa2 budget 20.0 ms per solve 3D RMSE: KIMS 0.187, LM 0.193, SFUISE 0.207, filter 0.197
ground truth
devkit ESKF (our run) (0.197 m)
SFUISE (0.207 m)
LM (0.193 m)
KIMS (0.187 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

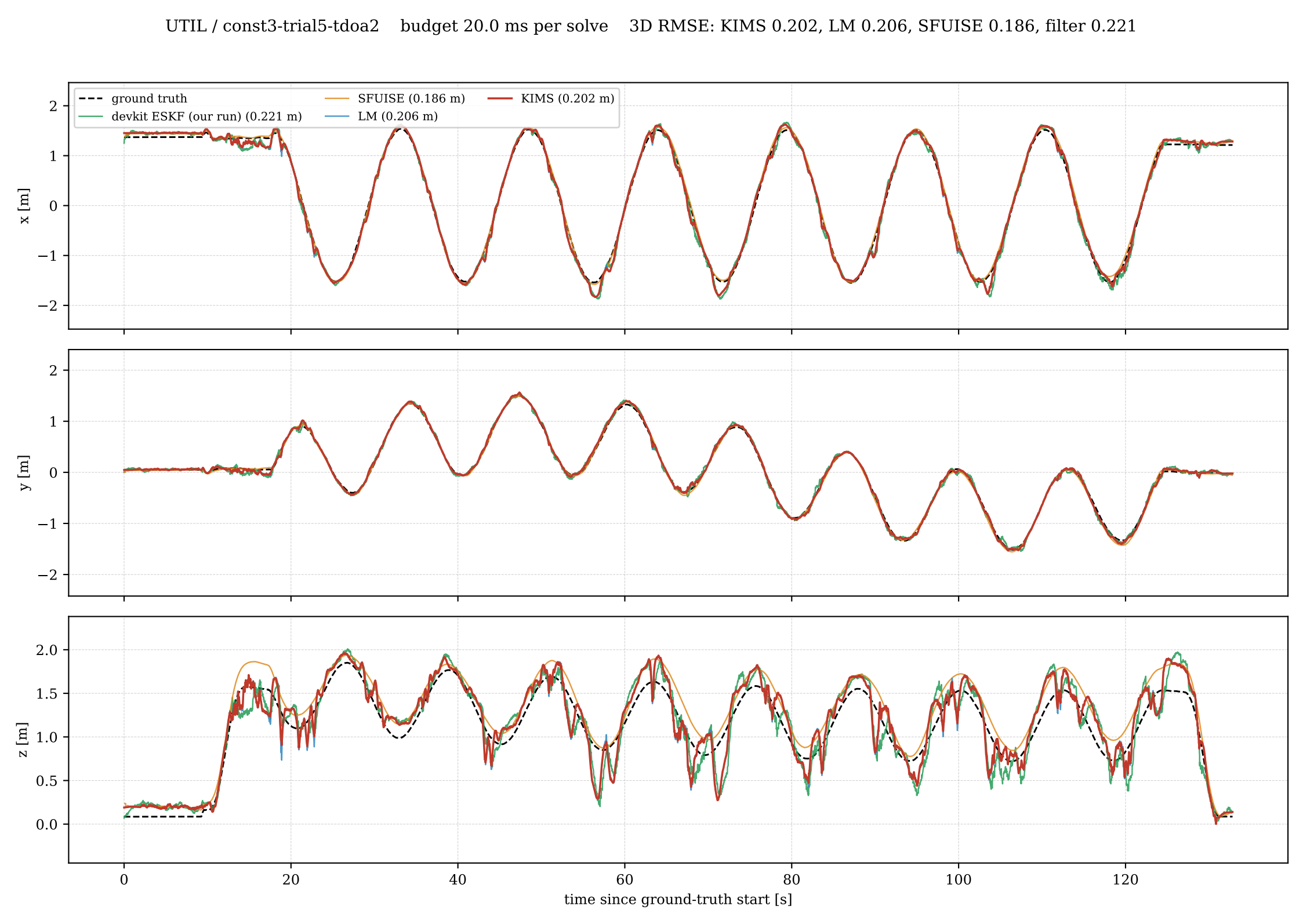

UTIL / const3-trial2-tdoa3 budget 20.0 ms per solve 3D RMSE: KIMS 0.169, LM 0.170, SFUISE 0.182, filter 0.219
ground truth
devkit ESKF (our run) (0.219 m)
SFUISE (0.182 m)
LM (0.170 m)
KIMS (0.169 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

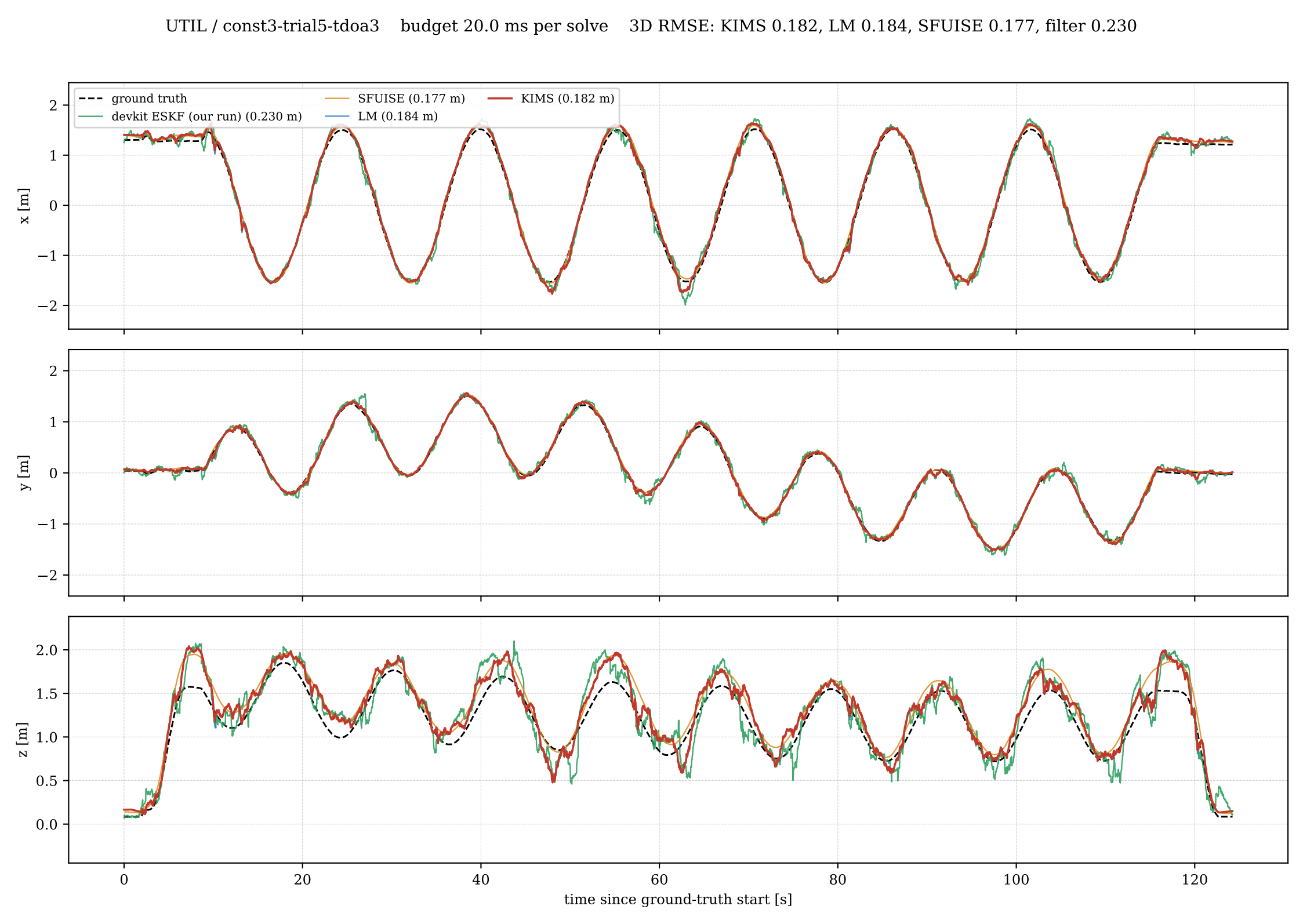

UTIL / const3-trial3-tdoa2 budget 20.0 ms per solve 3D RMSE: KIMS 0.187, LM 0.190, SFUISE 0.175, filter 0.189
ground truth
devkit ESKF (our run) (0.189 m)
SFUISE (0.175 m)
LM (0.190 m)
KIMS (0.187 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

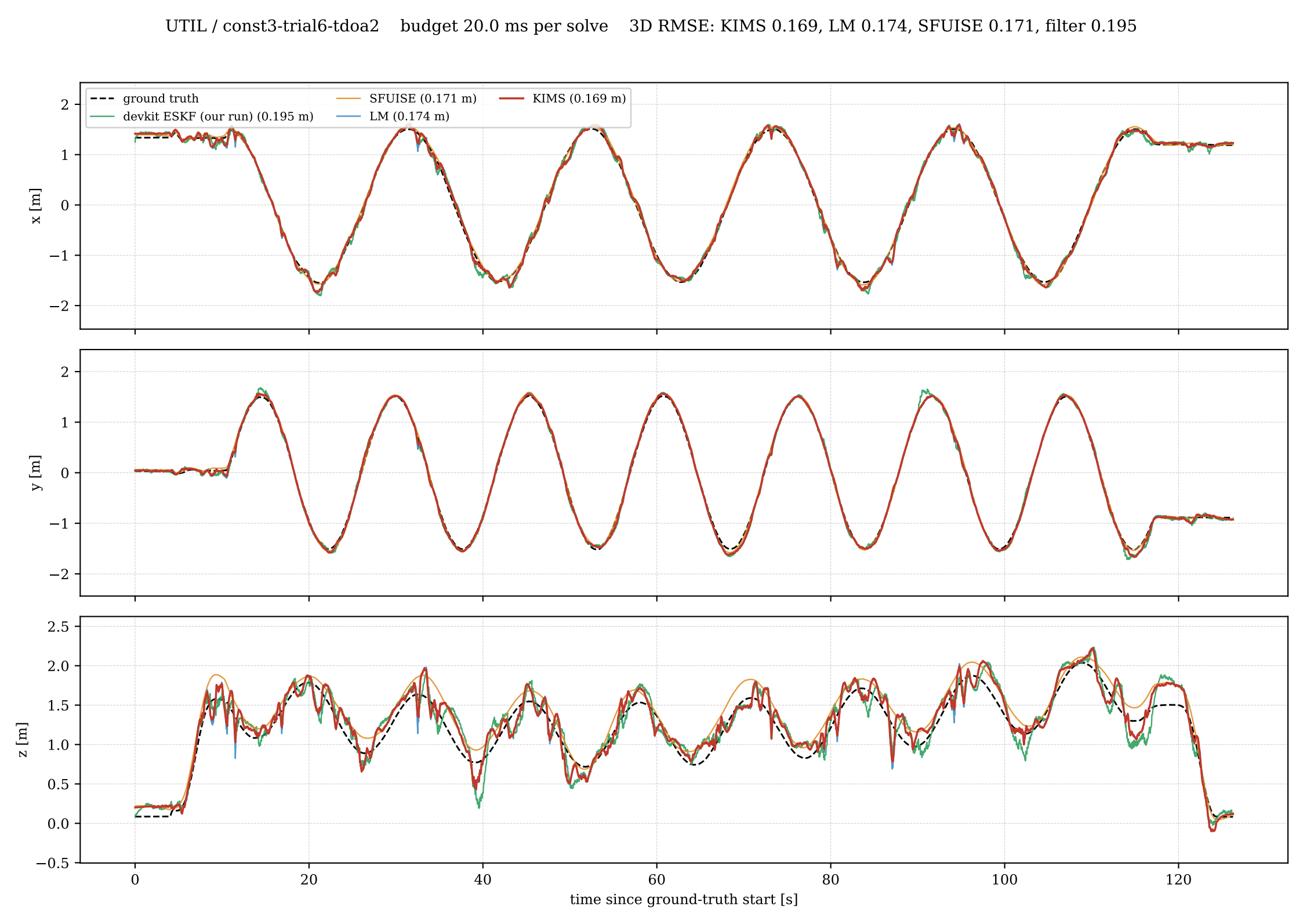

UTIL / const3-trial3-tdoa3 budget 20.0 ms per solve 3D RMSE: KIMS 0.175, LM 0.176, SFUISE 0.184, filter 0.208
ground truth
devkit ESKF (our run) (0.208 m)
SFUISE (0.184 m)
LM (0.176 m)
KIMS (0.175 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

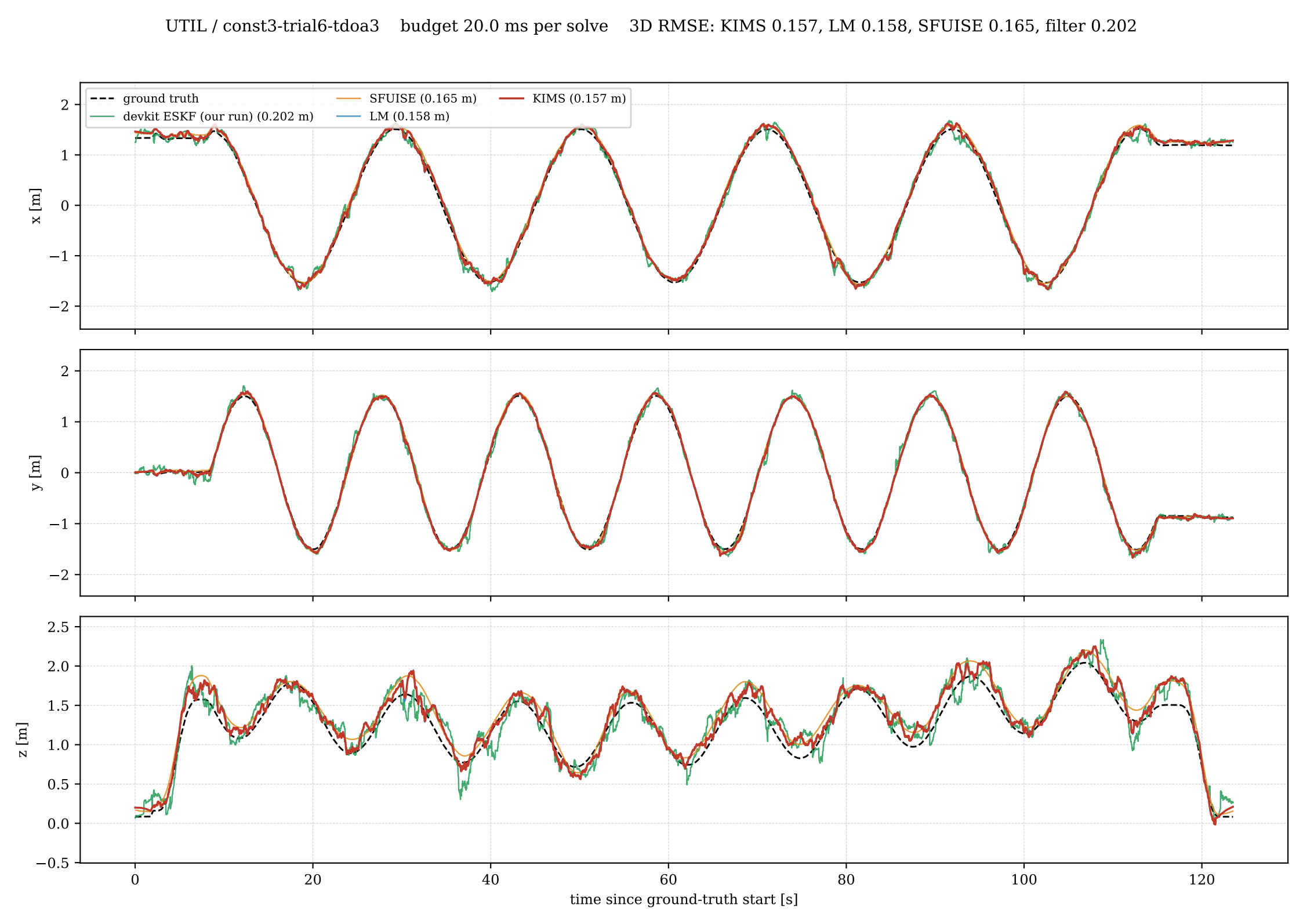

UTIL / const3-trial4-tdoa2 budget 20.0 ms per solve 3D RMSE: KIMS 0.197, LM 0.201, SFUISE 0.183, filter 0.217
ground truth
devkit ESKF (our run) (0.217 m)
SFUISE (0.183 m)
LM (0.201 m)
KIMS (0.197 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

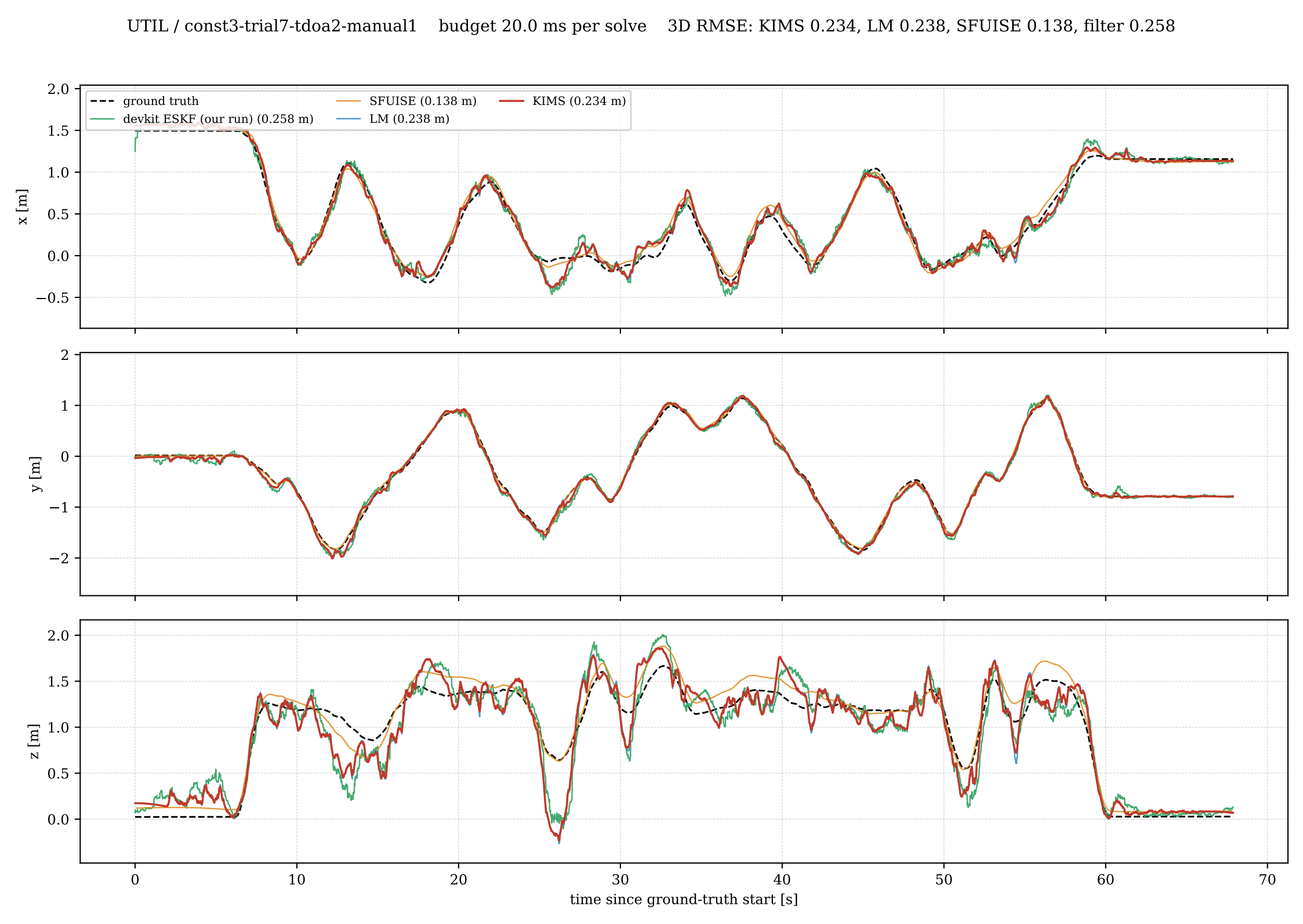

UTIL / const3-trial4-tdoa3 budget 20.0 ms per solve 3D RMSE: KIMS 0.160, LM 0.160, SFUISE 0.174, filter 0.199
ground truth
devkit ESKF (our run) (0.199 m)
SFUISE (0.174 m)
LM (0.160 m)
KIMS (0.160 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

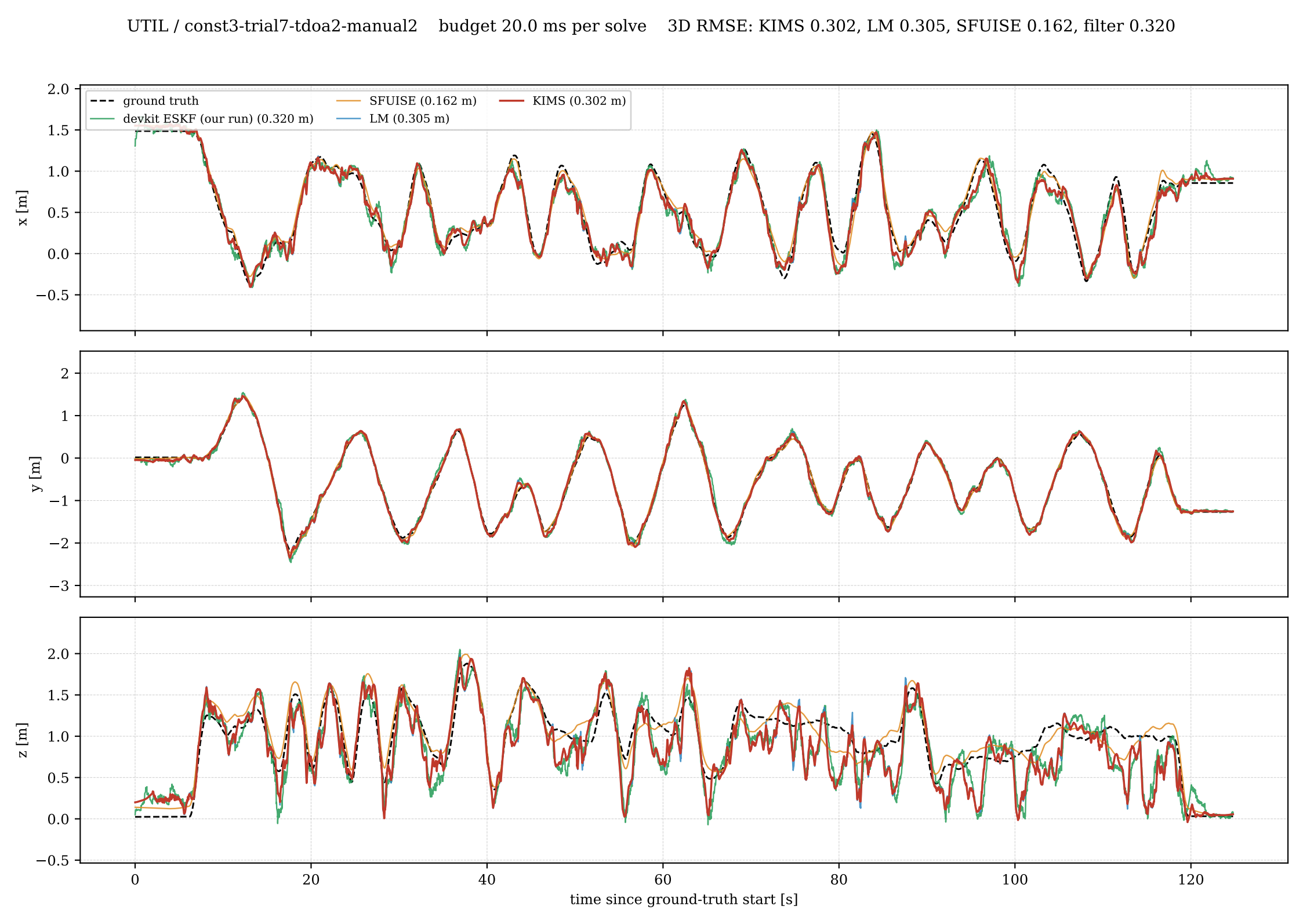

UTIL / const3-trial5-tdoa2 budget 20.0 ms per solve 3D RMSE: KIMS 0.202, LM 0.206, SFUISE 0.186, filter 0.221
ground truth
devkit ESKF (our run) (0.221 m)
SFUISE (0.186 m)
LM (0.206 m)
KIMS (0.202 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

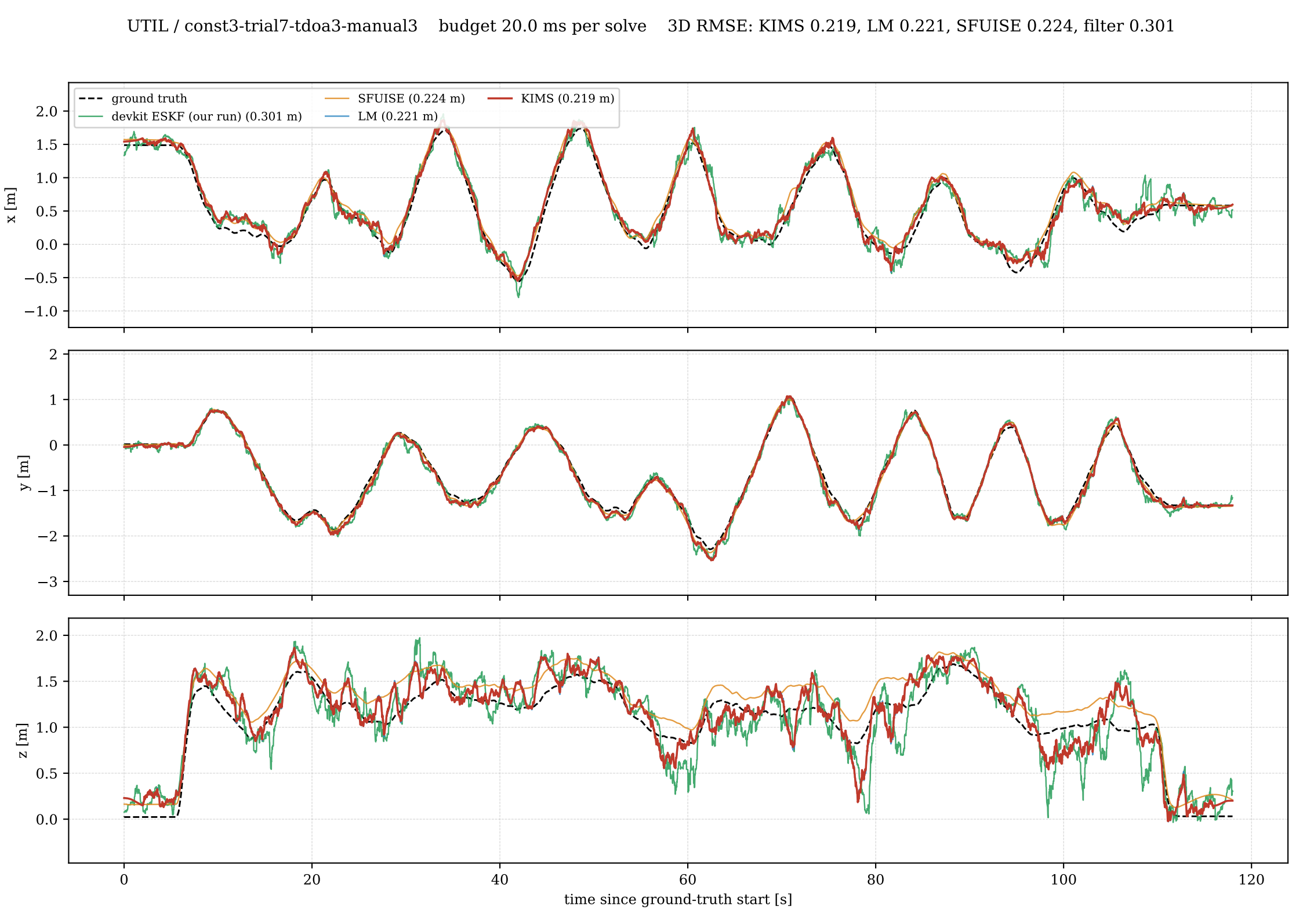

UTIL / const3-trial5-tdoa3 budget 20.0 ms per solve 3D RMSE: KIMS 0.182, LM 0.184, SFUISE 0.177, filter 0.230
ground truth
devkit ESKF (our run) (0.230 m)
SFUISE (0.177 m)
LM (0.184 m)
KIMS (0.182 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

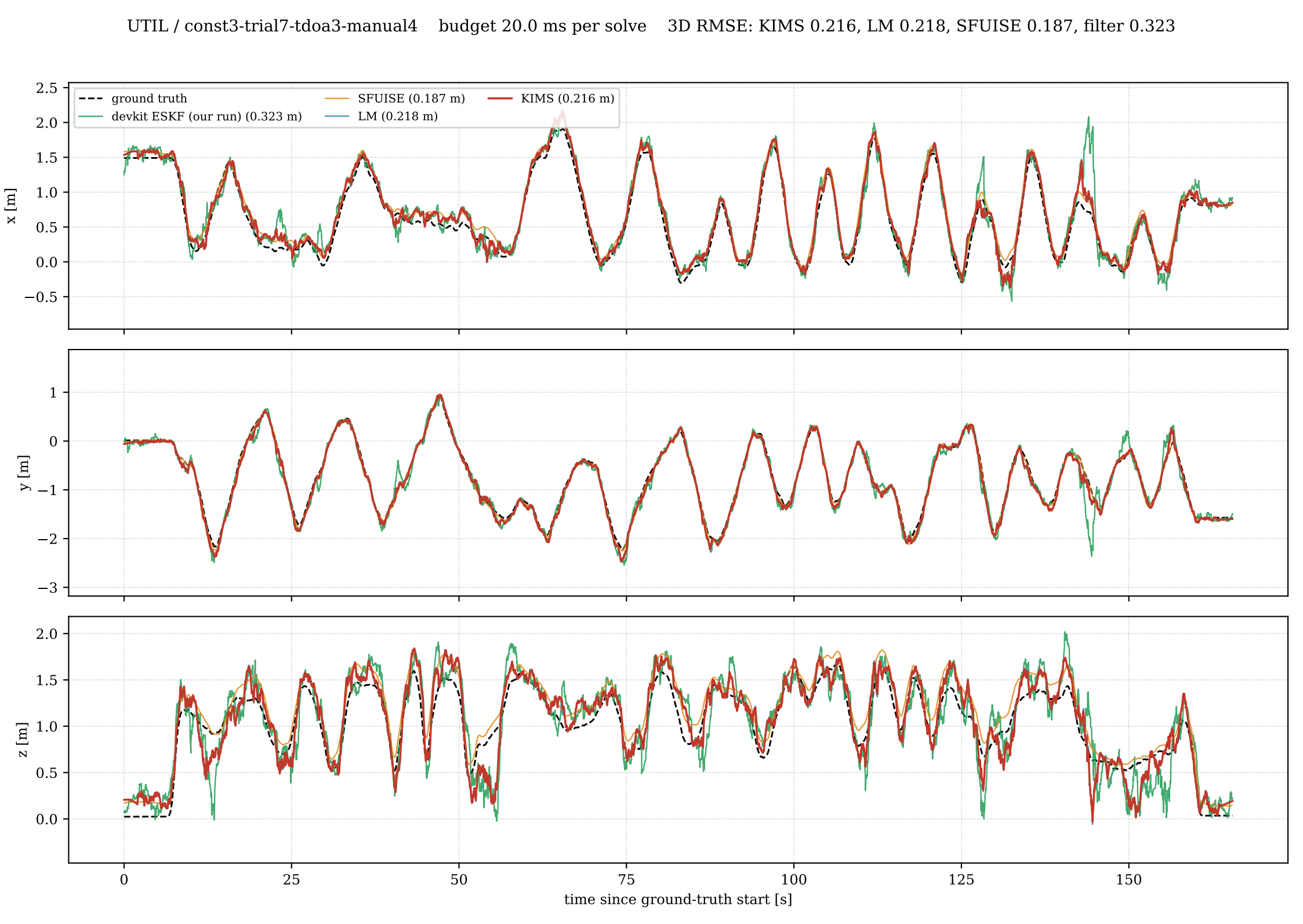

UTIL / const3-trial6-tdoa2 budget 20.0 ms per solve 3D RMSE: KIMS 0.169, LM 0.174, SFUISE 0.171, filter 0.195
ground truth
devkit ESKF (our run) (0.195 m)
SFUISE (0.171 m)
LM (0.174 m)
KIMS (0.169 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

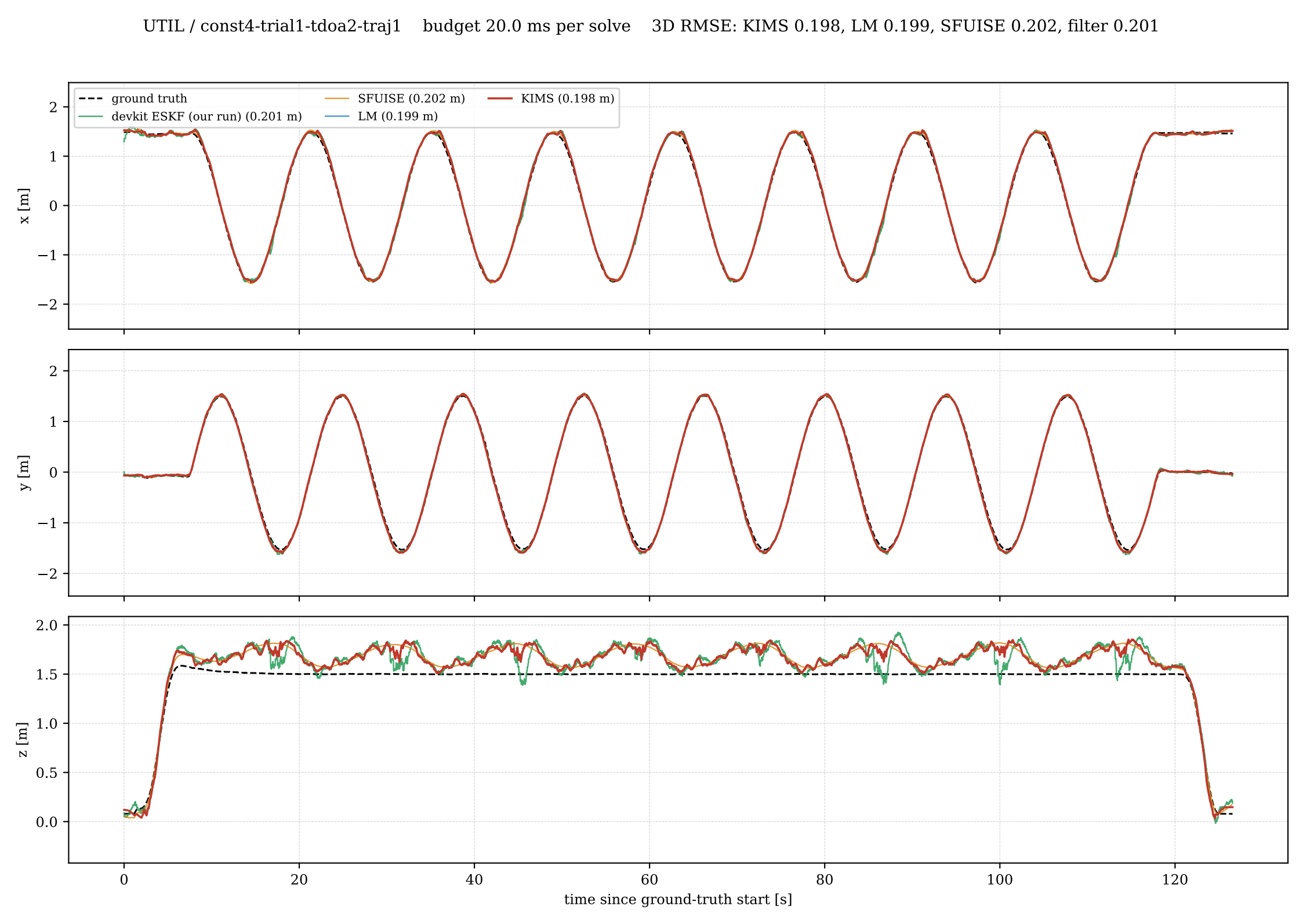

UTIL / const3-trial6-tdoa3 budget 20.0 ms per solve 3D RMSE: KIMS 0.157, LM 0.158, SFUISE 0.165, filter 0.202
ground truth
devkit ESKF (our run) (0.202 m)
SFUISE (0.165 m)
LM (0.158 m)
KIMS (0.157 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

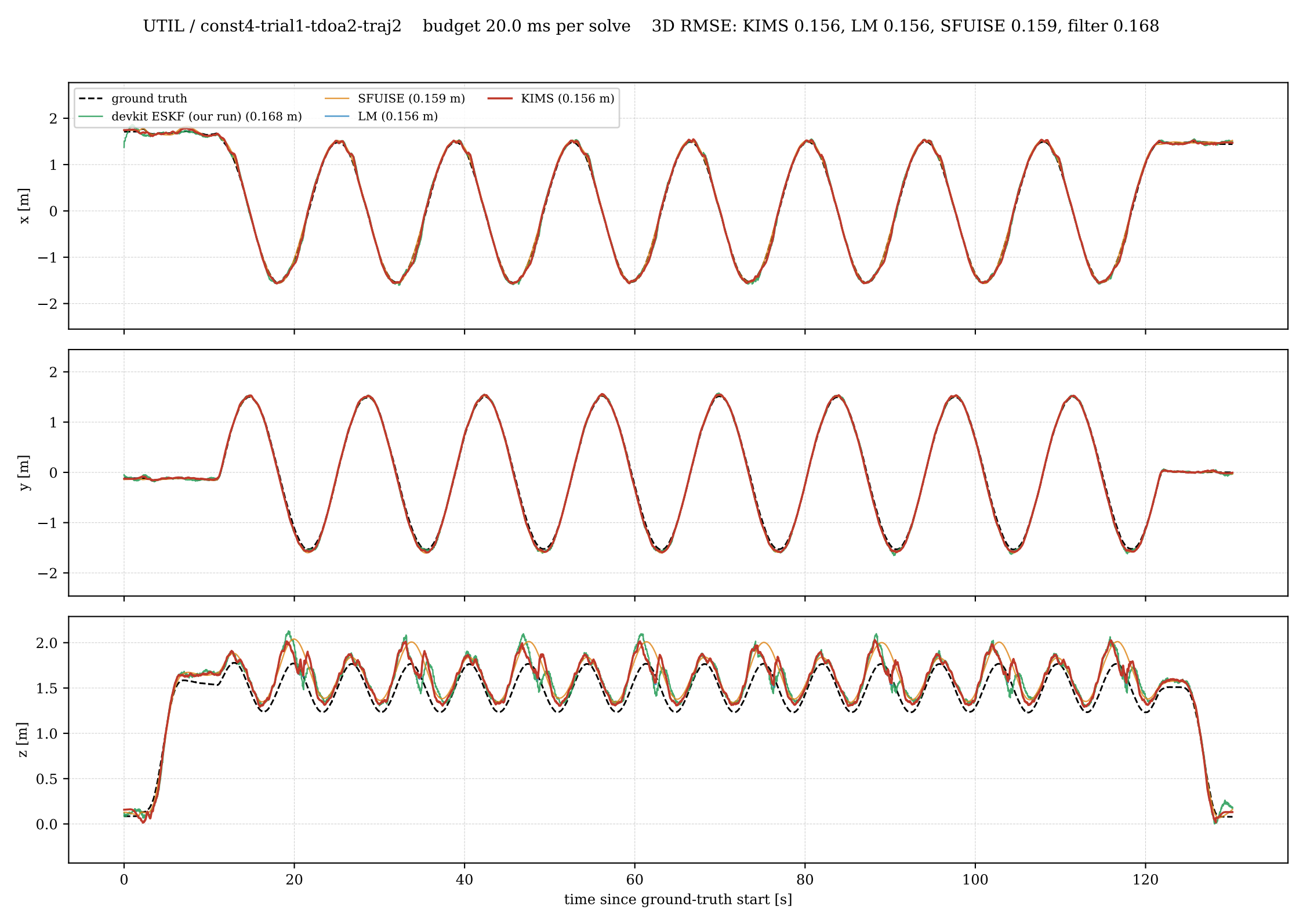

UTIL / const3-trial7-tdoa2-manual1 budget 20.0 ms per solve 3D RMSE: KIMS 0.234, LM 0.238, SFUISE 0.138, filter 0.258
ground truth
devkit ESKF (our run) (0.258 m)
SFUISE (0.138 m)
LM (0.238 m)
KIMS (0.234 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

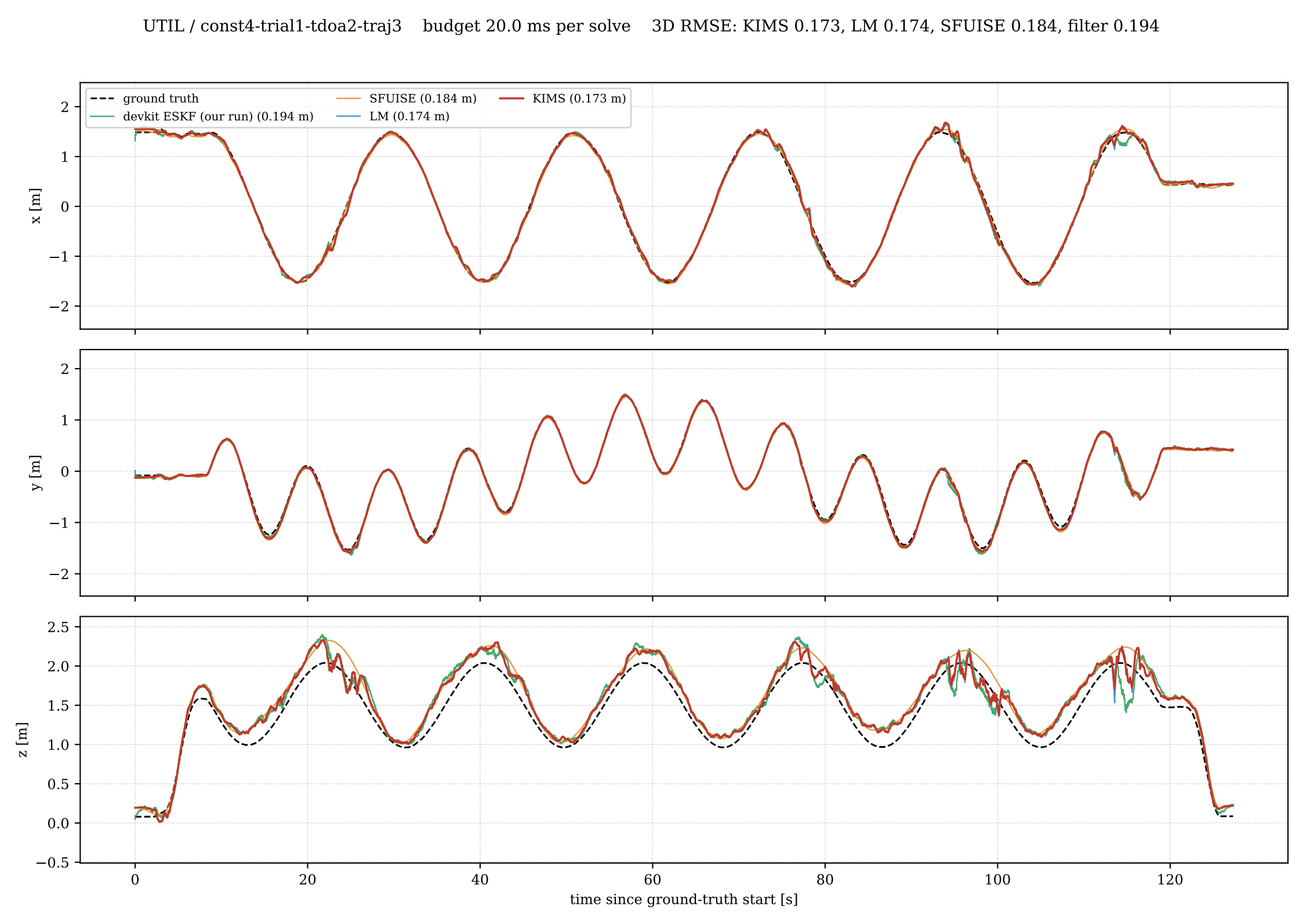

UTIL / const3-trial7-tdoa2-manual2 budget 20.0 ms per solve 3D RMSE: KIMS 0.302, LM 0.305, SFUISE 0.162, filter 0.320
ground truth
devkit ESKF (our run) (0.320 m)
SFUISE (0.162 m)
LM (0.305 m)
KIMS (0.302 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

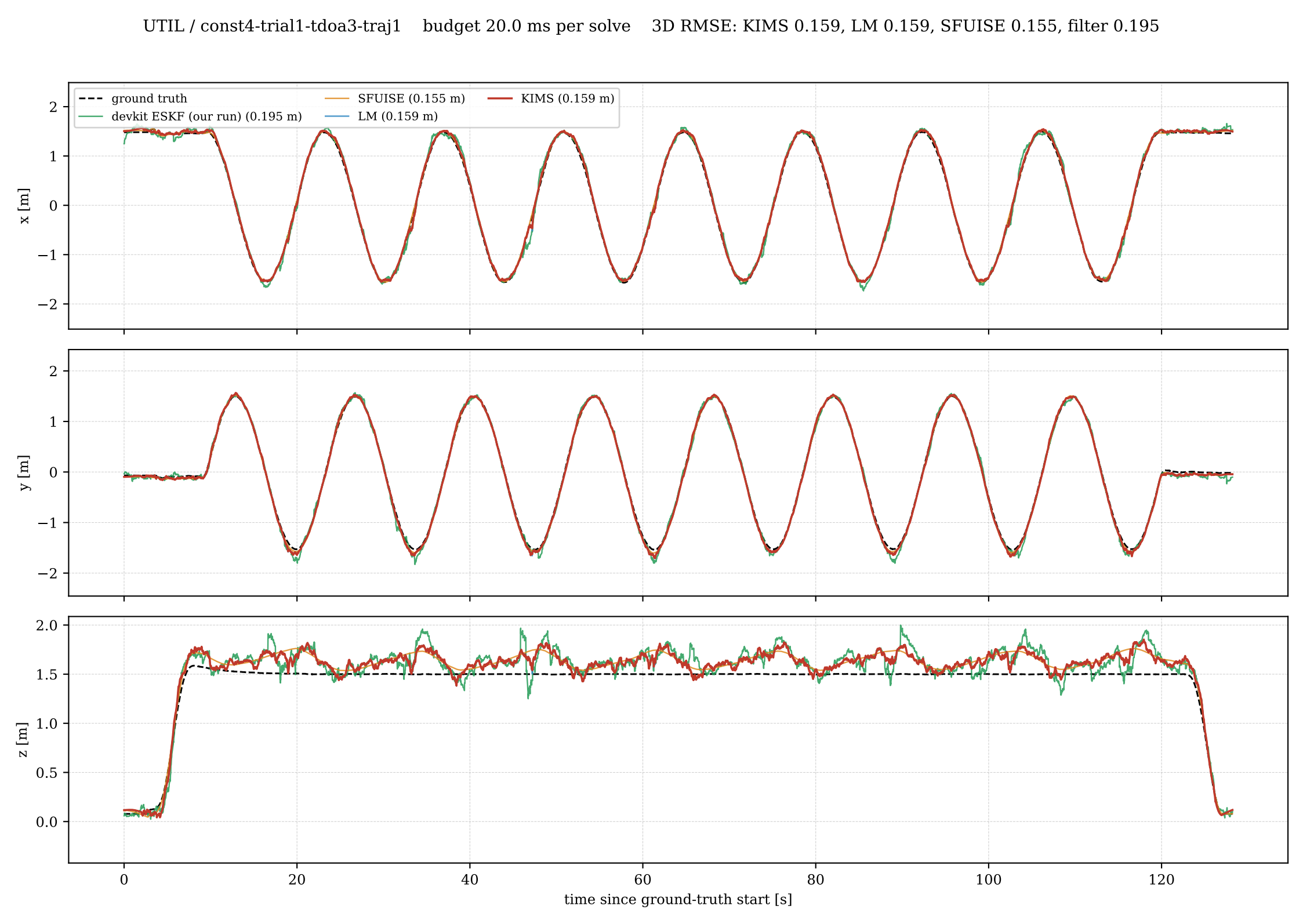

UTIL / const3-trial7-tdoa3-manual3 budget 20.0 ms per solve 3D RMSE: KIMS 0.219, LM 0.221, SFUISE 0.224, filter 0.301
ground truth
devkit ESKF (our run) (0.301 m)
SFUISE (0.224 m)
LM (0.221 m)
KIMS (0.219 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

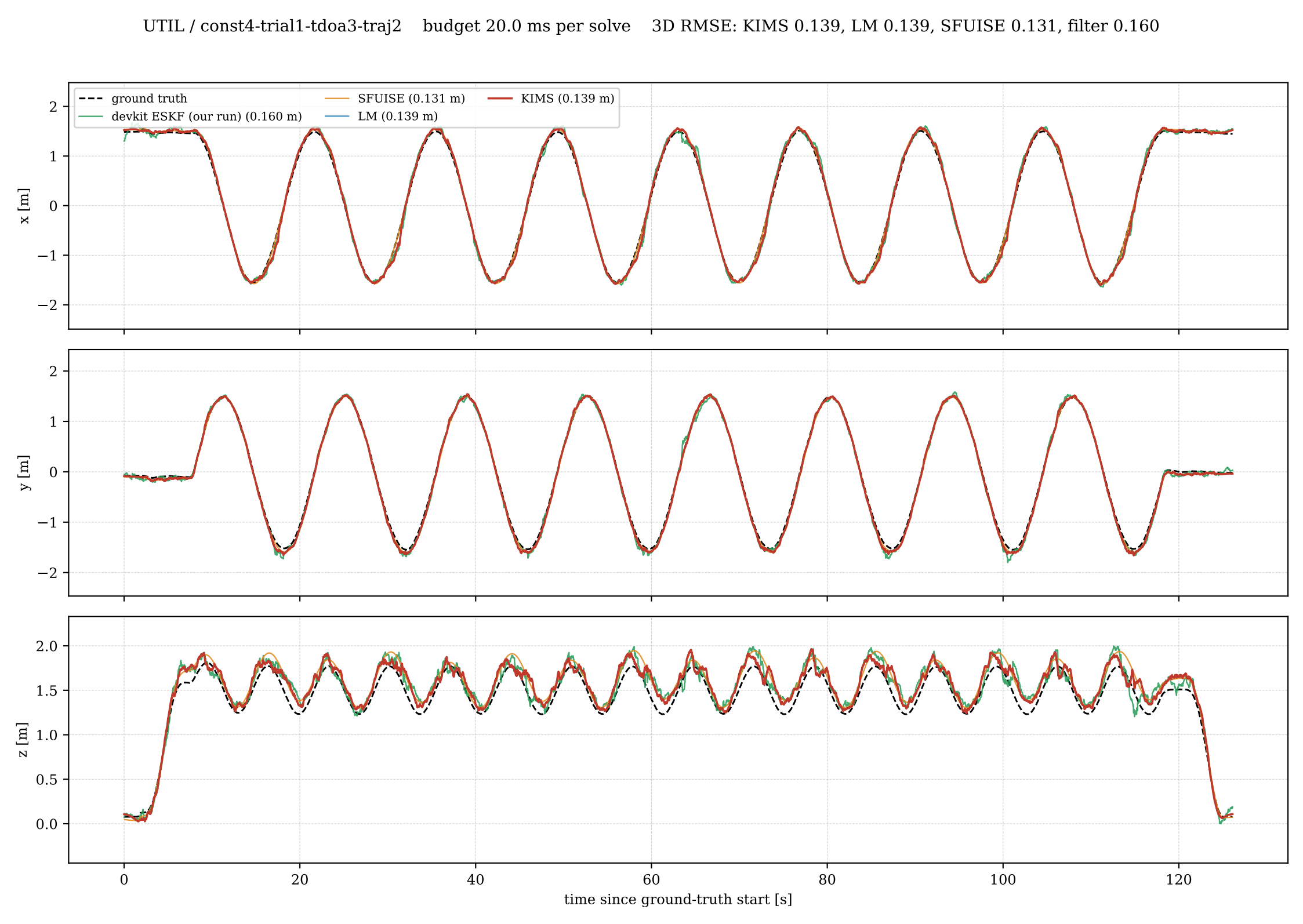

UTIL / const3-trial7-tdoa3-manual4 budget 20.0 ms per solve 3D RMSE: KIMS 0.216, LM 0.218, SFUISE 0.187, filter 0.323
ground truth
devkit ESKF (our run) (0.323 m)
SFUISE (0.187 m)
LM (0.218 m)
KIMS (0.216 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

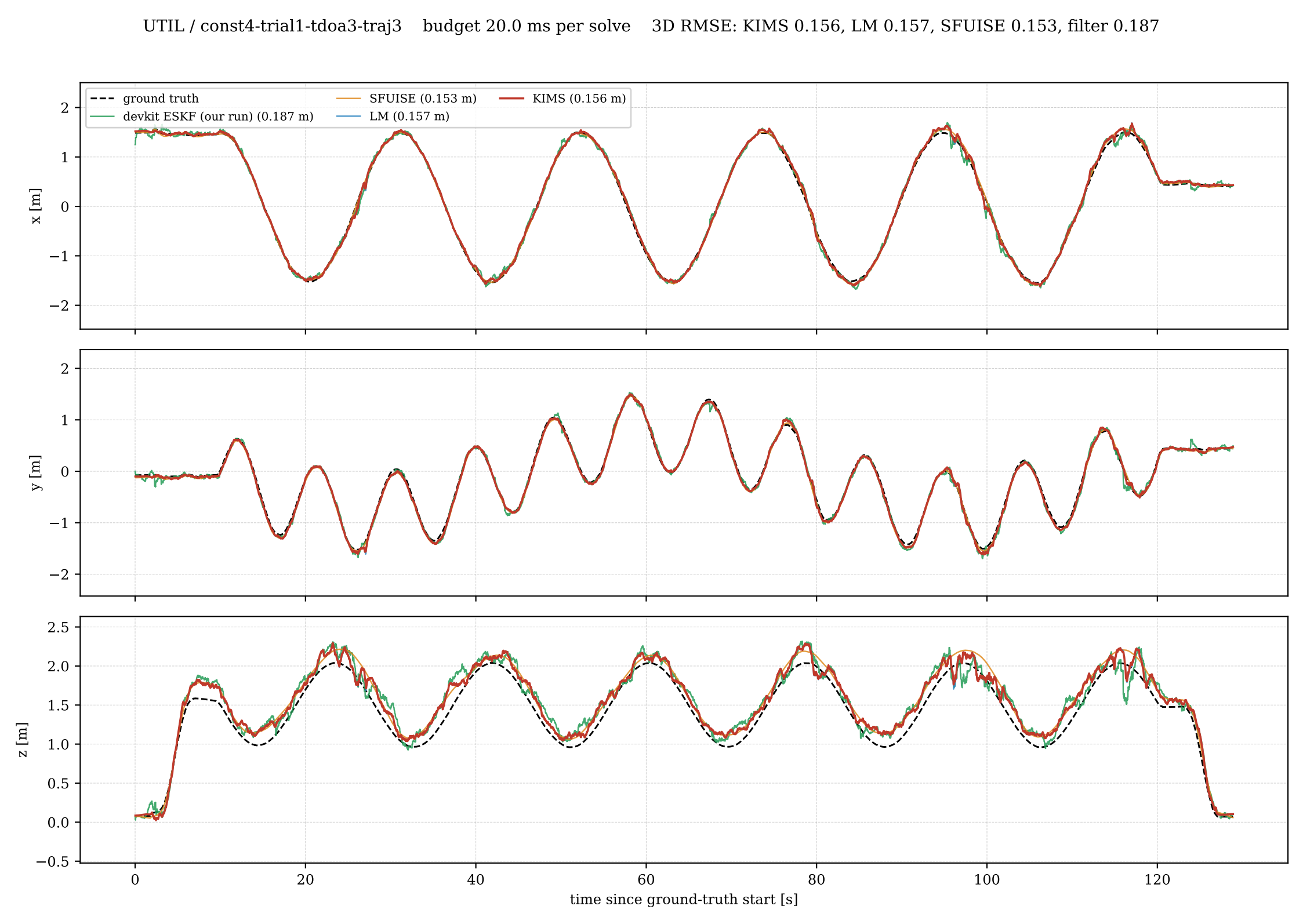

UTIL / const4-trial1-tdoa2-traj1 budget 20.0 ms per solve 3D RMSE: KIMS 0.198, LM 0.199, SFUISE 0.202, filter 0.201
ground truth
devkit ESKF (our run) (0.201 m)
SFUISE (0.202 m)
LM (0.199 m)
KIMS (0.198 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

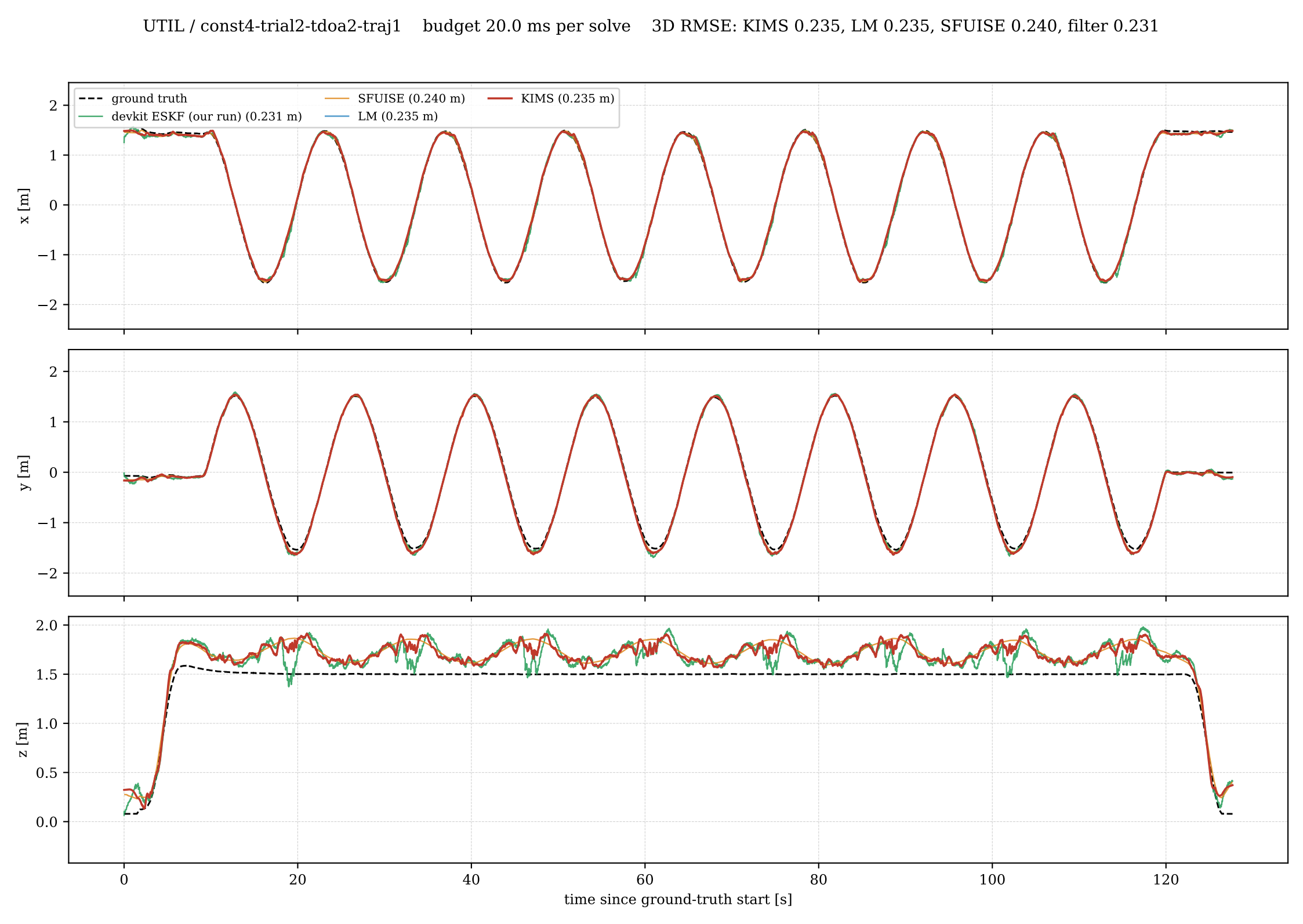

UTIL / const4-trial1-tdoa2-traj2 budget 20.0 ms per solve 3D RMSE: KIMS 0.156, LM 0.156, SFUISE 0.159, filter 0.168
ground truth
devkit ESKF (our run) (0.168 m)
SFUISE (0.159 m)
LM (0.156 m)
KIMS (0.156 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

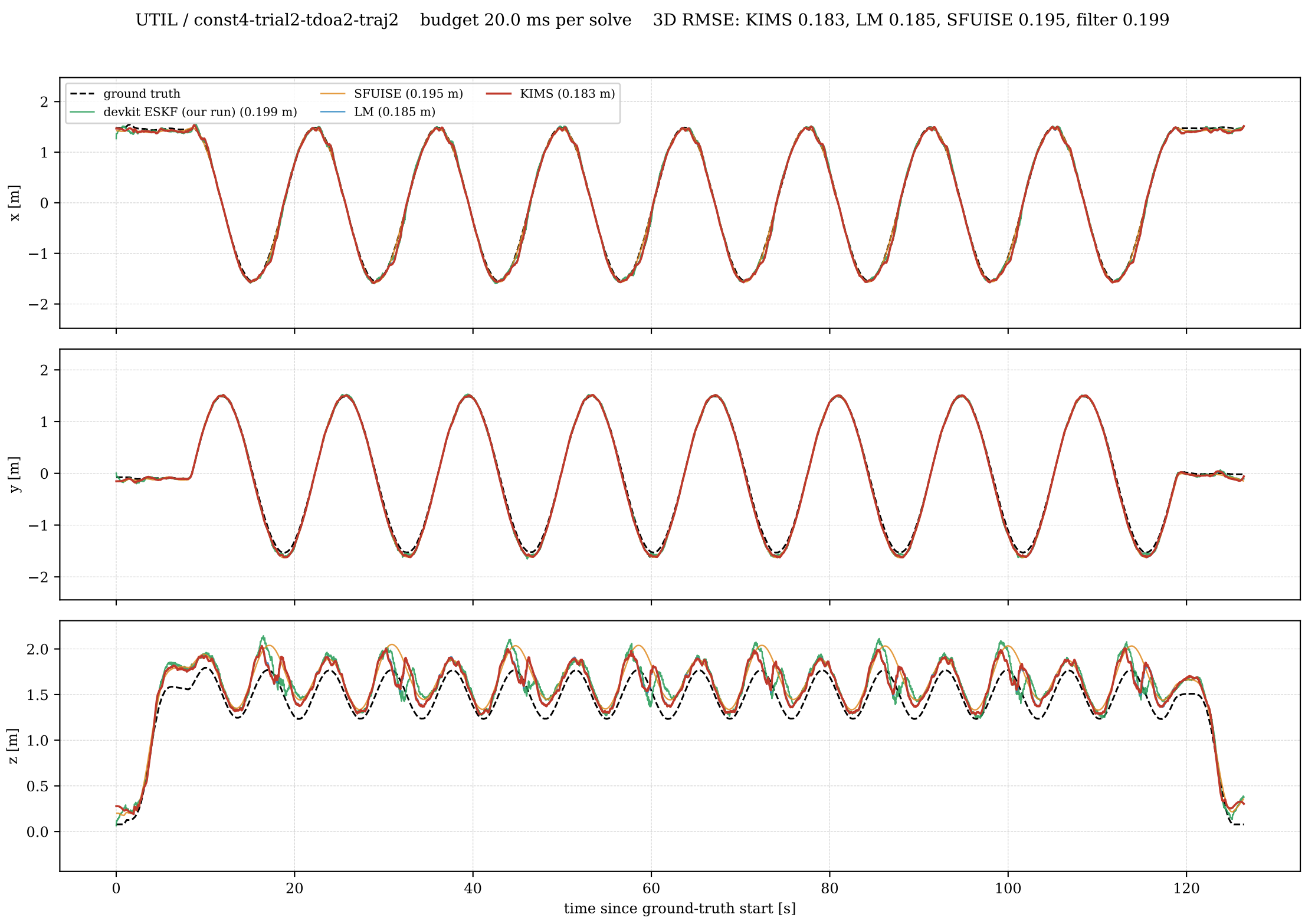

UTIL / const4-trial1-tdoa2-traj3 budget 20.0 ms per solve 3D RMSE: KIMS 0.173, LM 0.174, SFUISE 0.184, filter 0.194
ground truth
devkit ESKF (our run) (0.194 m)
SFUISE (0.184 m)
LM (0.174 m)
KIMS (0.173 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

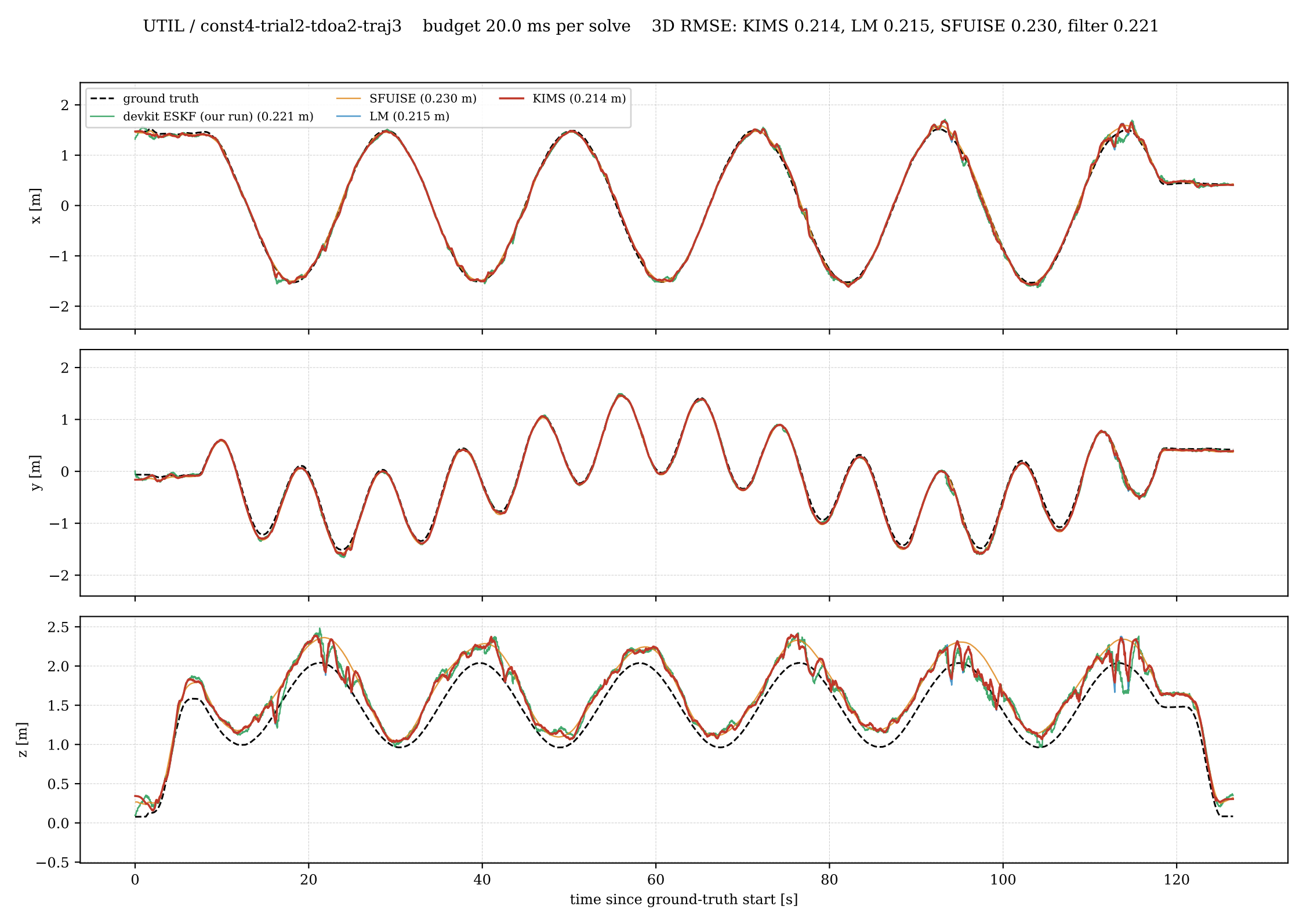

UTIL / const4-trial1-tdoa3-traj1 budget 20.0 ms per solve 3D RMSE: KIMS 0.159, LM 0.159, SFUISE 0.155, filter 0.195
ground truth
devkit ESKF (our run) (0.195 m)
SFUISE (0.155 m)
LM (0.159 m)
KIMS (0.159 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

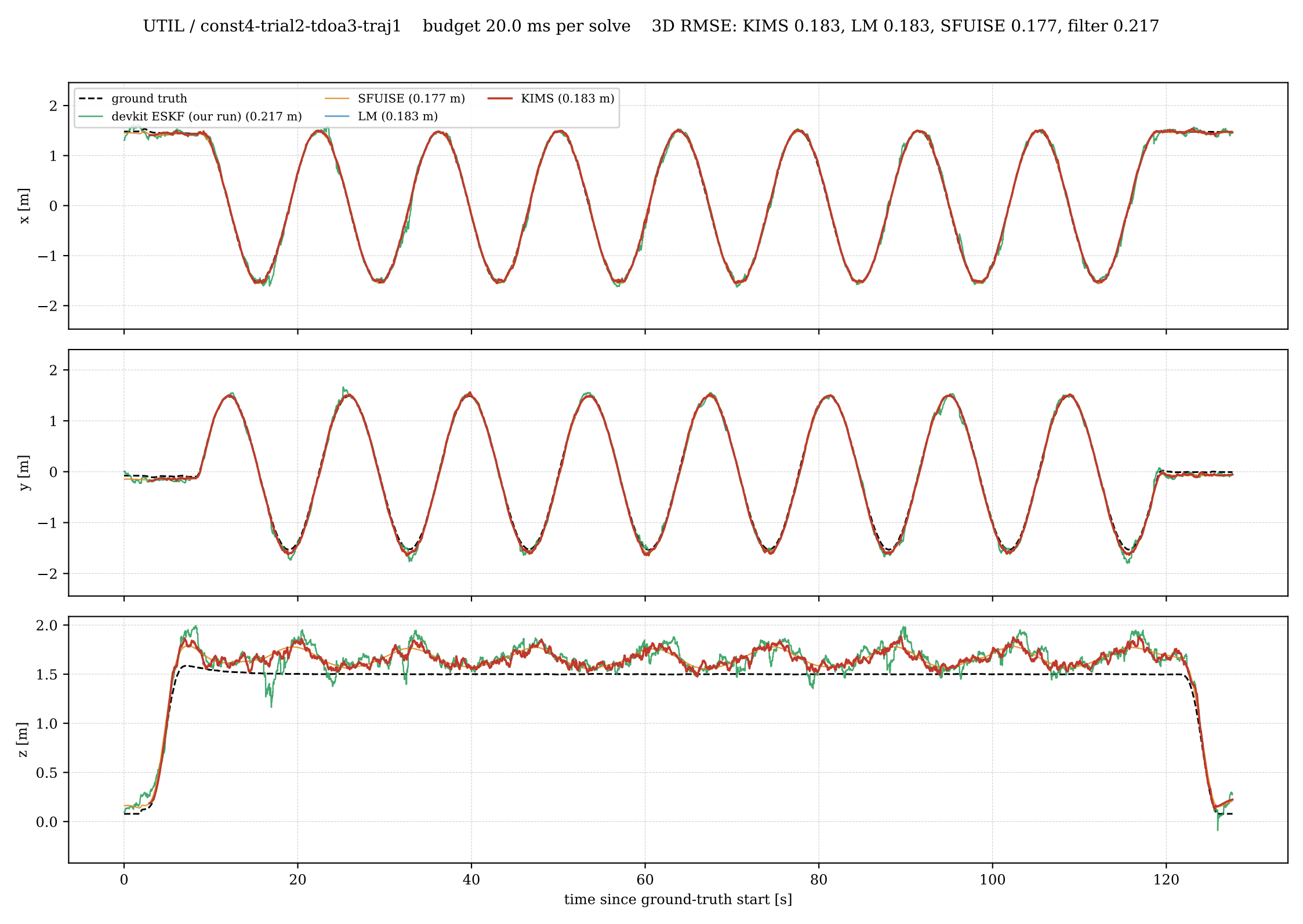

UTIL / const4-trial1-tdoa3-traj2 budget 20.0 ms per solve 3D RMSE: KIMS 0.139, LM 0.139, SFUISE 0.131, filter 0.160
ground truth
devkit ESKF (our run) (0.160 m)
SFUISE (0.131 m)
LM (0.139 m)
KIMS (0.139 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

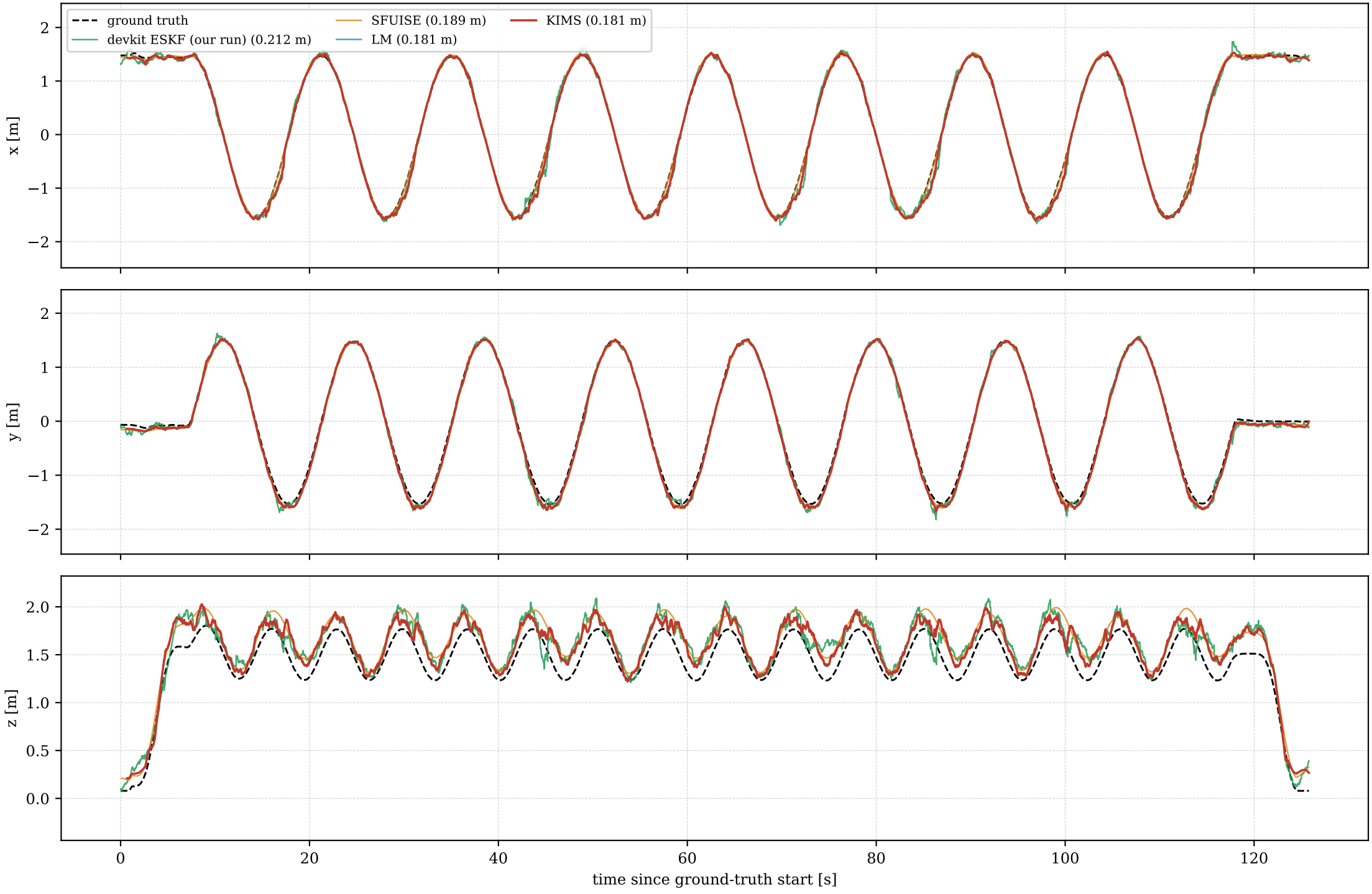

UTIL / const4-trial1-tdoa3-traj3 budget 20.0 ms per solve 3D RMSE: KIMS 0.156, LM 0.157, SFUISE 0.153, filter 0.187
ground truth
devkit ESKF (our run) (0.187 m)
SFUISE (0.153 m)
LM (0.157 m)
KIMS (0.156 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

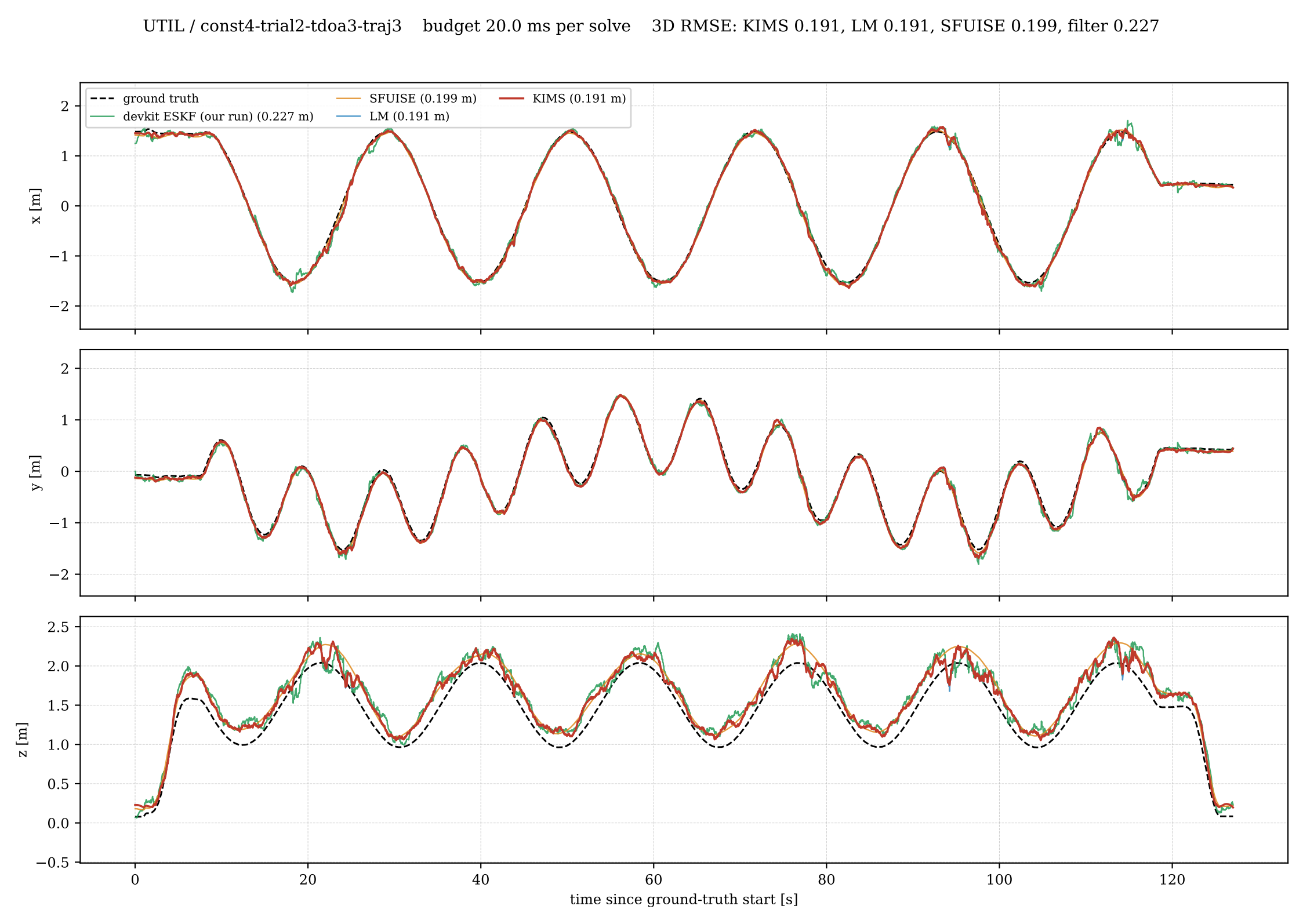

UTIL / const4-trial2-tdoa2-traj1 budget 20.0 ms per solve 3D RMSE: KIMS 0.235, LM 0.235, SFUISE 0.240, filter 0.231
ground truth
devkit ESKF (our run) (0.231 m)
SFUISE (0.240 m)
LM (0.235 m)
KIMS (0.235 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

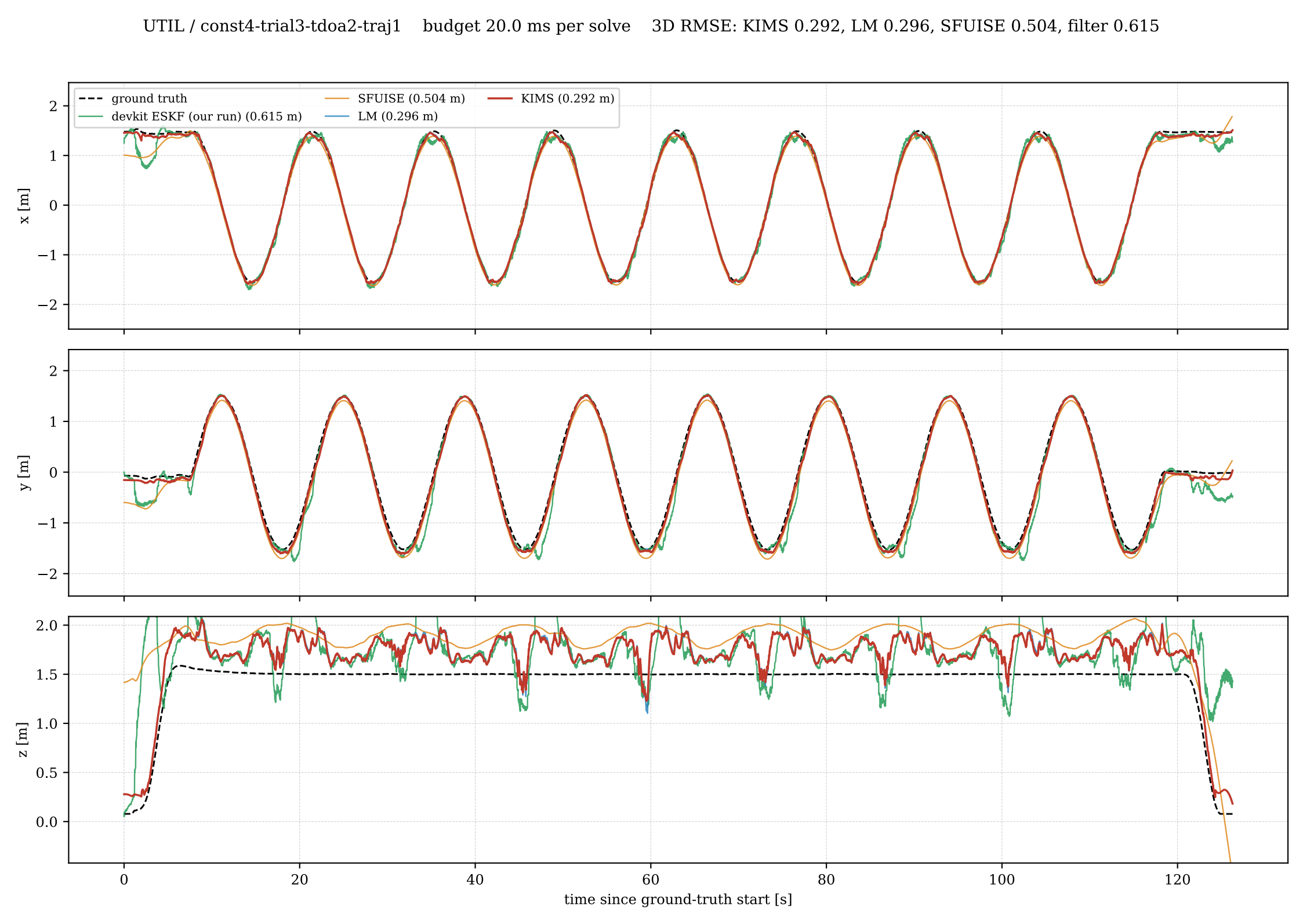

UTIL / const4-trial2-tdoa2-traj2 budget 20.0 ms per solve 3D RMSE: KIMS 0.183, LM 0.185, SFUISE 0.195, filter 0.199
ground truth
devkit ESKF (our run) (0.199 m)
SFUISE (0.195 m)
LM (0.185 m)
KIMS (0.183 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

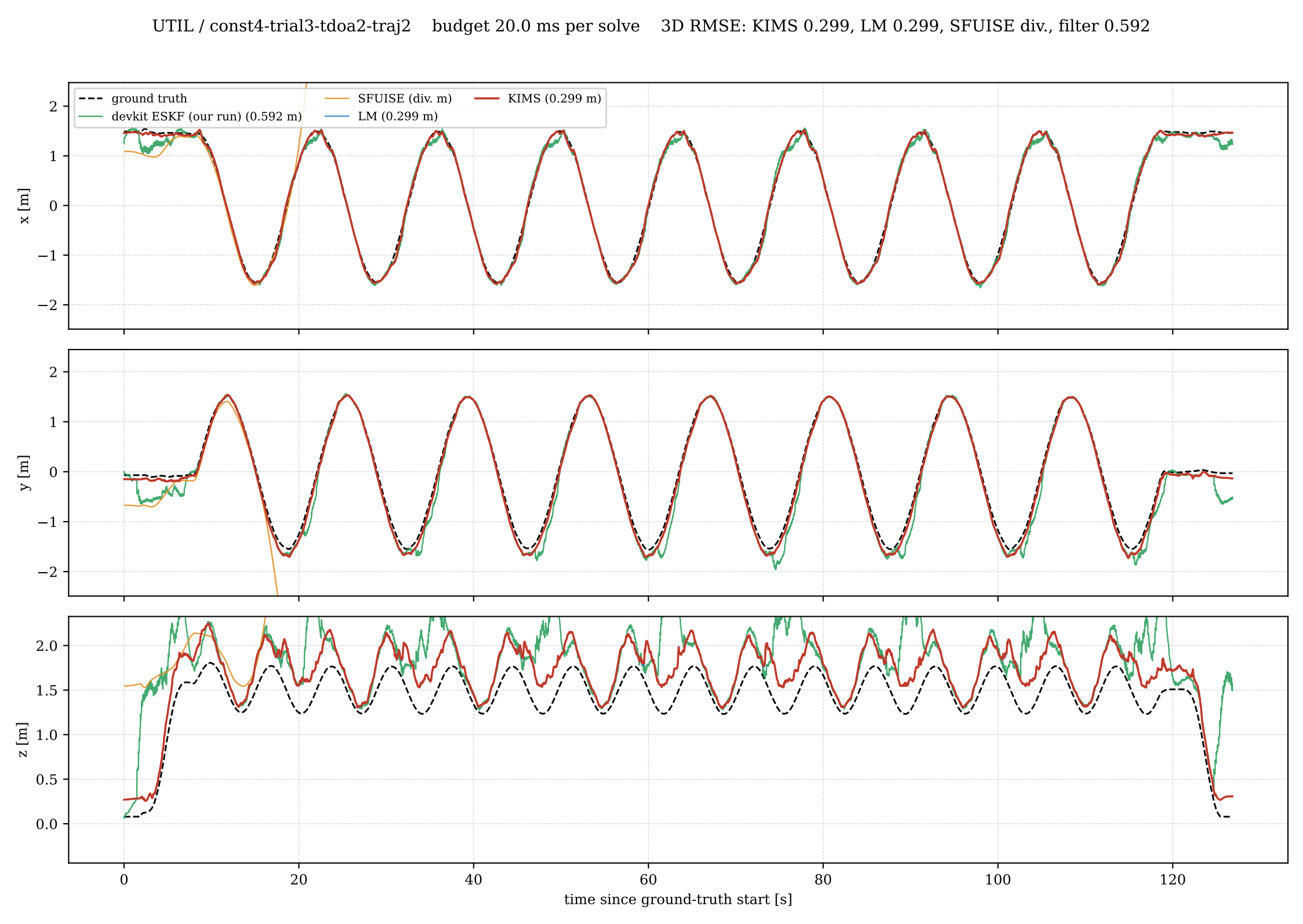

UTIL / const4-trial2-tdoa2-traj3 budget 20.0 ms per solve 3D RMSE: KIMS 0.214, LM 0.215, SFUISE 0.230, filter 0.221
ground truth
devkit ESKF (our run) (0.221 m)
SFUISE (0.230 m)
LM (0.215 m)
KIMS (0.214 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

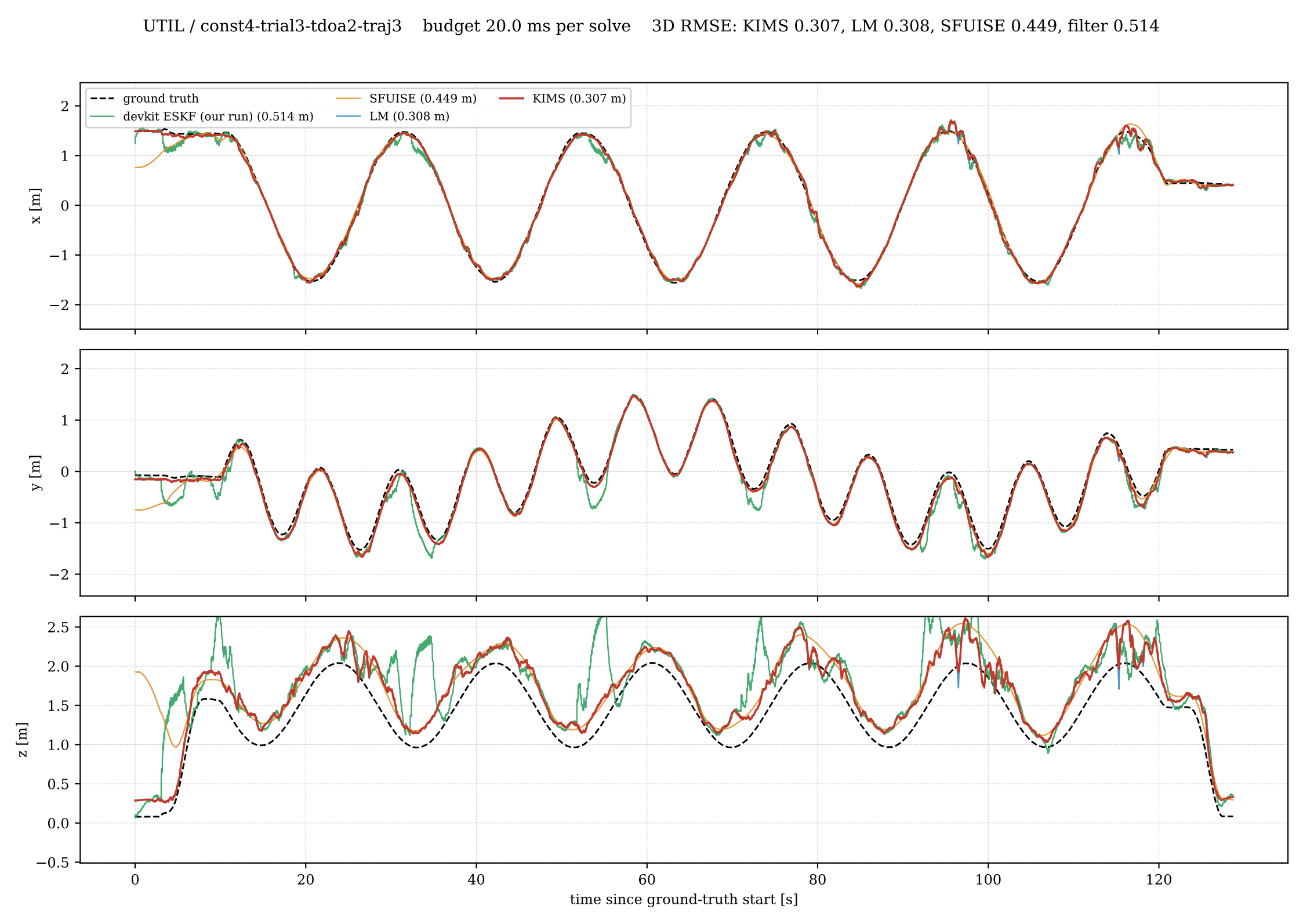

UTIL / const4-trial2-tdoa3-traj1 budget 20.0 ms per solve 3D RMSE: KIMS 0.183, LM 0.183, SFUISE 0.177, filter 0.217
ground truth
devkit ESKF (our run) (0.217 m)
SFUISE (0.177 m)
LM (0.183 m)
KIMS (0.183 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

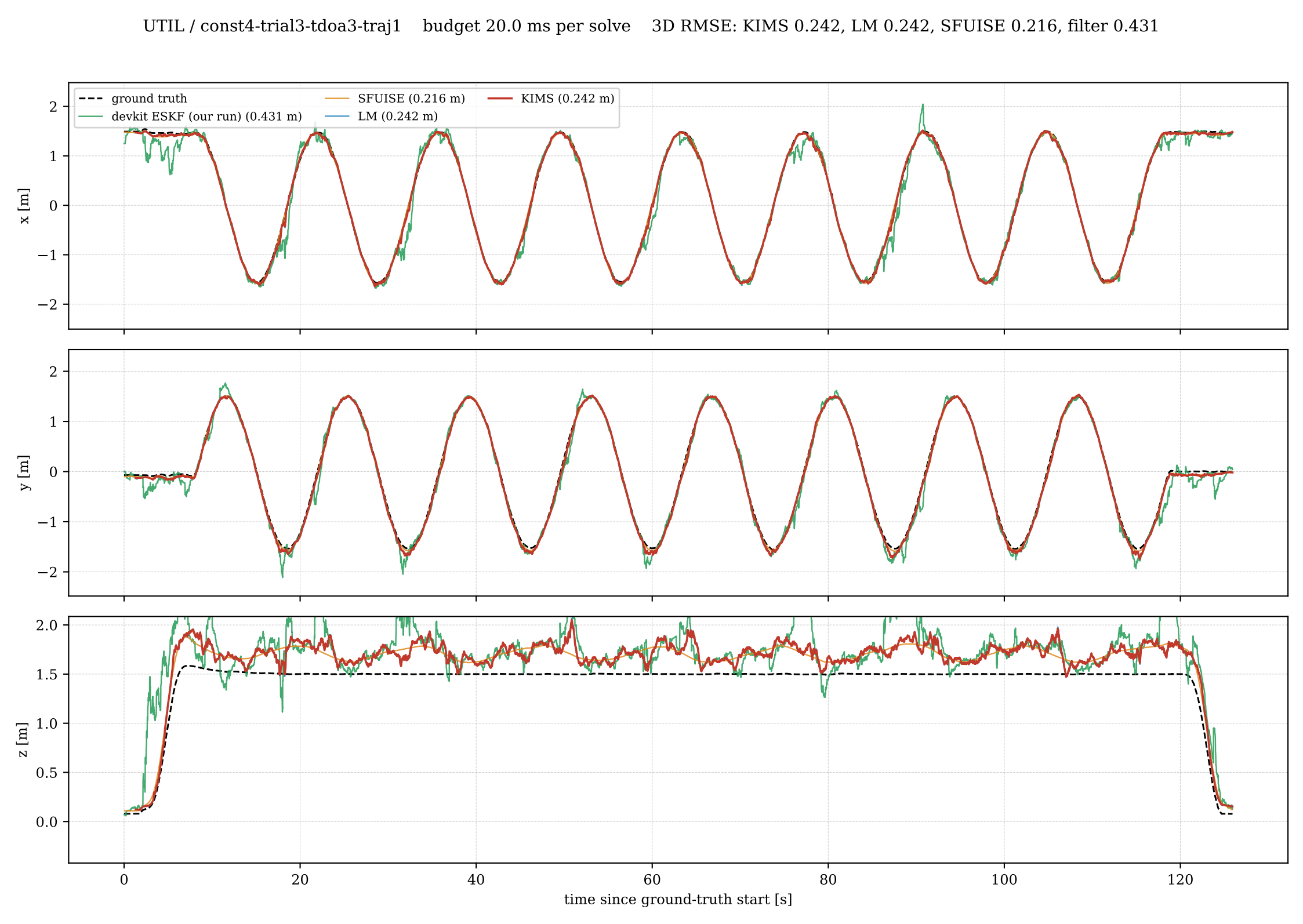

UTIL / const4-trial2-tdoa3-traj2 budget 20.0 ms per solve 3D RMSE: KIMS 0.181, LM 0.181, SFUISE 0.189, filter 0.212
ground truth
devkit ESKF (our run) (0.212 m)
SFUISE (0.189 m)
LM (0.181 m)
KIMS (0.181 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

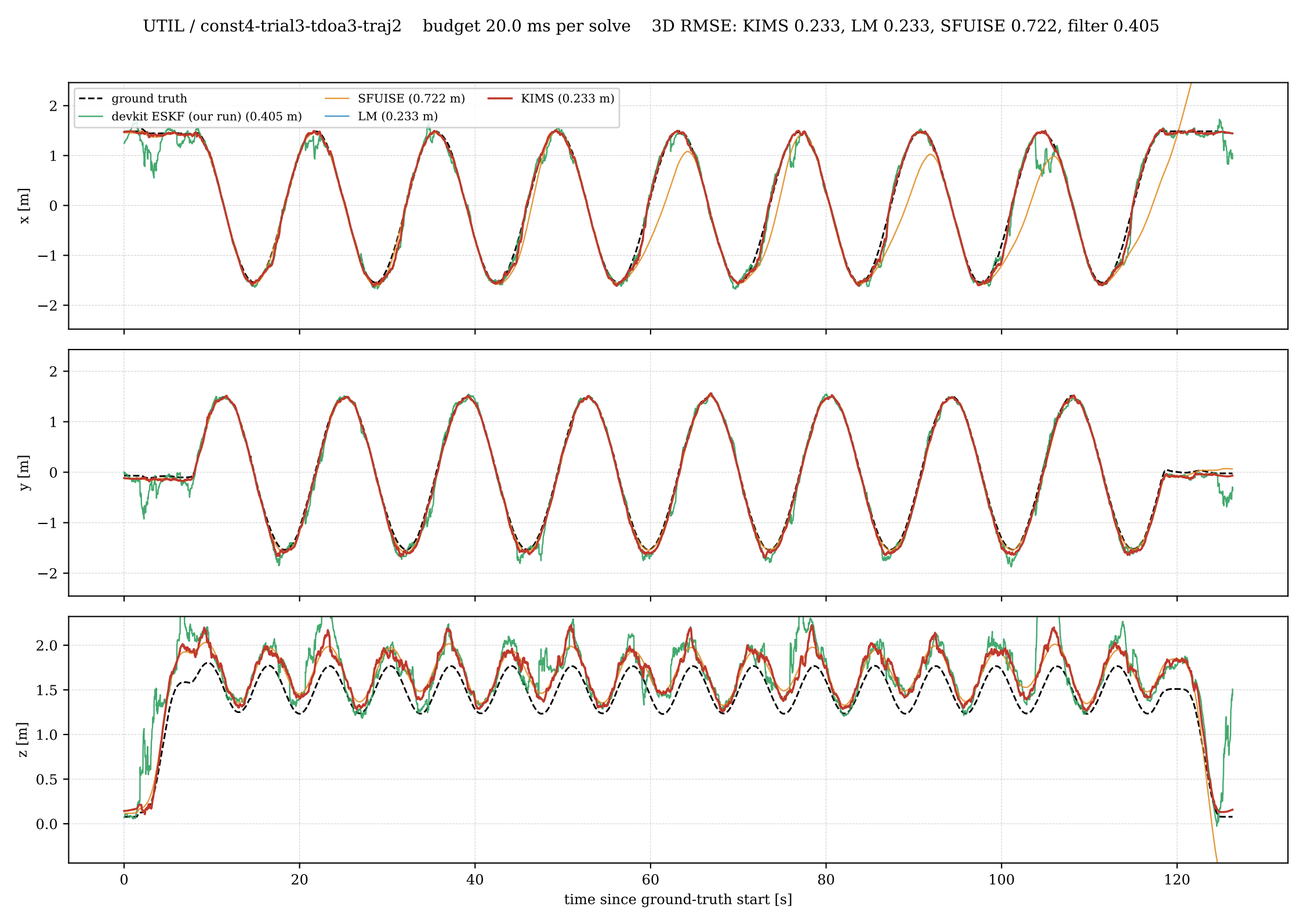

UTIL / const4-trial2-tdoa3-traj3 budget 20.0 ms per solve 3D RMSE: KIMS 0.191, LM 0.191, SFUISE 0.199, filter 0.227
ground truth
devkit ESKF (our run) (0.227 m)
SFUISE (0.199 m)
LM (0.191 m)
KIMS (0.191 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

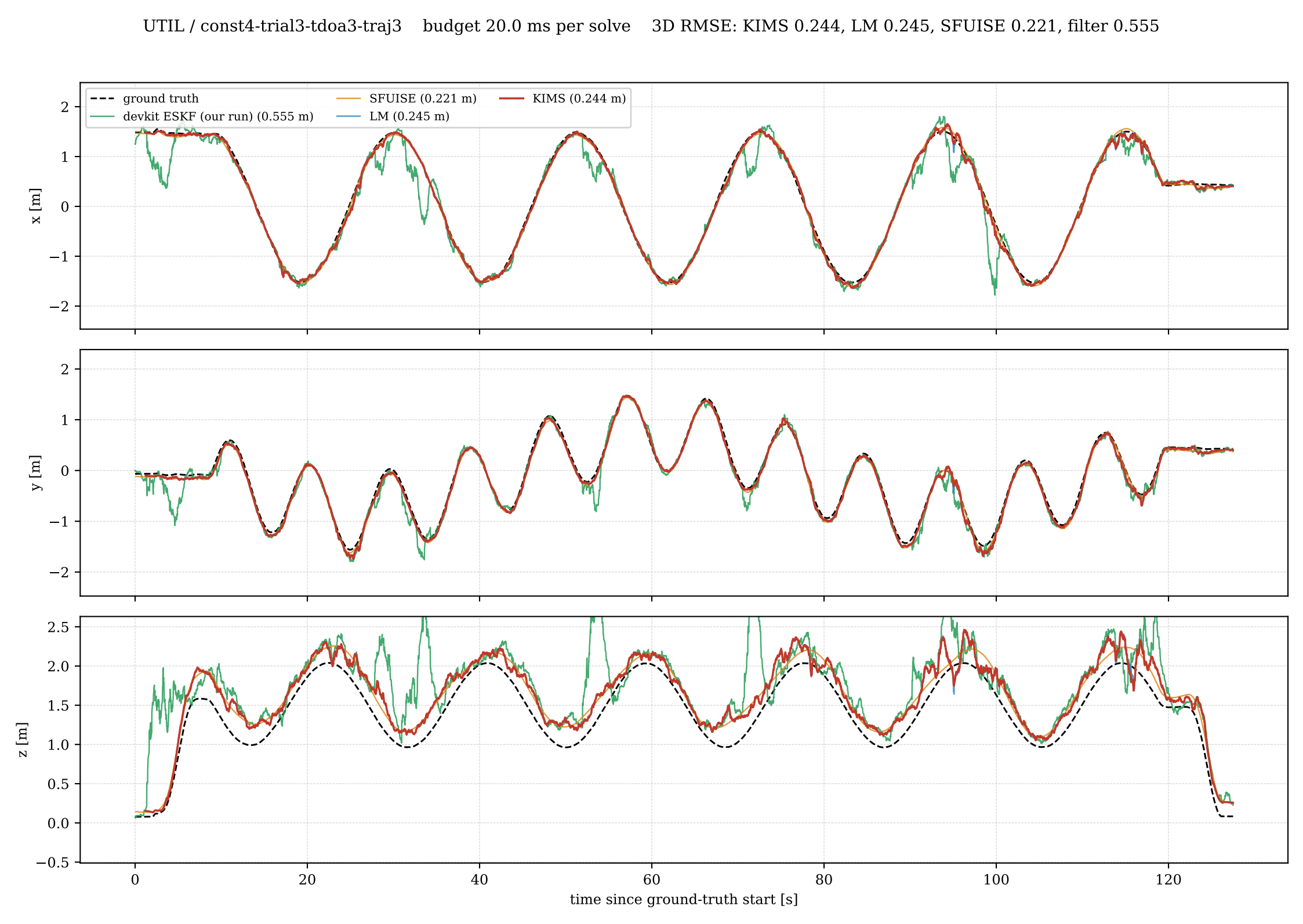

UTIL / const4-trial3-tdoa2-traj1 budget 20.0 ms per solve 3D RMSE: KIMS 0.292, LM 0.296, SFUISE 0.504, filter 0.615
ground truth
devkit ESKF (our run) (0.615 m)
SFUISE (0.504 m)
LM (0.296 m)
KIMS (0.292 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

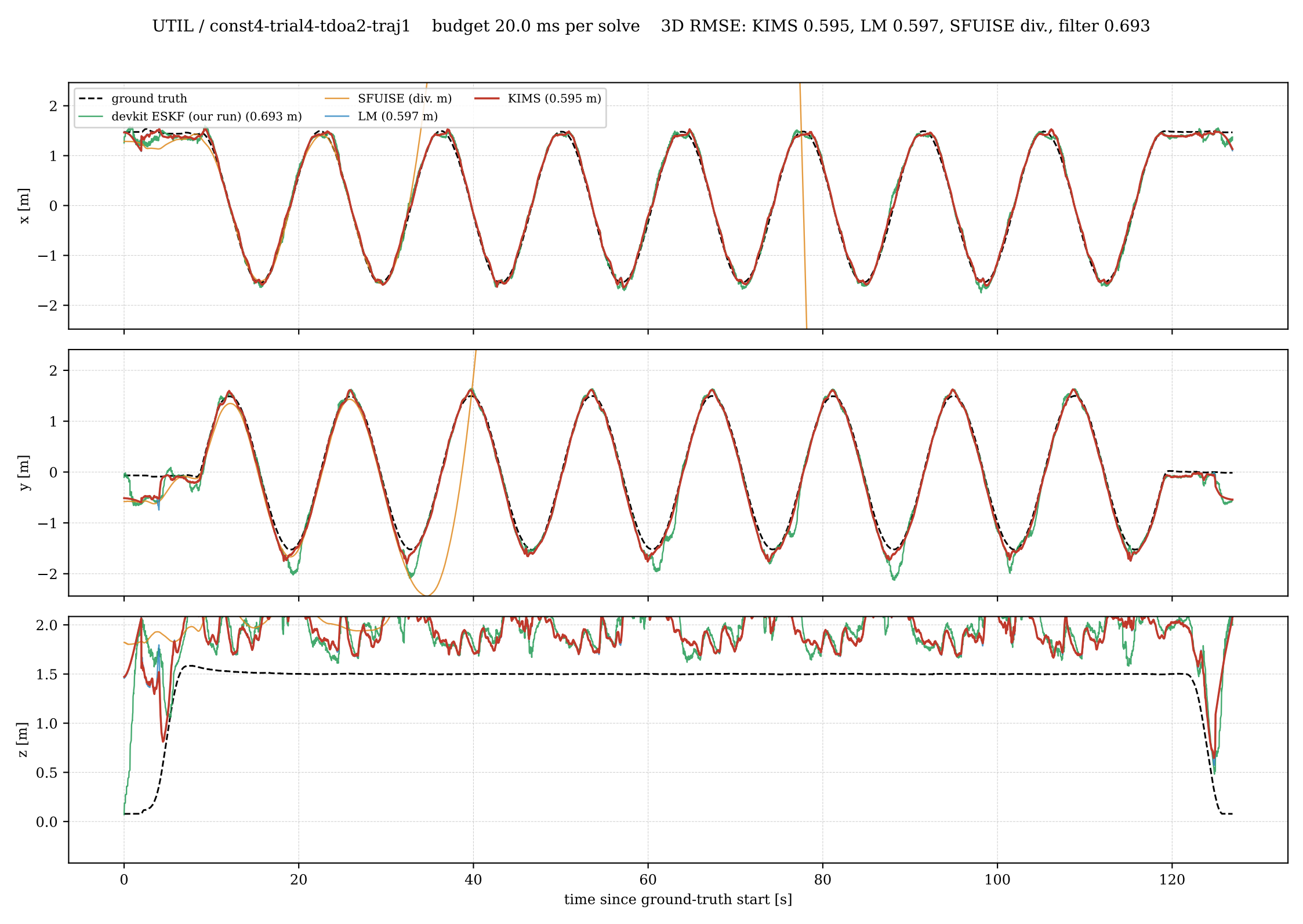

UTIL / const4-trial3-tdoa2-traj2 budget 20.0 ms per solve 3D RMSE: KIMS 0.299, LM 0.299, SFUISE div., filter 0.592
ground truth
devkit ESKF (our run) (0.592 m)
SFUISE (div. m)
LM (0.299 m)
KIMS (0.299 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

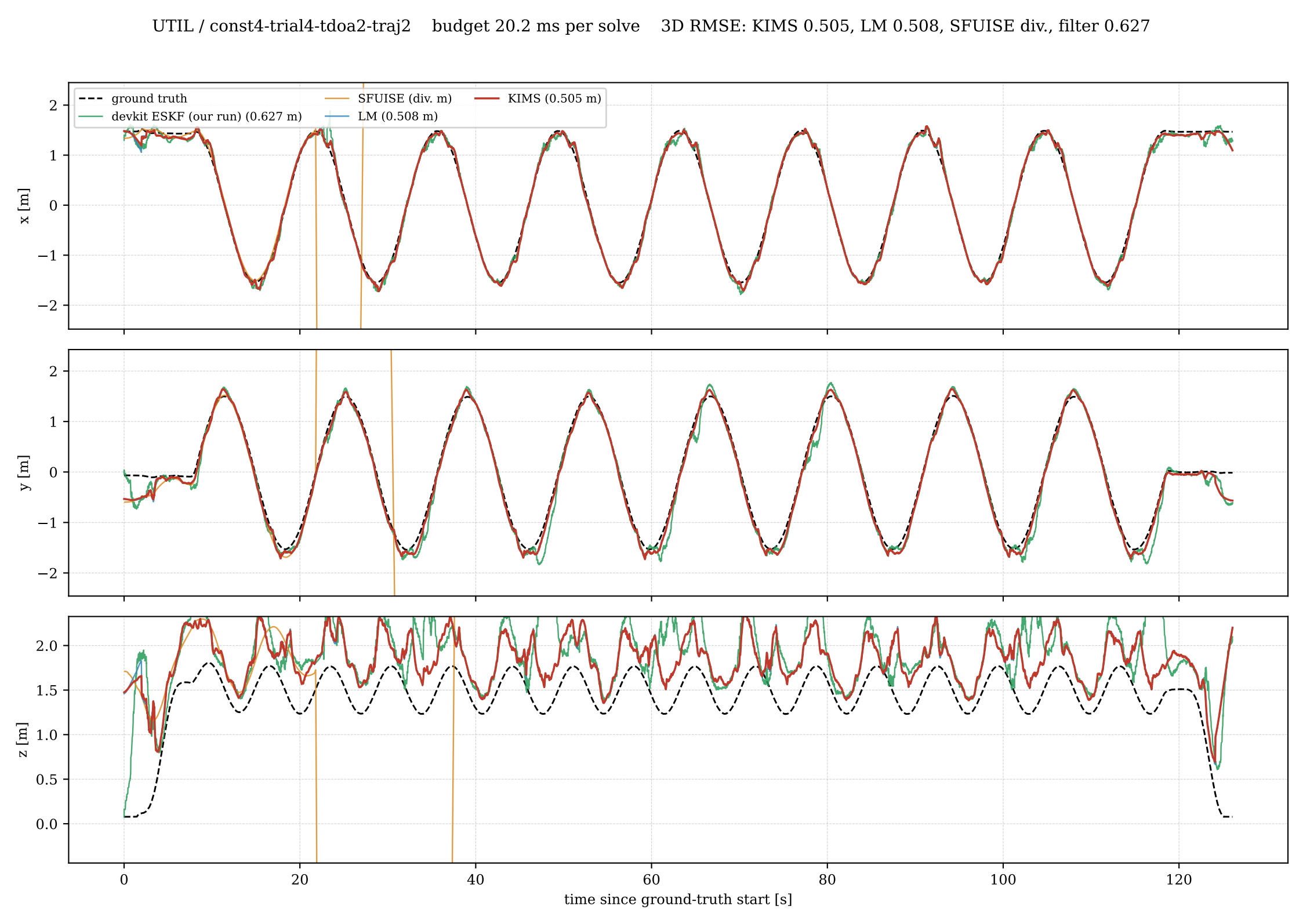

UTIL / const4-trial3-tdoa2-traj3 budget 20.0 ms per solve 3D RMSE: KIMS 0.307, LM 0.308, SFUISE 0.449, filter 0.514
ground truth
devkit ESKF (our run) (0.514 m)
SFUISE (0.449 m)
LM (0.308 m)
KIMS (0.307 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

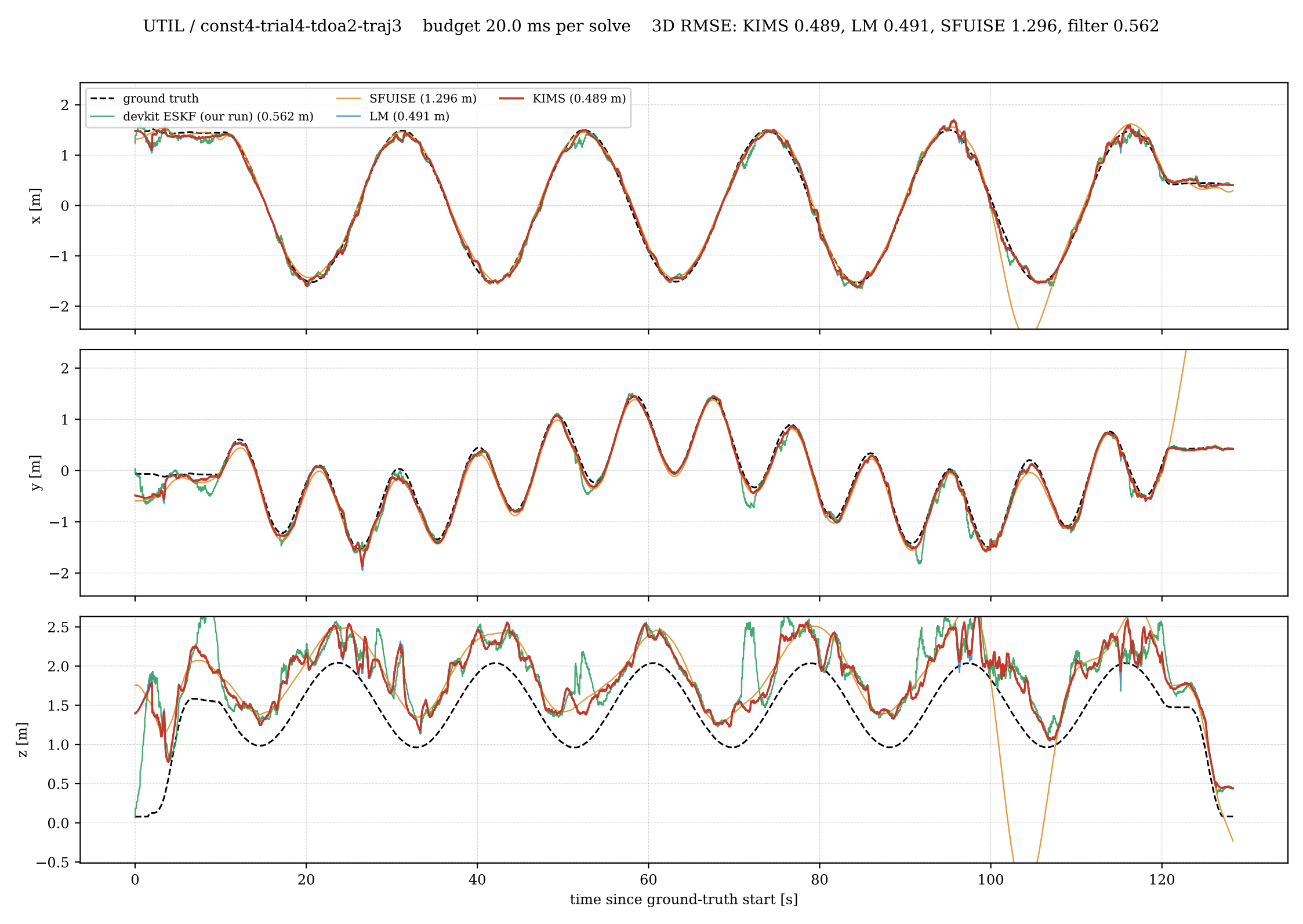

UTIL / const4-trial3-tdoa3-traj1 budget 20.0 ms per solve 3D RMSE: KIMS 0.242, LM 0.242, SFUISE 0.216, filter 0.431
ground truth
devkit ESKF (our run) (0.431 m)
SFUISE (0.216 m)
LM (0.242 m)
KIMS (0.242 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

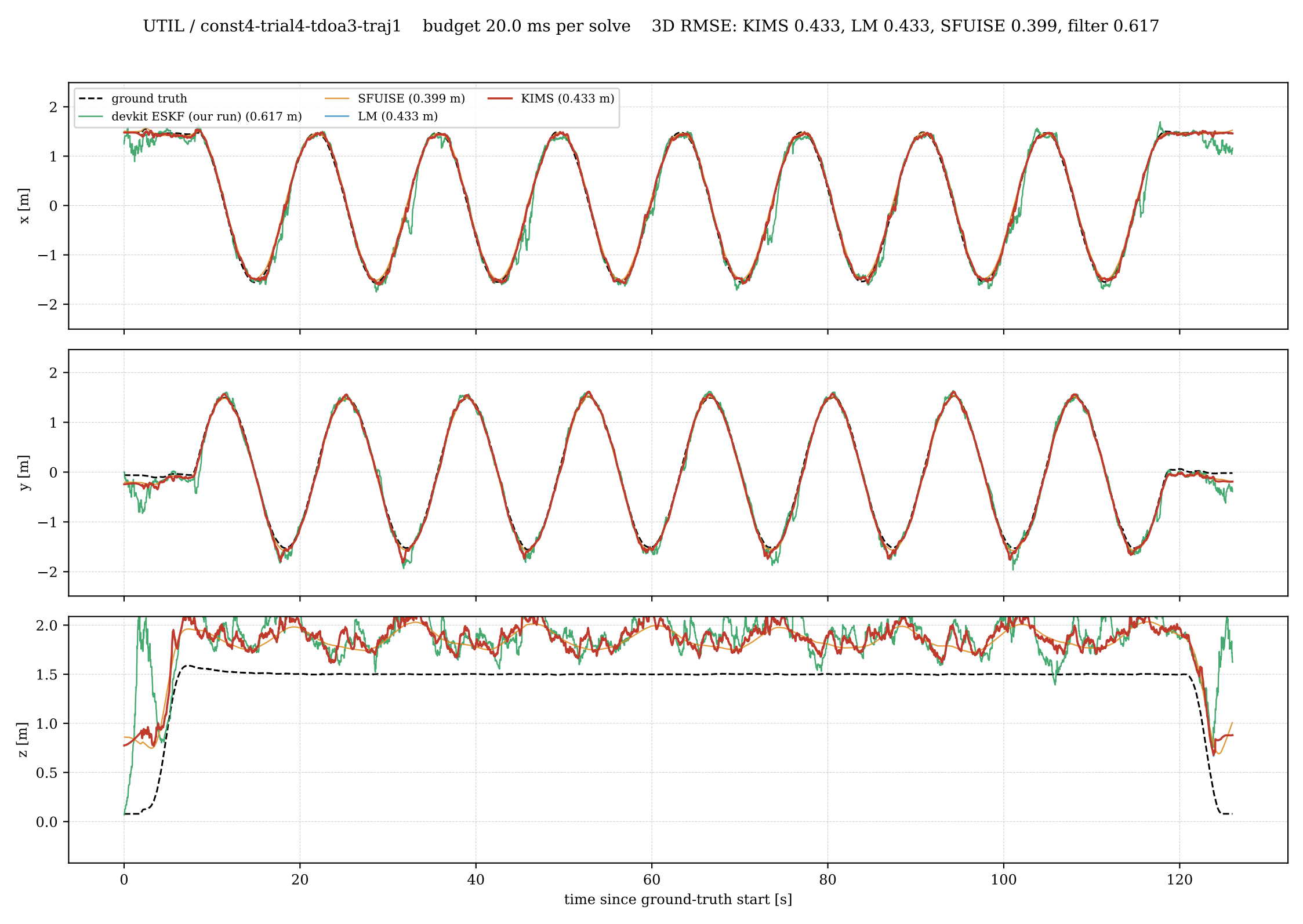

UTIL / const4-trial3-tdoa3-traj2 budget 20.0 ms per solve 3D RMSE: KIMS 0.233, LM 0.233, SFUISE 0.722, filter 0.405
ground truth
devkit ESKF (our run) (0.405 m)
SFUISE (0.722 m)
LM (0.233 m)
KIMS (0.233 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

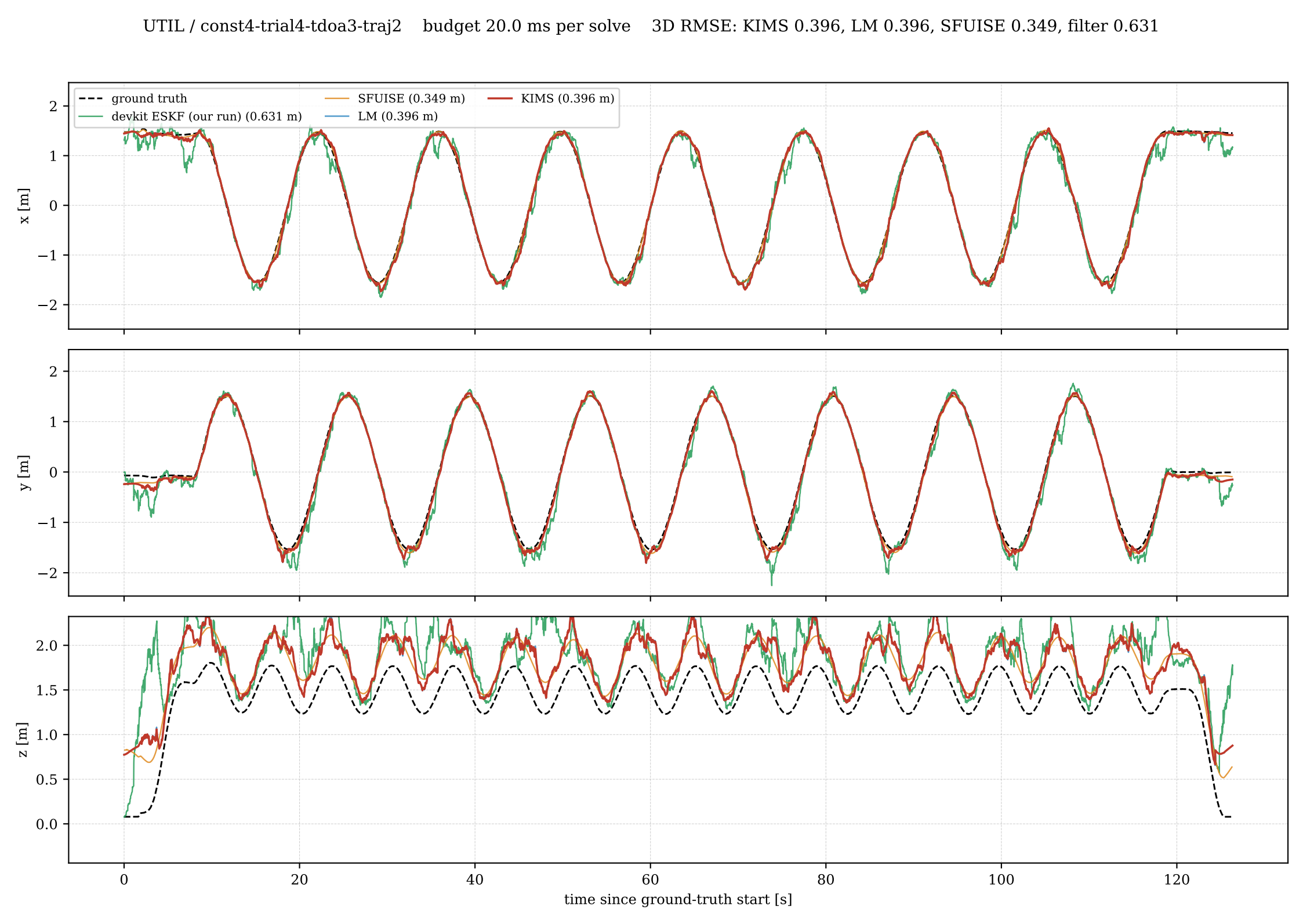

UTIL / const4-trial3-tdoa3-traj3 budget 20.0 ms per solve 3D RMSE: KIMS 0.244, LM 0.245, SFUISE 0.221, filter 0.555
ground truth
devkit ESKF (our run) (0.555 m)
SFUISE (0.221 m)
LM (0.245 m)
KIMS (0.244 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

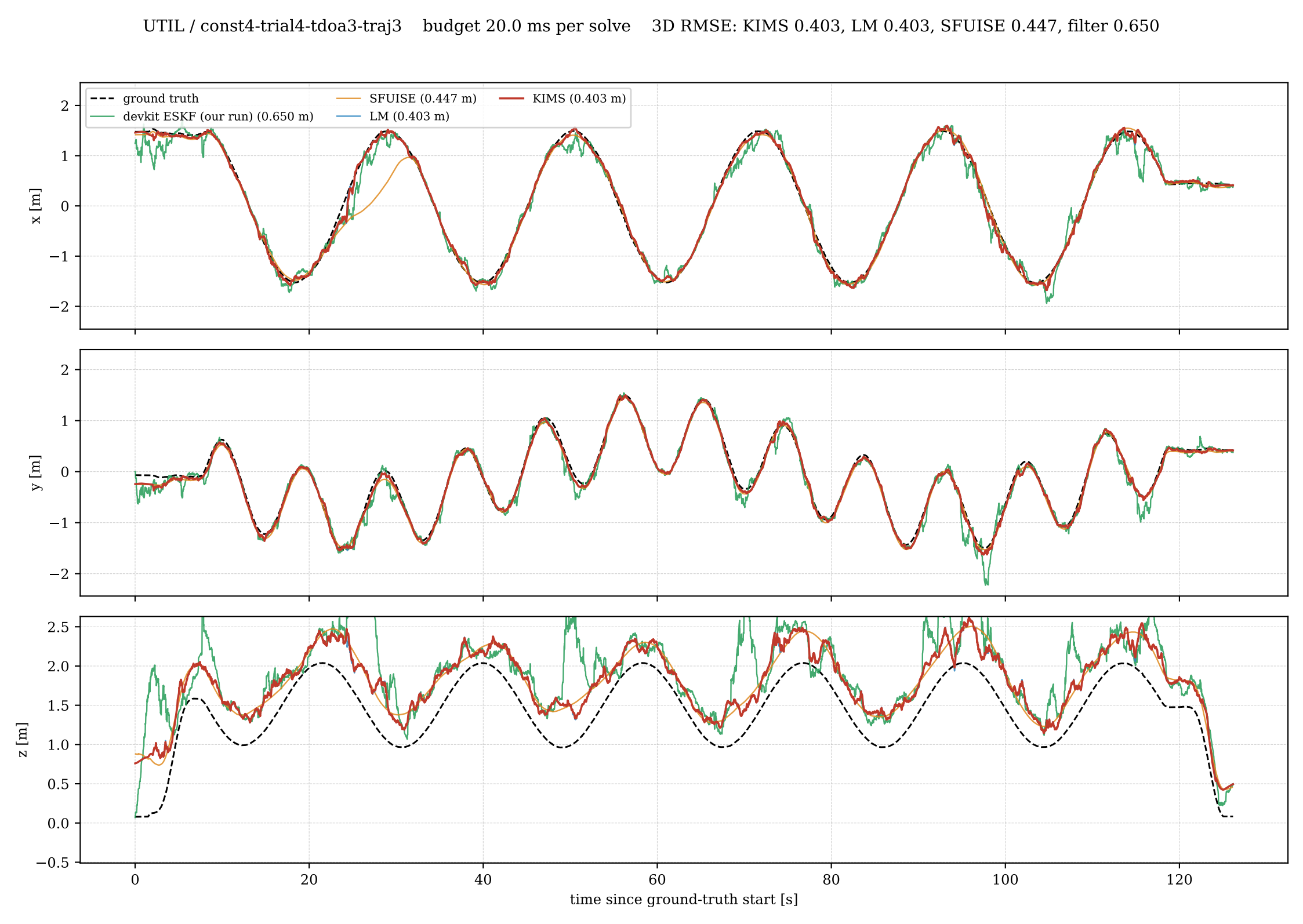

UTIL / const4-trial4-tdoa2-traj1 budget 20.0 ms per solve 3D RMSE: KIMS 0.595, LM 0.597, SFUISE div., filter 0.693
ground truth
devkit ESKF (our run) (0.693 m)
SFUISE (div. m)
LM (0.597 m)
KIMS (0.595 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

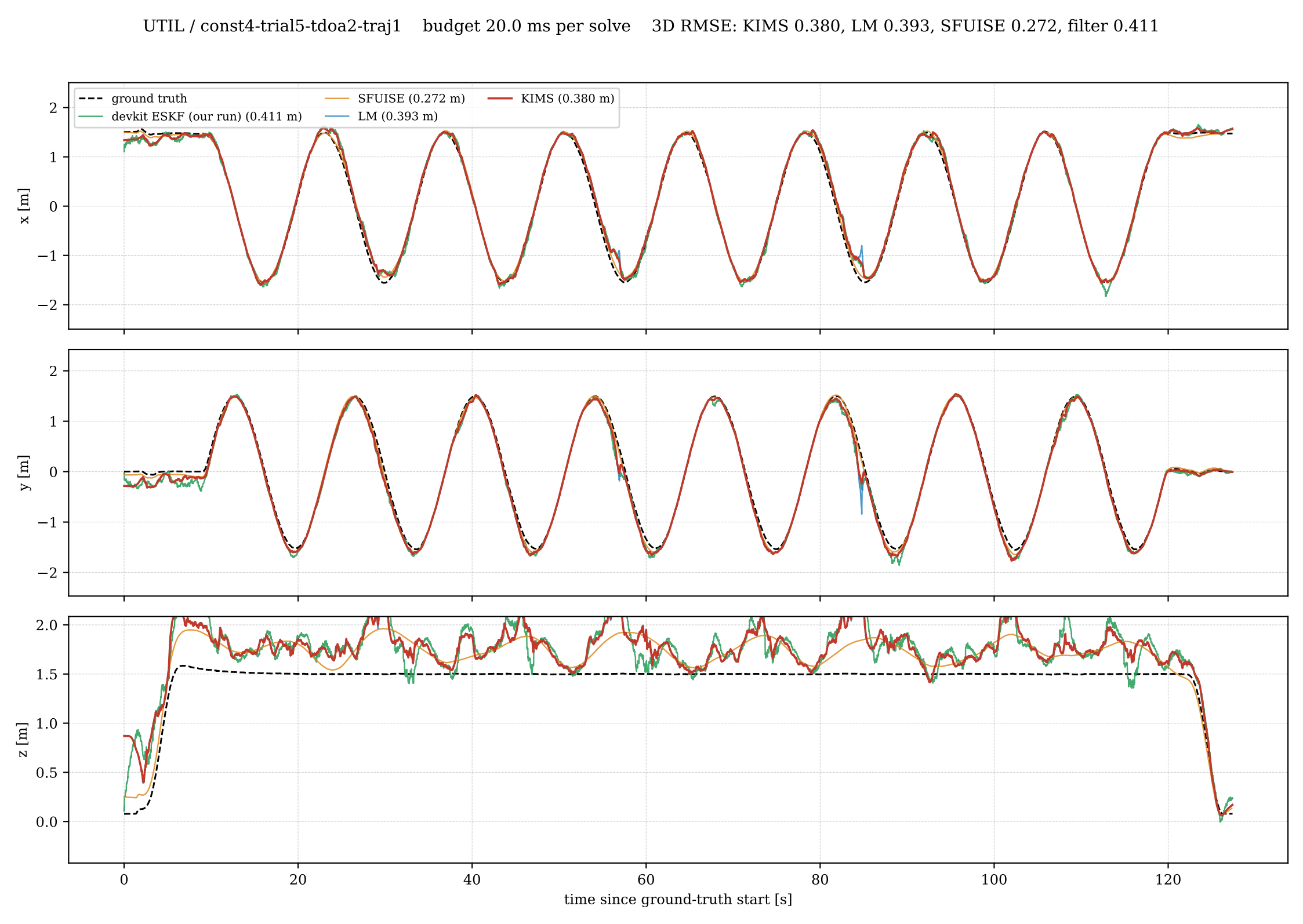

UTIL / const4-trial4-tdoa2-traj2 budget 20.2 ms per solve 3D RMSE: KIMS 0.505, LM 0.508, SFUISE div., filter 0.627
ground truth
devkit ESKF (our run) (0.627 m)
SFUISE (div. m)
LM (0.508 m)
KIMS (0.505 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

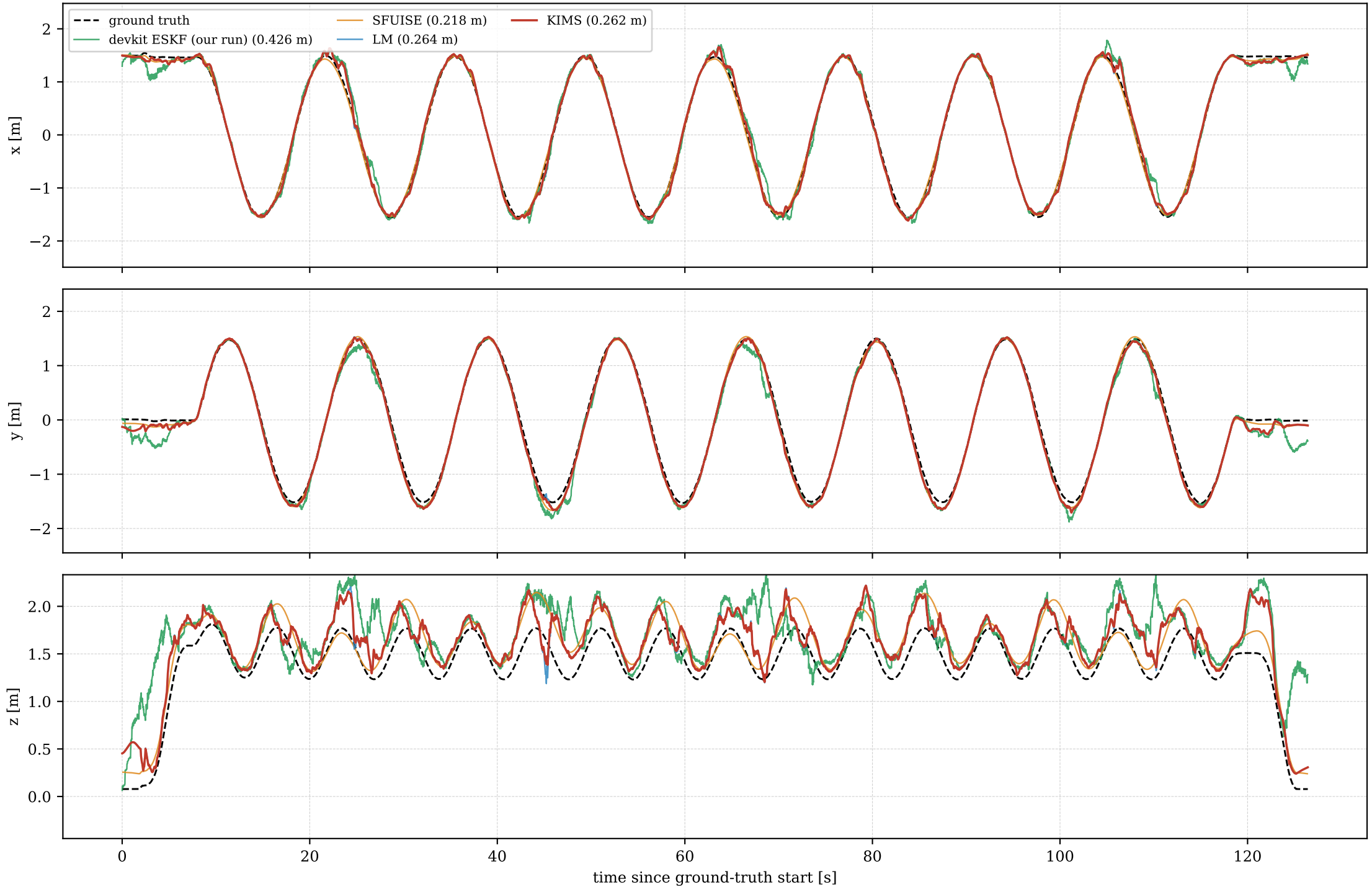

UTIL / const4-trial4-tdoa2-traj3 budget 20.0 ms per solve 3D RMSE: KIMS 0.489, LM 0.491, SFUISE 1.296, filter 0.562
ground truth
devkit ESKF (our run) (0.562 m)
SFUISE (1.296 m)
LM (0.491 m)
KIMS (0.489 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

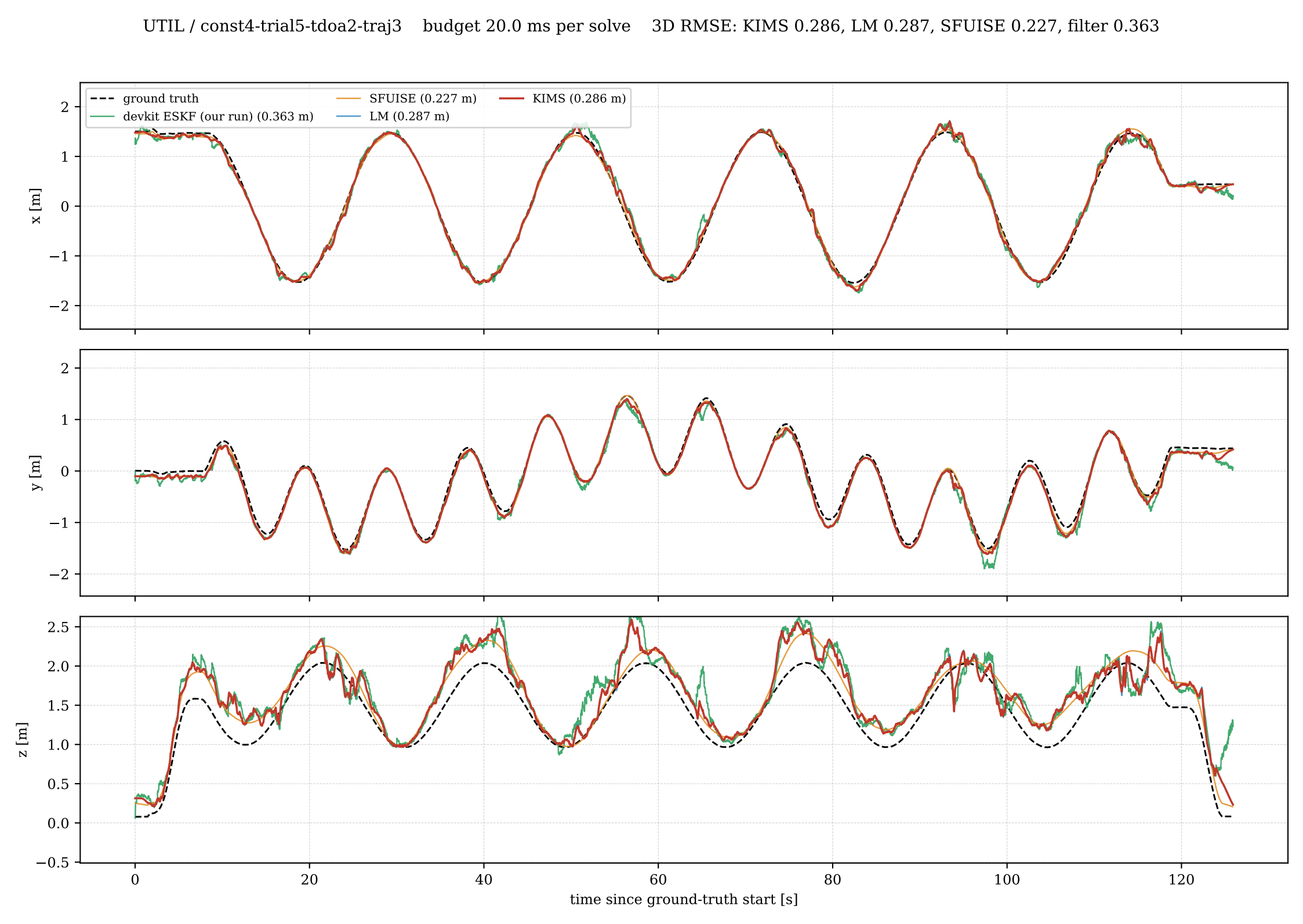

UTIL / const4-trial4-tdoa3-traj1 budget 20.0 ms per solve 3D RMSE: KIMS 0.433, LM 0.433, SFUISE 0.399, filter 0.617
ground truth
devkit ESKF (our run) (0.617 m)
SFUISE (0.399 m)
LM (0.433 m)
KIMS (0.433 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

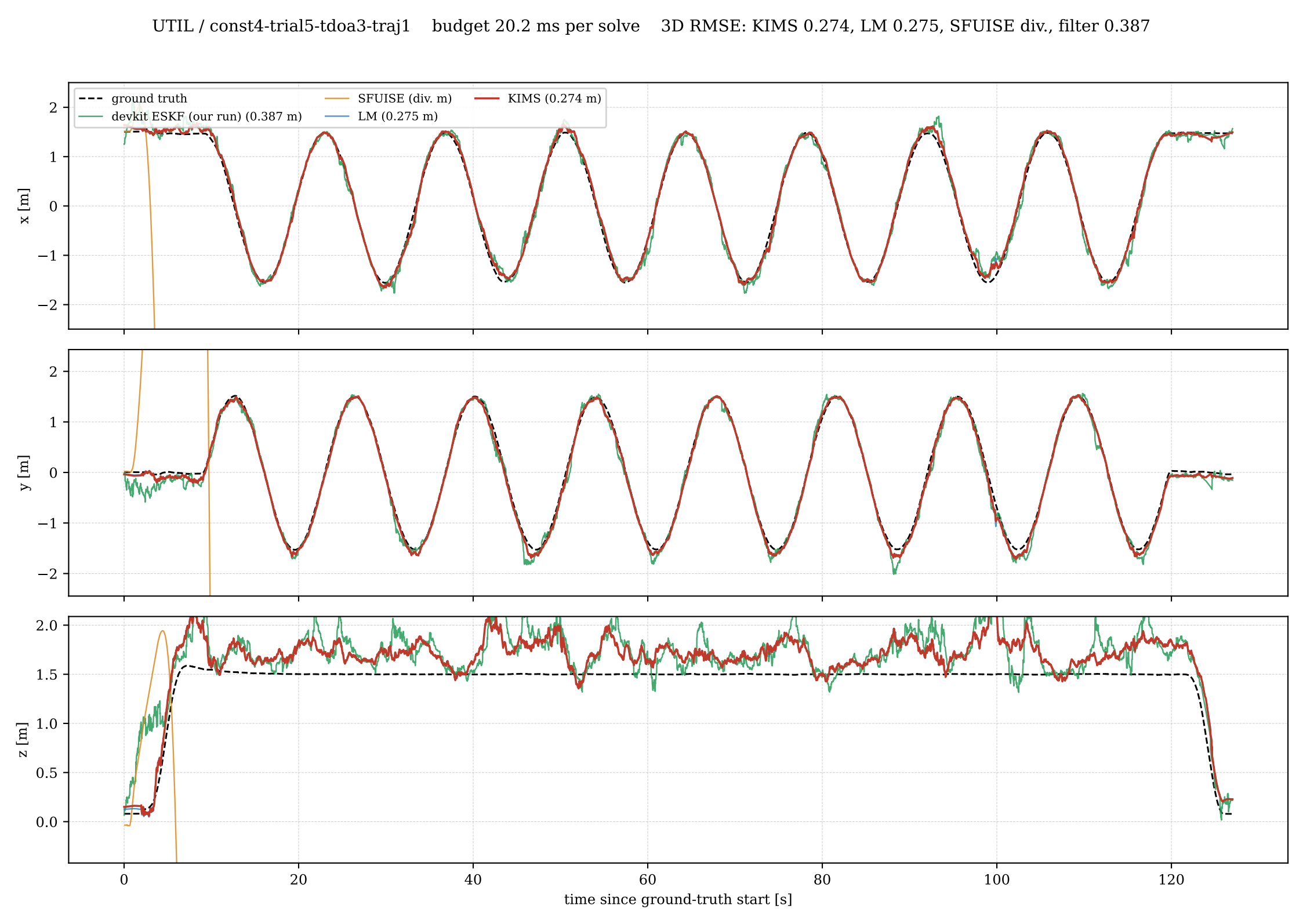

UTIL / const4-trial4-tdoa3-traj2 budget 20.0 ms per solve 3D RMSE: KIMS 0.396, LM 0.396, SFUISE 0.349, filter 0.631
ground truth
devkit ESKF (our run) (0.631 m)
SFUISE (0.349 m)
LM (0.396 m)
KIMS (0.396 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

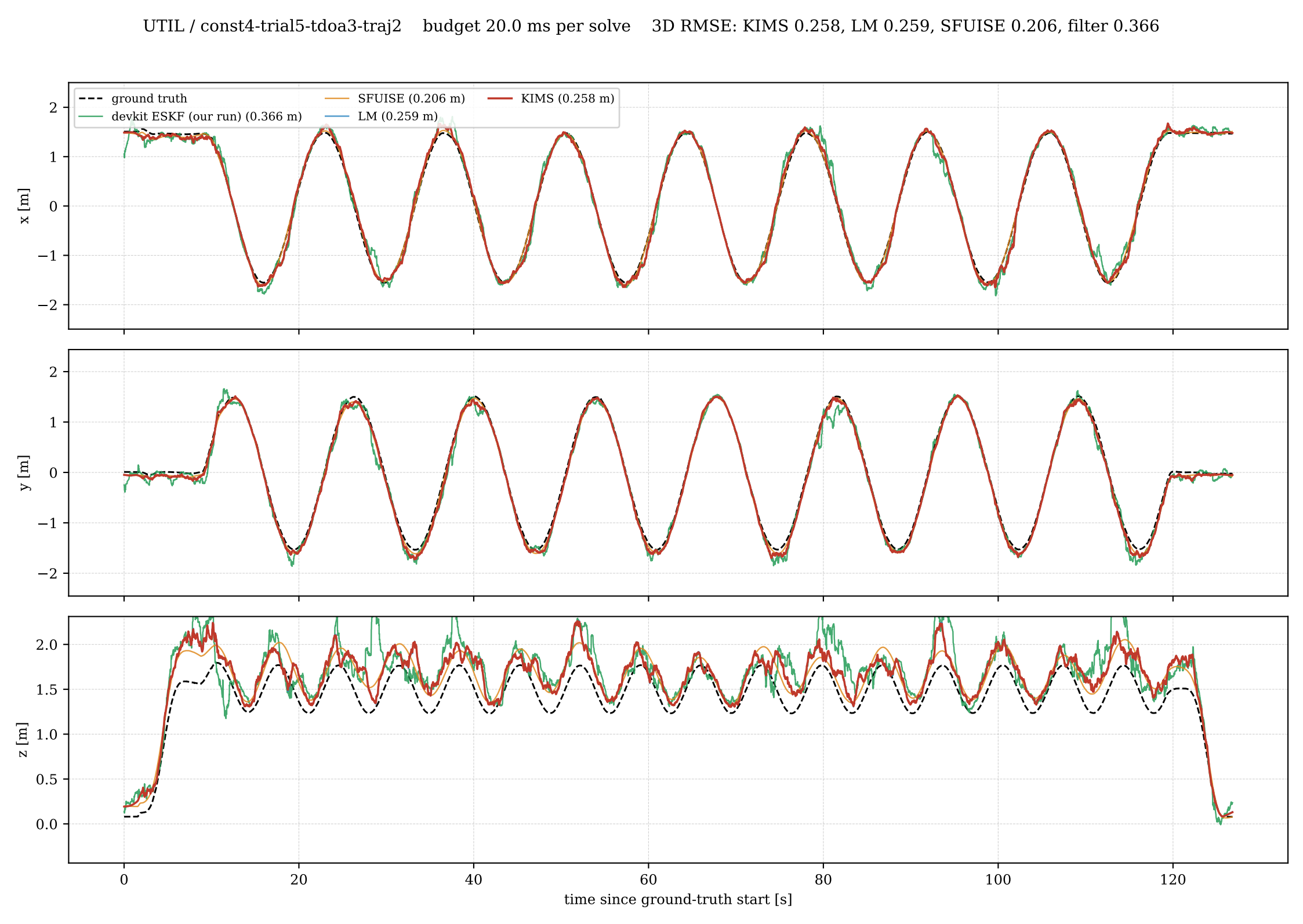

UTIL / const4-trial4-tdoa3-traj3 budget 20.0 ms per solve 3D RMSE: KIMS 0.403, LM 0.403, SFUISE 0.447, filter 0.650
ground truth
devkit ESKF (our run) (0.650 m)
SFUISE (0.447 m)
LM (0.403 m)
KIMS (0.403 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

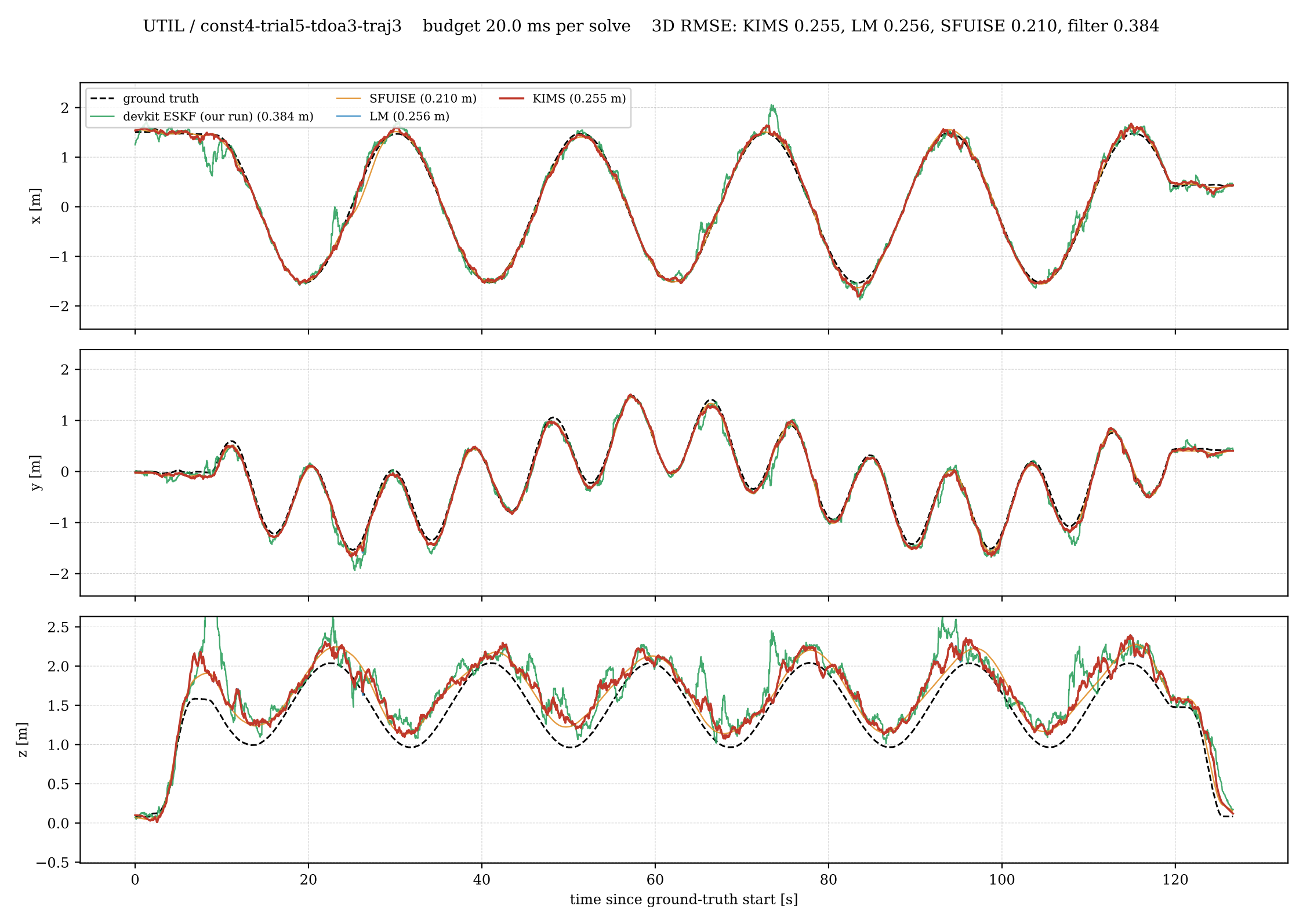

UTIL / const4-trial5-tdoa2-traj1 budget 20.0 ms per solve 3D RMSE: KIMS 0.380, LM 0.393, SFUISE 0.272, filter 0.411
ground truth
devkit ESKF (our run) (0.411 m)
SFUISE (0.272 m)
LM (0.393 m)
KIMS (0.380 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

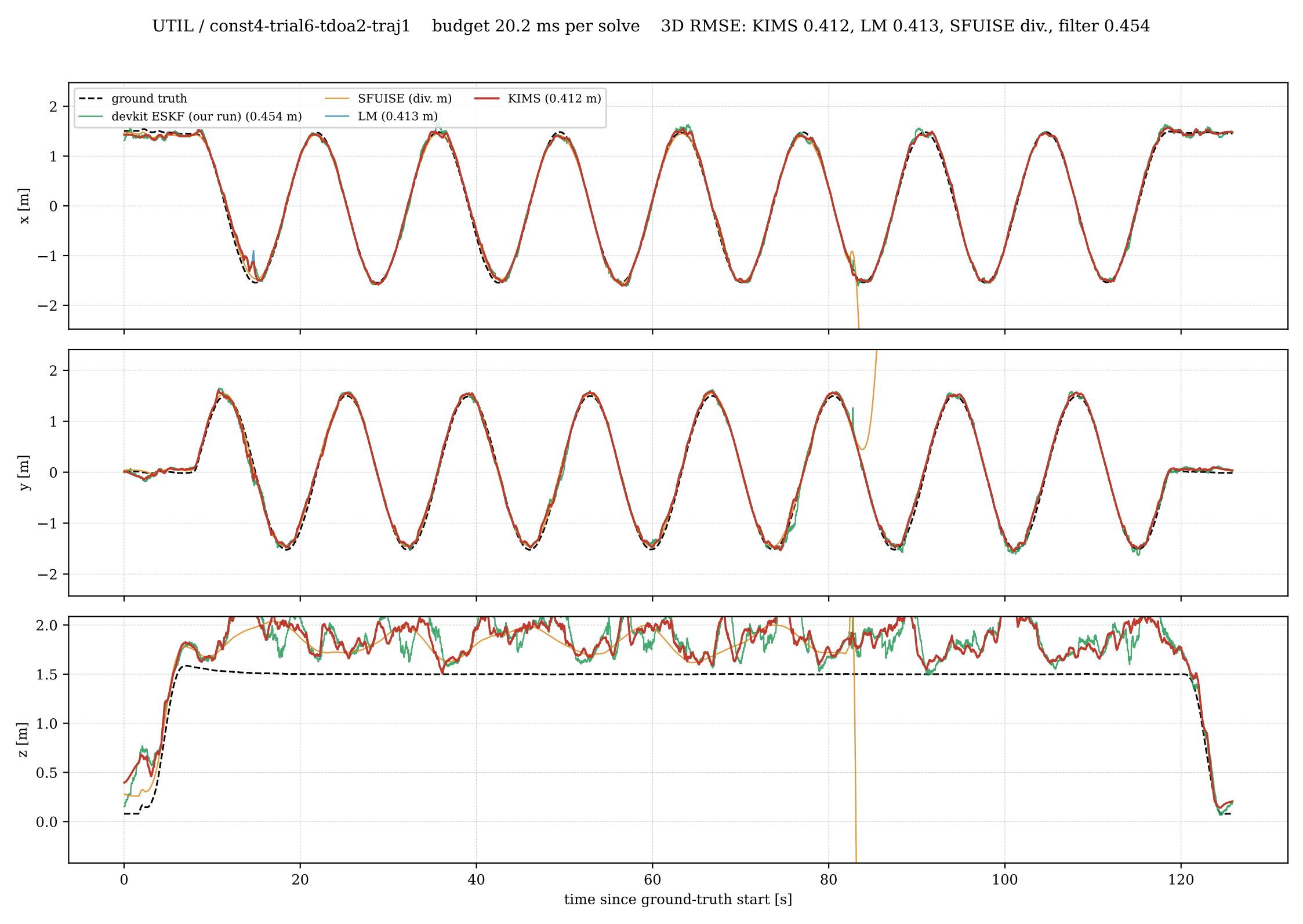

UTIL / const4-trial5-tdoa2-traj2 budget 20.0 ms per solve 3D RMSE: KIMS 0.262, LM 0.264, SFUISE 0.218, filter 0.426
ground truth
devkit ESKF (our run) (0.426 m)
SFUISE (0.218 m)
LM (0.264 m)
KIMS (0.262 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

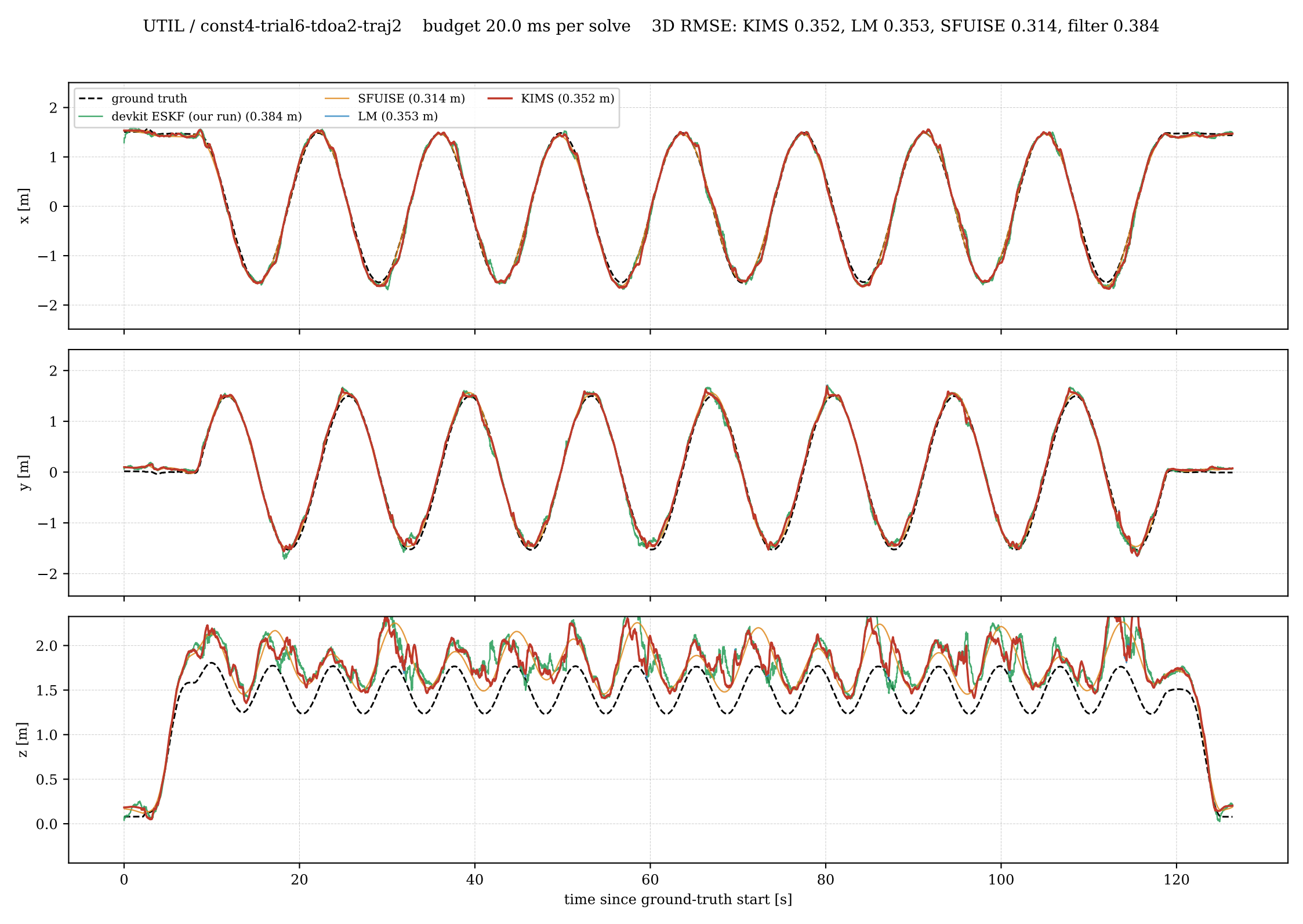

UTIL / const4-trial5-tdoa2-traj3 budget 20.0 ms per solve 3D RMSE: KIMS 0.286, LM 0.287, SFUISE 0.227, filter 0.363
ground truth
devkit ESKF (our run) (0.363 m)
SFUISE (0.227 m)
LM (0.287 m)
KIMS (0.286 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

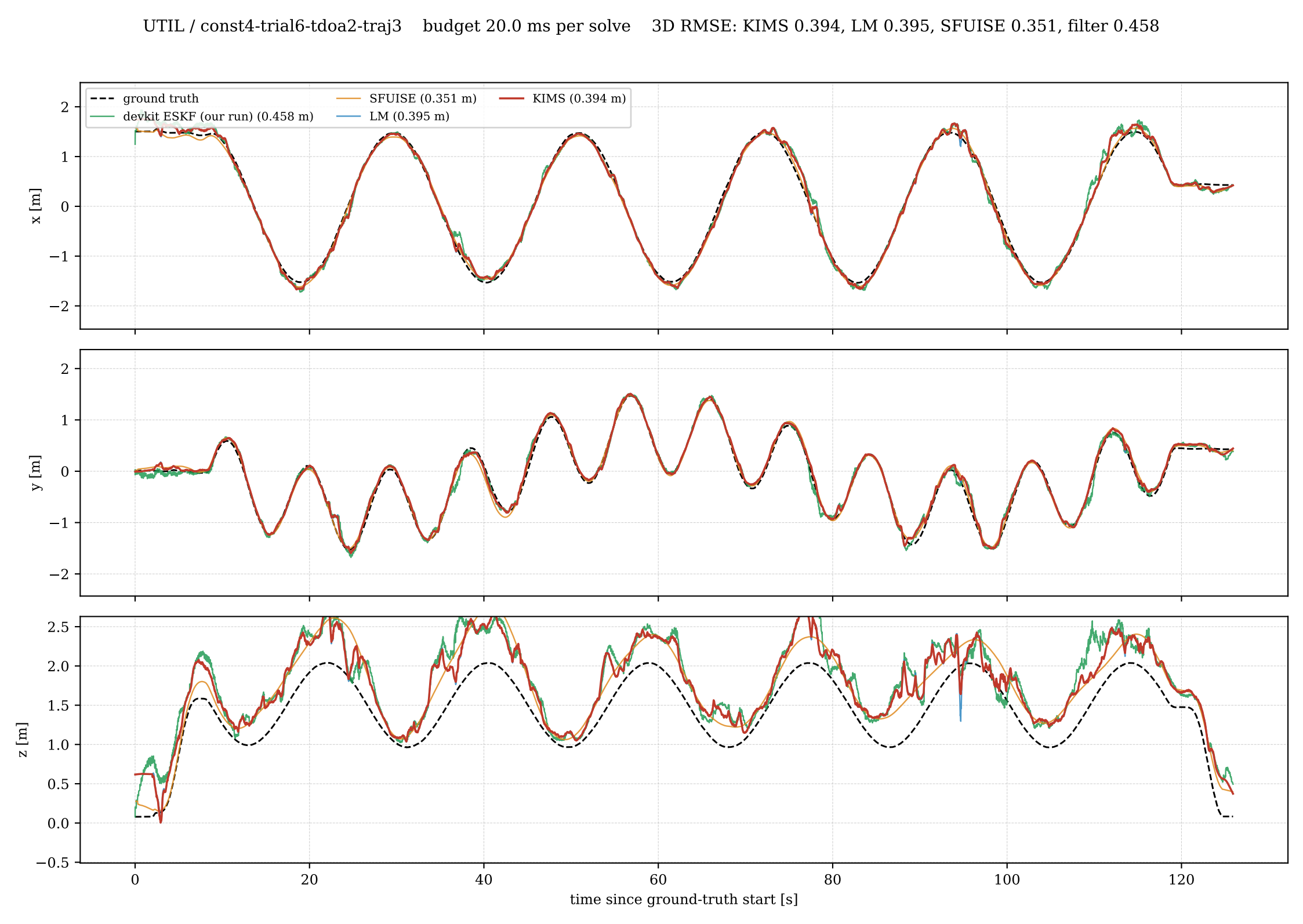

UTIL / const4-trial5-tdoa3-traj1 budget 20.2 ms per solve 3D RMSE: KIMS 0.274, LM 0.275, SFUISE div., filter 0.387
ground truth
devkit ESKF (our run) (0.387 m)
SFUISE (div. m)
LM (0.275 m)
KIMS (0.274 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

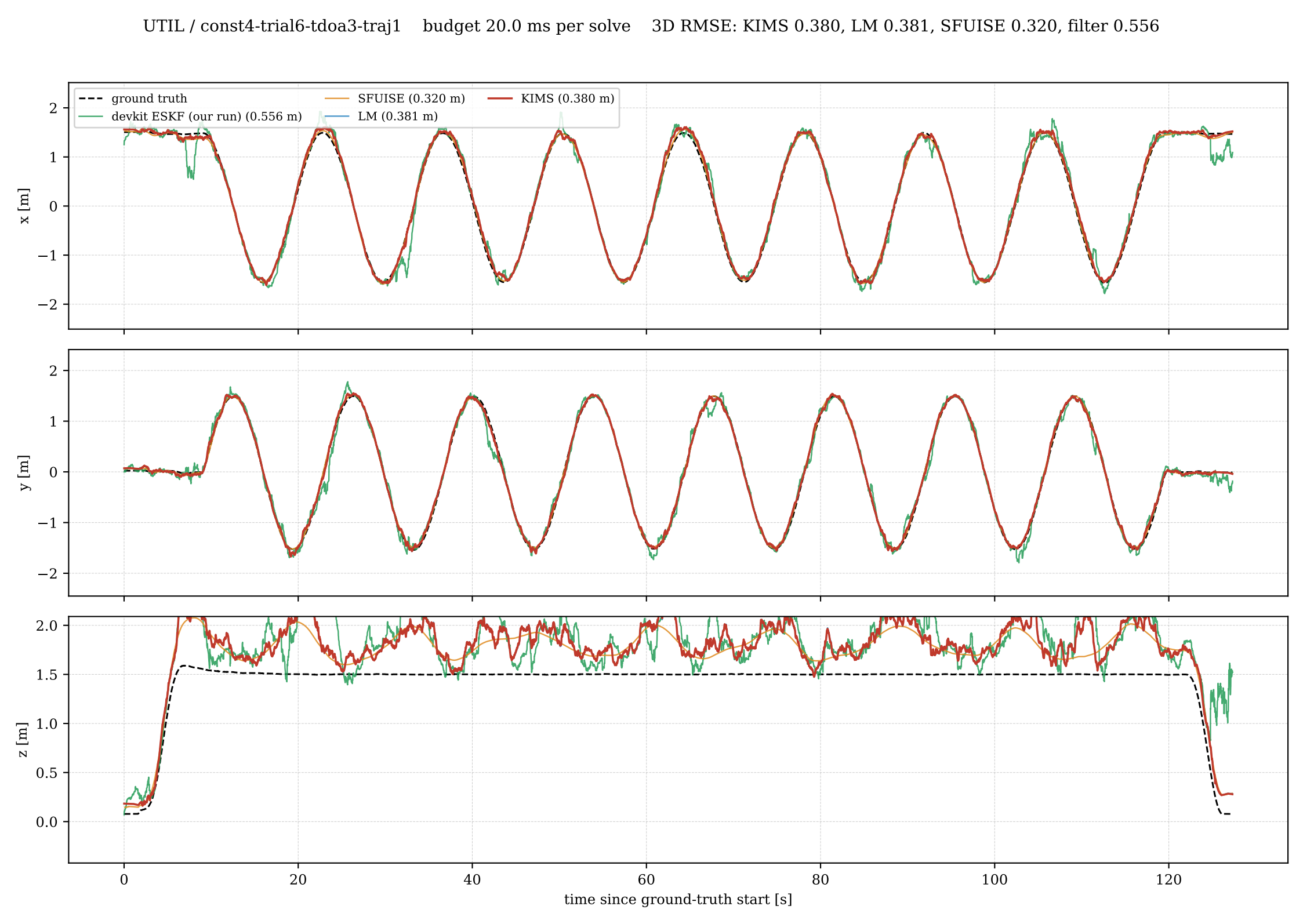

UTIL / const4-trial5-tdoa3-traj2 budget 20.0 ms per solve 3D RMSE: KIMS 0.258, LM 0.259, SFUISE 0.206, filter 0.366
ground truth
devkit ESKF (our run) (0.366 m)
SFUISE (0.206 m)
LM (0.259 m)
KIMS (0.258 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

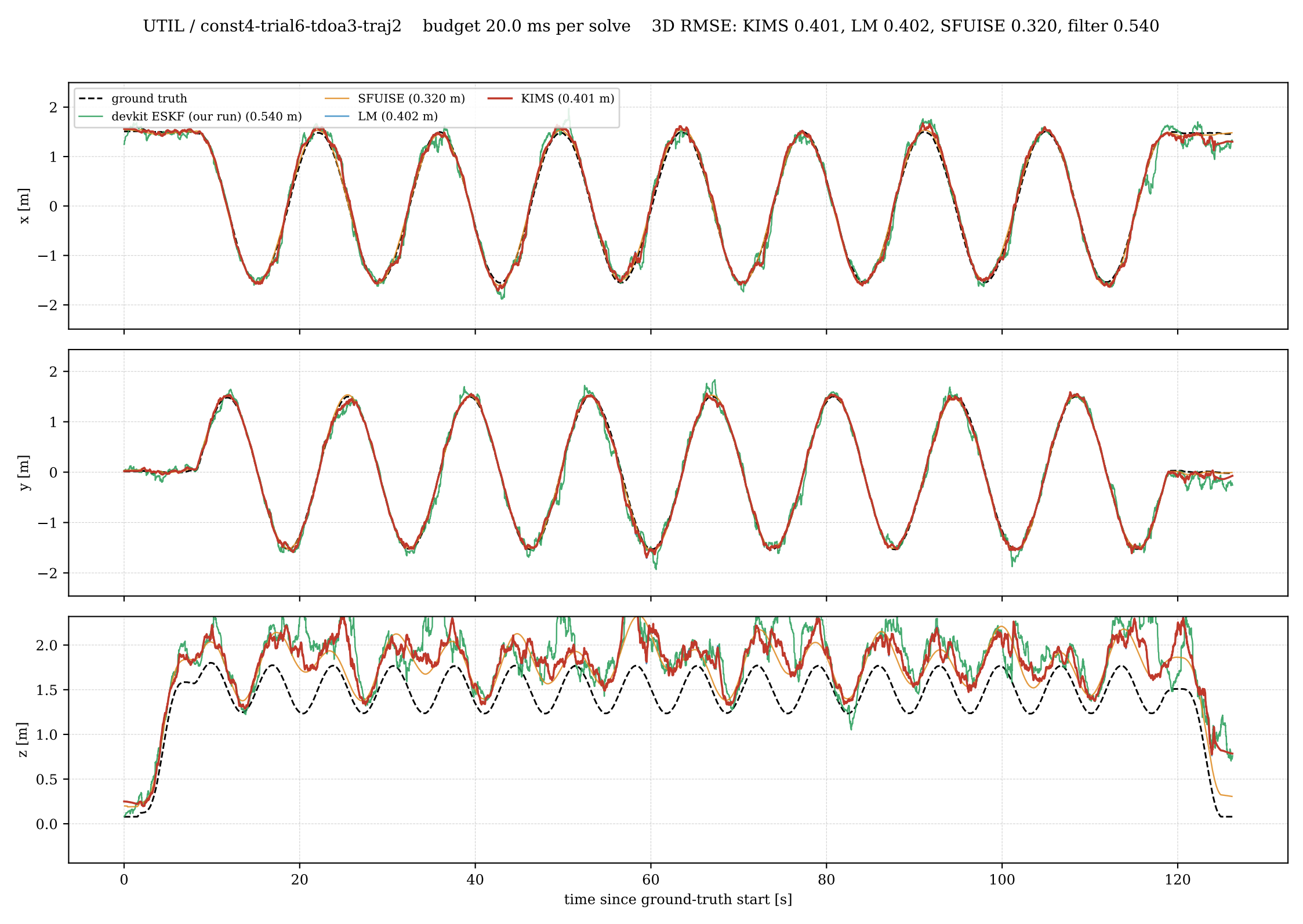

UTIL / const4-trial5-tdoa3-traj3 budget 20.0 ms per solve 3D RMSE: KIMS 0.255, LM 0.256, SFUISE 0.210, filter 0.384
ground truth
devkit ESKF (our run) (0.384 m)
SFUISE (0.210 m)
LM (0.256 m)
KIMS (0.255 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

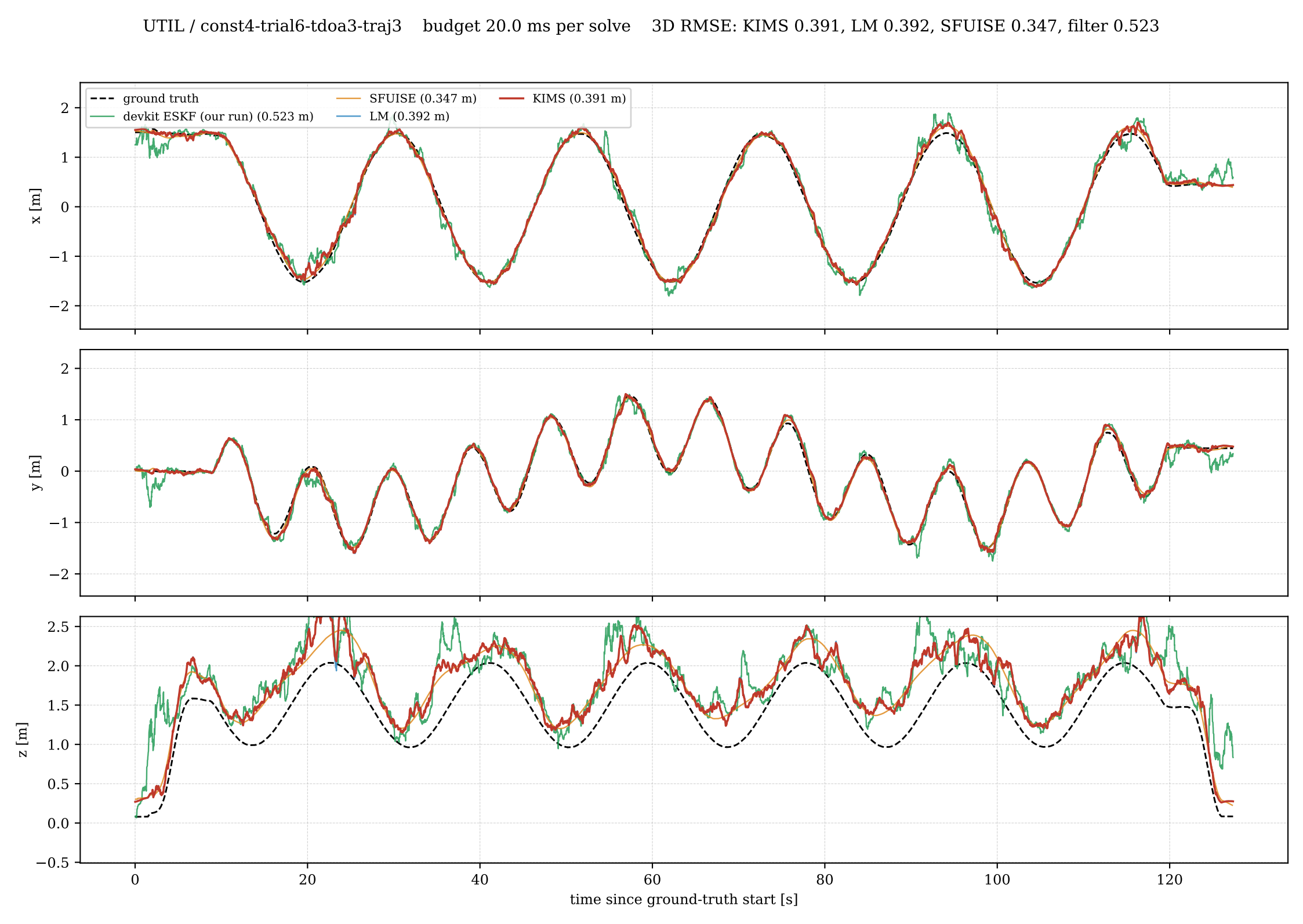

UTIL / const4-trial6-tdoa2-traj1 budget 20.2 ms per solve 3D RMSE: KIMS 0.412, LM 0.413, SFUISE div., filter 0.454
ground truth
devkit ESKF (our run) (0.454 m)
SFUISE (div. m)
LM (0.413 m)
KIMS (0.412 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

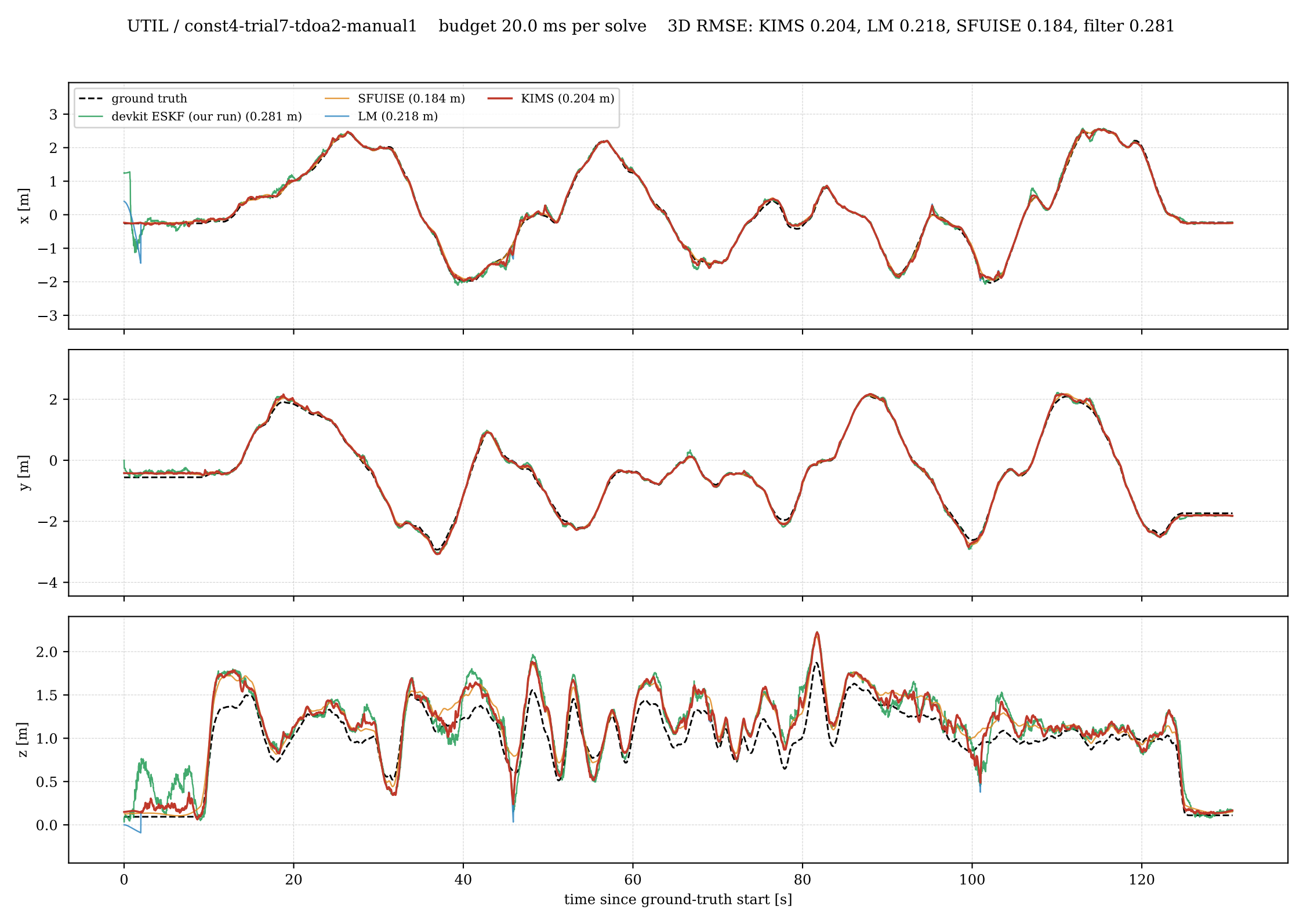

UTIL / const4-trial6-tdoa2-traj2 budget 20.0 ms per solve 3D RMSE: KIMS 0.352, LM 0.353, SFUISE 0.314, filter 0.384
ground truth
devkit ESKF (our run) (0.384 m)
SFUISE (0.314 m)
LM (0.353 m)
KIMS (0.352 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

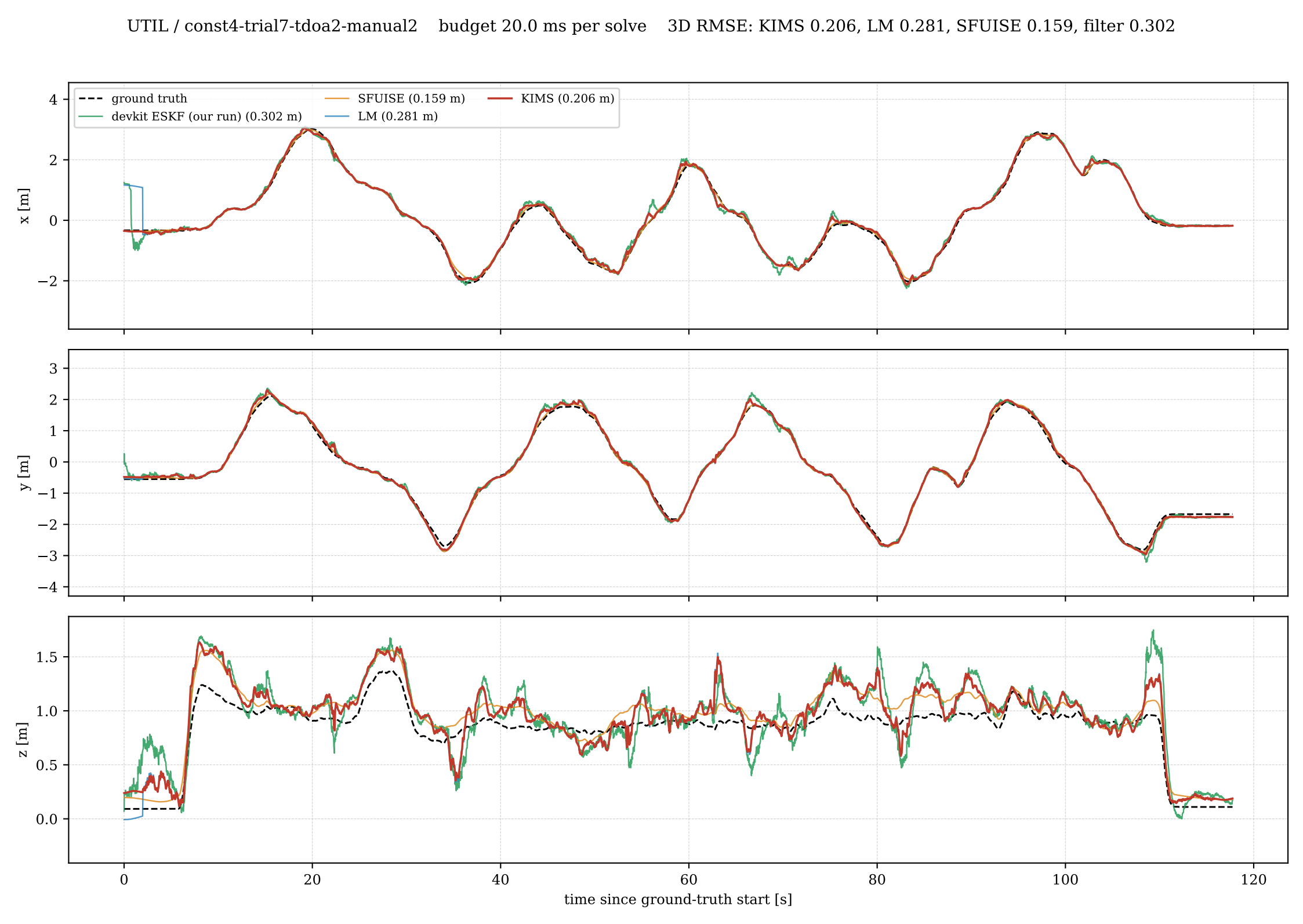

UTIL / const4-trial6-tdoa2-traj3 budget 20.0 ms per solve 3D RMSE: KIMS 0.394, LM 0.395, SFUISE 0.351, filter 0.458
ground truth
devkit ESKF (our run) (0.458 m)
SFUISE (0.351 m)
LM (0.395 m)
KIMS (0.394 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]

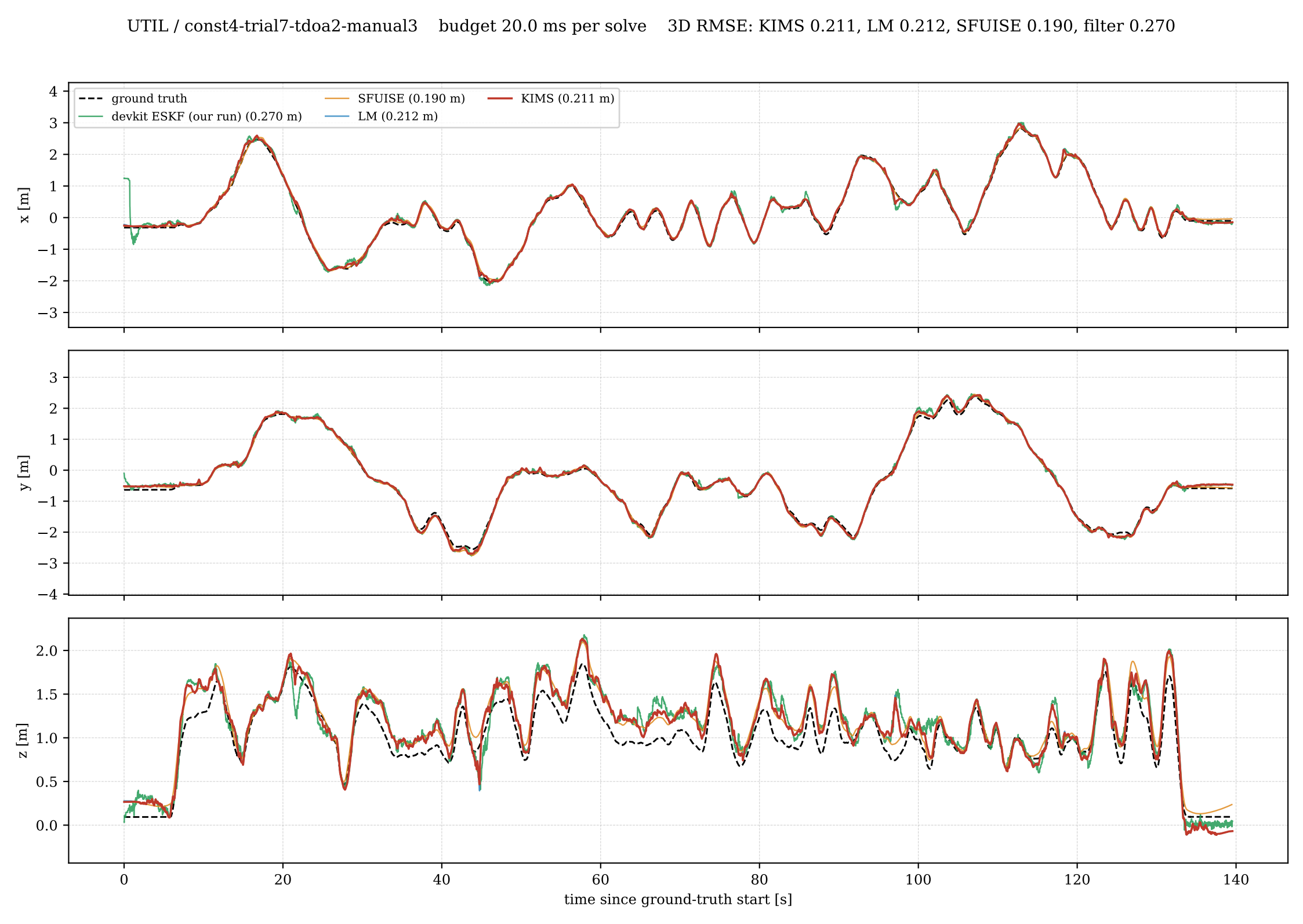

UTIL / const4-trial6-tdoa3-traj1 budget 20.0 ms per solve 3D RMSE: KIMS 0.380, LM 0.381, SFUISE 0.320, filter 0.556
ground truth
devkit ESKF (our run) (0.556 m)
SFUISE (0.320 m)
LM (0.381 m)
KIMS (0.380 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]


UTIL / const4-trial6-tdoa3-traj2 budget 20.0 ms per solve 3D RMSE: KIMS 0.401, LM 0.402, SFUISE 0.320, filter 0.540
ground truth
devkit ESKF (our run) (0.540 m)
SFUISE (0.320 m)
LM (0.402 m)
KIMS (0.401 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]


UTIL / const4-trial6-tdoa3-traj3 budget 20.0 ms per solve 3D RMSE: KIMS 0.391, LM 0.392, SFUISE 0.347, filter 0.523
ground truth
devkit ESKF (our run) (0.523 m)
SFUISE (0.347 m)
LM (0.392 m)
KIMS (0.391 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]


UTIL / const4-trial7-tdoa2-manual1 budget 20.0 ms per solve 3D RMSE: KIMS 0.204, LM 0.218, SFUISE 0.184, filter 0.281
ground truth
devkit ESKF (our run) (0.281 m)
SFUISE (0.184 m)
LM (0.218 m)
KIMS (0.204 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]


UTIL / const4-trial7-tdoa2-manual2 budget 20.0 ms per solve 3D RMSE: KIMS 0.206, LM 0.281, SFUISE 0.159, filter 0.302
ground truth
devkit ESKF (our run) (0.302 m)
SFUISE (0.159 m)
LM (0.281 m)
KIMS (0.206 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]


UTIL / const4-trial7-tdoa2-manual3 budget 20.0 ms per solve 3D RMSE: KIMS 0.211, LM 0.212, SFUISE 0.190, filter 0.270
ground truth
devkit ESKF (our run) (0.270 m)
SFUISE (0.190 m)
LM (0.212 m)
KIMS (0.211 m)
x [m]
y [m]
z [m]
time since ground-truth start [s]